Newcomb's Paradox as a Postselected Closed Timelike Curve

Z. Hunter [1, 2, *]

[1] *Ontologic Labs*
[2] *Universitat de Barcelona, Barcelona, Spain*

(Dated: 20 August 2026)

Newcomb's paradox is a classical thought experiment in decision theory: an agent takes either an unobserved box alone (one-boxing), or that box together with a second box holding a small, already observed reward (two-boxing). The unobserved box contains a large reward only if a predictor had earlier forecast one-boxing. We model forecast and act as binary quantum records and impose consistency through a postselected closed-timelike-curve (P-CTC) final boundary. A parity-dependent Bell interaction weights forecast–act agreement and mismatch histories differently; its ideal limit removes forecast errors, and opening the Bell contraction yields an equivalent forward-time heralded parity circuit. Dephasing the consistency setting produces a noisy P-CTC with the same record weights, so forecast errors are realized physically as decoherence. For an unbiased forecast prior, one-boxing has the larger expected payoff once the accepted forecast is more accurate than a threshold set by the reward ratio; at standard stakes, a forecast barely better than a coin flip suffices. Symmetrizing the two records gives two Newcomb problems side by side, and weighting each history by its survival probability changes the symmetric payoff table from Prisoner's Dilemma to Stag Hunt, with distinct equilibrium and risk-dominance thresholds. A classical rejection filter reproduces every payoff-basis ordering, while complementary-basis measurements distinguish the coherent, noisy, and classical implementations. These results separate the decision-theoretic consequences of postselected consistency from its quantum signatures, recasting Newcomb's predictor as a finite-accuracy time-travel consistency condition and, in the symmetric construction, as a game between parallel worldlines of the same agent.

I. INTRODUCTION

William Newcomb posed the problem; Nozick [1] brought it to philosophy, and Gardner's *Scientific American* columns [2, 3] made it widely known. A transparent box contains a small, already observed reward m, while an unobserved opaque box contains either a much larger reward M or nothing. In the standard million-versus-thousand example, $M = \$1{,}000{,}000$ and $m = \$1{,}000$. An agent chooses either to take only the opaque box, one-boxing, or to take both boxes, two-boxing. Before the choice is made, a predictor, denoted Ω, has already filled the opaque box if it forecast one-boxing and left it empty if it forecast two-boxing. Taking both boxes always adds the transparent reward, so two-boxing dominates: it pays more for either fixed box content. Yet a sufficiently reliable predictor makes one-boxers far more likely to face a filled opaque box. That tension between dominance and correlation is Newcomb's paradox. The main families of proposed resolutions differ in which relation between act and forecast is held fixed when the decision is evaluated [4–9]. The disagreement can be stated in terms of the joint probability distribution assigned to forecast and act; Wolpert and Benford [10] distinguish two probabilistic structures relating the two (Sec. III C). Underneath it lies a physical issue: how can a forecast made earlier remain reliably correlated with an act performed later? The construction below realizes that correlation as a postselected consistency condition and makes its strength continuously tunable.

A physicist may decline to take a position on the existence of free will and base his decision on the result of a quantum measurement, much as Schrödinger's cat's destiny depends on a measurement outcome. We therefore model the act as a qubit X. The physicist controls how X is prepared; the recorded act is its final computational-basis outcome, the readout in the fixed binary basis $\{|0\rangle, |1\rangle\}$, carried out as one-boxing or two-boxing. The forecast is stored in a second binary register Y. The predictor is granted every classical fact about the setup, including the preparation of X; the single thing denied it is the realized act itself. This restriction is Newcomb's temporal premise in quantum form. Classically, once a bit x exists, Ω can copy it and implement a policy $y(x)$ without changing what x will be. With a decision qubit, a nondemolition copy of the computational-basis value can preserve the later one-boxing and two-boxing probabilities. For a coherent preparation, however, a perfectly readable early record removes the coherence between the two act values, the quantity probed by the complementary-basis measurements below. An early classical fact about the act would already be such a record; Assumption 1 in Sec. II denies Ω that record and makes its physical cost explicit.

Once the predictor is forbidden from reading out X early, prediction consistency must be enforced indirectly, by acting on the joint state of the two records. Newcomb's predictor and the grandfather paradox then become two versions of the same physical question: how a theory represents a consistency constraint between a record and an event. In the grandfather paradox, inconsistent histories are those in which an event on the temporal loop prevents its own cause. In Newcomb's problem, they can be read

* z@ontologiclabs.com

as the mismatch histories, in which the predictor records one act and the agent later performs the other. In a literal retrocausal reading, Ω would send information to the past.[1]

Closed-timelike-curve (CTC) models provide the language for that constraint. These models divide the world into a chronology-respecting system and a CTC system that meet through a localized interaction. Deutsch's model requires the state of the CTC system to be a fixed point of that interaction [18]. In Newcomb-like situations, however, the central object is the correlation between the CTC system and the external records, which the Deutsch map can wash out [19]. Postselected closed timelike curves (P-CTCs), proposed in unpublished work of Bennett and Schumacher [20], independently by Svetlichny [21], and developed by Lloyd and collaborators, instead model the CTC through postselected quantum teleportation [22, 24]. One half of a Bell pair interacts with the chronology-respecting system, after which the two Bell registers are projected back onto the state in which they were prepared; the Bell pair stands in for the loop. This final Bell projection defines a trace-decreasing, pre-normalized output branch. Conditional on that outcome, the output state is the nonlinear P-CTC update, $\rho \mapsto C_{\mathcal{U}}\rho C_{\mathcal{U}}^{\dagger}/\operatorname{Tr}(C_{\mathcal{U}}\rho C_{\mathcal{U}}^{\dagger})$, with $C_{\mathcal{U}}$ the Bell projection of the interaction $\mathcal{U}$.

Section II implements this final boundary on the joint forecast–act record. A parity-dependent Bell interaction assigns different amplitudes to agreement and mismatch, and cutting open the contracted Bell line gives a forward-time parity-ancilla circuit with the same selected operator on XY; we call this selected map the parity filter. Nonzero forecast error also has a physical realization as noise: Sec. II C constructs a binary noisy P-CTC in which bit-flip noise on the CTC system produces the forecast errors, an instance of the noisy P-CTCs of Ji, Lloyd, and Wilde [38]. The coherent parity filter and the noisy channel can therefore share the same agreement and mismatch acceptance probabilities while acting differently on superpositions of agreement and mismatch. Section II derives the Bell contraction, its forward-time opening, and this binary noisy realization; Sec. VII generalizes the noise to dephasing of the consistency setting and derives its consequences for the forecast–act correlation, coherence, classicality, and causal order. Appendix A fixes the Bell-boundary conventions.

The Bell contraction and the opened parity circuit are two drawings of one construction: they induce the same parity filter on the two records. Conditioning on the corresponding recorded outcome in the opened circuit therefore reproduces the Bell-boundary result in an ordinary laboratory experiment. Whether a spacetime realization could enforce such a final boundary is a separate question, discussed in Sec. VIII B.

Time-travel stories have long imagined futures whose weight is felt before they arrive, from the worldlines of *Steins;Gate* to the return of the Roaring Kitty in 2024.[2] President Donald Trump stated that "everything is computer" [26], but could it be that everything is closed timelike curve?[3] Taken literally, each would set some ceiling on the Universe's computational power. Lloyd argues the Universe can be regarded as a quantum computer [27]; extending that picture, let Q be the conjecture that this ceiling is BQP.[4] If selected final boundaries are also physical and act only as postselection, the ceiling would become PostBQP, which Aaronson proved equal to PP [30], a class that already contains the NP-complete problems, and a single qubit traversing a P-CTC suffices to perform any postselected quantum measurement with certainty [33]. Call Q_{post} the conjecture that nature supplies this postselection resource, such as in a Universe as a postselected quantum computer, so that its ceiling is PostBQP = PP. The containment BQP $\subseteq$ PP is proved and widely believed to be strict, though a strict separation would imply P $\neq$ PSPACE and is itself unproven [34–36]. Whether Q_{post}'s ceiling sits strictly above Q's therefore rests on postselection enlarging BQP, and on final boundary conditions that nature may supply. The boundary studied here is as small as Newcomb's question allows: two binary records, the earlier forecast and the later act, tied by a single consistency condition; in the opened circuit it amounts to postselecting a single ancilla qubit.

The present construction can perhaps be best explained with reference to the short film *Stag Hunt* [37], as Ji, Lloyd, and Wilde explain theirs with reference to *Interstellar* [38]. The film is the kernel of the present analysis: reaching a Stag Hunt through Newcomb's problem and a postselected closed timelike curve.[5] We read the mechanism

[1] Newcomb and the Benford of Ref. [10] also coauthored the paper naming the tachyonic antitelephone [17].

[2] The Roaring Kitty is a central figure of the 2020-ongoing great meme-stock sagas involving GameStop Corp. and pre-bankruptcy Bed Bath & Beyond Inc. His 2024 return took the form of a weekslong stream of 110 most excellent memes, which followers decoded as a timeline running in reverse towards an awaited final boundary condition. By the time this paragraph was written, another recurring figure from those sagas was serving as acting U.S. Director of National Intelligence [25]. These recent developments highlight the importance of time travel in national security and financial markets.

[3] A P-CTC loop is an entangled pair with a selected final boundary (Sec. II), locally indistinguishable from ordinary forward evolution; with entanglement ubiquitous, the question reduces to whether nature supplies such boundaries, as the black-hole final-state conjectures suggest [64].

[4] Here Q stands for quantum. The complexity class BQP, the problems a quantum computer solves efficiently, is the upshot of the quantum extended Church–Turing thesis, that efficient physical computation and uniform polynomial-size quantum circuits simulate one another; Deutsch's physical Church–Turing principle is the earlier, weaker form, asking only that every physical process be simulable in principle, not efficiently [28]. Bostrom asks a related question, whether a computer outside the Universe is running it as a simulation [29]; Q concerns the Universe's own evolution.

[5] Intertitles track two protagonists as Alice and Bob cooperating

of the film as the postselected consistency condition just described. In the film the characters seal a room and obtain a box; we read the box as the unobserved box of Newcomb's problem and the sealing as isolation from the noise that would otherwise both remove the act register's coherence and lower the consistency of the forecast, the two effects of dephasing developed in Sec. VII. In the present analysis, treating both records as decisions of one agent turns the symmetric game of Secs. IV and V into a self-consistency game against another version of oneself. Section V develops this reading in congruence with the film's exchange of dead for alive: the two worldlines may code death and survival oppositely, and consistency after decoding then reads $y = x$ or $y = x \oplus 1$, with the film's closing pull-back to a field of histories read there as the ensemble of histories that survive the consistency condition.

That consistency condition is enforced by the parity filter, a postselected parity check on the forecast record Y and the act register X. It retains agreement histories ($x = y$) at rate p_{match} and mismatch histories ($x \neq y$, forecast errors) at rate p_{mismatch}. Let A (for acceptance) denote the event that the check is passed and the branch retained, so that $p_{\mathrm{match}} = \Pr(\mathsf{A} \mid x = y)$ and $p_{\mathrm{mismatch}} = \Pr(\mathsf{A} \mid x \neq y)$; records conditioned on A are the accepted histories. In the opened circuit, A is an ordinary heralding outcome, and the forecast–act correlation exists only in the subensemble it identifies; the P-CTC reading casts the accepted record as a forecast in hand before the act. The odds ratio of the two rates,

$$\Gamma := \frac{p_{\mathrm{match}}}{p_{\mathrm{mismatch}}}, \tag{1}$$

is the *consistency factor*: how many times more likely an agreement history is to survive selection than a mismatch history. The payoff analysis below needs only this odds ratio. At $\Gamma = 1$ the postselection supplies no consistency bias; as $\Gamma \to \infty$ mismatch histories are eliminated and the accepted forecast becomes exact: the infallible-predictor limit of Newcomb's problem and, were one to identify the predictor with the agent, Novikov's consistency principle [71] in full. Finite Γ interpolates between the two, retaining forecast errors at relative weight Γ^{-1}. In the noisy realization, a noiseless CTC interaction would enforce a perfect forecast, $y = x$ in every accepted history, and forecast error is a bit flip of probability ℓ: a correct forecast survives only when the loop qubit is not flipped and a mismatch survives only when it is, so $p_{\mathrm{match}} = 1 - \ell$, $p_{\mathrm{mismatch}} = \ell$, and $\Gamma = (1 - \ell)/\ell$ is the odds against a noisy P-CTC error. On this reading, a time machine need not be all or nothing: dephasing the consistency setting lowers Γ continuously from the ideal limit, while a random relative phase between agreement and mismatch histories costs the act register's coherence without moving Γ (Sec. VII).

The payoff analysis reads both registers in the computational basis, which we also call the record or payoff basis. On a definite record the filter only reweights, a match by p_{match} and a mismatch by p_{mismatch}; because that is all it does, for product inputs diagonal in this basis the payoff ordering of any two act preparations depends on the filter through Γ alone, and a classical rejection filter with the same weights reproduces every payoff-basis ordering (Sec. II E). No gate ever writes a value into either register. When the ancilla outcome is ignored, the opened circuit does not reweight agreement relative to mismatch; conditioning on A produces the consistency bias (Sec. II B). Whatever is quantum about the construction must therefore live off the payoff diagonal, in the information the payoff table discards.

We compare three filters matched on the two weights: the coherent parity filter, the binary noisy P-CTC, and the classical rejection filter. Matched this way, they have identical payoff-basis statistics and differ only in the coherence they keep. Split XY into an agreement sector ($x = y$) and a mismatch sector ($x \neq y$). We call *local coherence* the coherence a single register shows on its own, measured by the Pauli X, σ_x, on either register: on the act register it is the surviving superposition of one-boxing and two-boxing, which a readable early act record would erase (Assumption 1). The *joint coherence* instead shows up only when X and Y are measured together: the phase relation between $|00\rangle$ and $|11\rangle$, or between $|01\rangle$ and $|10\rangle$, invisible to either register alone. The coherent filter keeps both kinds of coherence and the classical rejection filter keeps neither, while the binary noisy P-CTC keeps only the joint coherence. Section VI and Appendix H derive the measurements that distinguish the three at finite Γ.

On a definite record (x, y) the opaque box pays M when the forecast is one-boxing and nothing otherwise, and two-boxing adds the transparent reward m. Write it $u(x, y)$. The same u is scored across the selected branch in two ways. Conditioning on acceptance gives the accepted-history payoff $\bar{U}_{\mathrm{acc}}$, the expected payoff among accepted histories. Keeping the branch weight instead gives the selected payoff yield $\bar{U}_{\mathrm{sel}} = \Pr(\mathsf{A})\bar{U}_{\mathrm{acc}}$: the payoff averaged over every run, with a discarded run scored as zero. $\bar{U}_{\mathrm{acc}}$ is what an agent inside the branch sees, to whom only accepted runs happen; $\bar{U}_{\mathrm{sel}}$ is what an outside bookkeeper sees, counting every attempt and scoring a discard as no win (Sec. III). The selected payoff yield is best motivated in the symmetric game of Secs. IV and V, where the two records become two players, only their payoffs relative to

---

or defecting. One of them recounts a paradox about a father and his son: there is a single elixir, and whoever drinks it dies. The son means to drink it; the father, cast as the predictor, builds his prediction by replay, looping back and altering events until his son's death is averted, a success that, as in the grandfather paradox, removes his reason to have returned at all. Spinoza is invoked to read Alice and Bob as a single entity on parallel worldlines; the Bhagavad Gita arrives through Oppenheimer's broadcast, its Sanskrit *kāla* reading both Death and Time; much of the late dialogue repurposes *Hamlet*, "to be and not to be" turning the soliloquy's either/or into a superposition; and the exchange "What is a body in a wavefunction anyway?"–"Universal memory" completes the film's essence: Being, Death, and Time.

each other count, and the yield's acceptance-weighting of each $u(x, y)$ builds the symmetric table that passes from Prisoner's Dilemma to Stag Hunt. The two scorings can part only when $\Pr(\mathsf{A})$ depends on the act preparation; at the unbiased forecast prior it does not, and the two agree at every Γ.

Lewis [11] read a Prisoner's Dilemma as two Newcomb problems placed side by side, and Sobel [12] asked which Newcomb problems have that structure. Our analysis builds the same symmetry into the selected payoffs in two steps. Section IV first gives the forecast register a Newcomb payoff of its own, the mirror image of the agent's under exchange of the two record labels: writing u_1 for the choosing register's payoff and u_2 for the forecast register's, $u_2(x, y) = u_1(y, x)$. Section V then promotes Y from a passive record to a decision register in its own right, so the two Newcomb problems sit in one symmetric table in which each player's action is the other's forecast, and reads the two players as parallel worldlines of a single agent.

Classifying the symmetric table draws on a few standard definitions from game theory. A normal-form game assigns each player a payoff for every profile, a pair of choices, one per player. A Nash equilibrium is a profile at which neither player gains by changing its own choice, and it is strict when every such change strictly loses [39]. A choice strictly dominates another when it pays more against every choice of the other player. In a Prisoner's Dilemma, defection strictly dominates cooperation, so mutual defection is the unique equilibrium even though mutual cooperation pays both players more. In a Stag Hunt, both diagonal profiles are equilibria, and the risk-dominant equilibrium is the one whose choice is the better reply against an opponent equally likely to play either option [41, 42]. In the Newcomb table, one-boxing plays the role of cooperation and two-boxing of defection, so coordination means mutual one-boxing. These two games classify the symmetric selected-payoff table, carrying the Lewis–Sobel classification one type further: as Γ increases, the table passes from Prisoner's Dilemma to Stag Hunt when mutual one-boxing first becomes an equilibrium, and crosses a second threshold when that equilibrium becomes risk-dominant, while ordinary two-box dominance at each fixed forecast holds at every Γ. Schmidt-Petri [16] also obtains a coordination verdict by extending Newcomb's problem to repeated play; here the two transitions occur in a single application of the filter.

Compared with earlier P-CTC constructions and quantum treatments of Newcomb's problem [22, 24, 95–97] (Sec. VIII A), the main results are:

1. *Classical sufficiency.* Every payoff-basis comparison depends only on Γ, and a classical rejection filter with the same weights reproduces every ordering, so the three implementations give the same verdict. Restricted to payoff-basis statistics, the construction is standard Newcomb with a correlated predictor whose accepted forecast is correct with reliability $q = \Gamma/(\Gamma+1)$: a one-parameter physical realization of the standard finite-accuracy problem. At the unbiased forecast prior the two scorings agree and one-boxing has the larger payoff once the distinguishability $\mathcal{D} = (\Gamma-1)/(\Gamma+1)$ exceeds m/M, at the standard stakes only $1/1000$, a predictor barely better than a coin flip.
2. *Prisoner's Dilemma to Stag Hunt.* Treating both records as decisions carries the symmetric selected-payoff table from Prisoner's Dilemma to Stag Hunt as Γ grows, mutual one-boxing first becoming a Nash equilibrium and then risk-dominant at the thresholds derived in Sec. IV. Between them, mutual one-boxing is already an equilibrium of the pair while the single-agent verdict is still two-boxing.
3. *Coherence diagnostics.* At equal Γ the payoff statistics coincide, while the local and joint coherences separate the filters. On the equal-superposition diagnostic input of Sec. VI, the coherent filter carries the transverse visibility $\mathcal{V}$, measuring the surviving single-register coherence, and saturates $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$; both quantum outputs are entangled for $\Gamma > 1$, their concurrence [90] equal to the $\mathcal{D}$ that sets every game threshold; the classical output is separable, and only the coherent output keeps the local coherence.
4. *Dephasing, causal order, and classicality.* A symmetric spread of the consistency setting removes the local coherence between agreement and mismatch, while widening the spread reduces the accepted forecast–act bias. When agreement and mismatch reach equal accepted weight, whether selection succeeds no longer depends on their parity; an ordering in which the selection outcome lies earlier than both records becomes compatible with ordinary locally sequential causality, satisfying parity erasure [56]. Simultaneously, the corresponding three-record tensor can be oriented with all three records as inputs without requiring the tensor to feed back into itself. Removing the remaining joint coherence gives the classical rejection filter.
5. *Retrocausal capacity.* At the unbiased forecast prior the accepted map from act X to forecast Y is a binary symmetric channel, and the retrocausal-capacity definition of Ji, Lloyd, and Wilde [38] gives $C_{\mathrm{retro}} = 2Q_{\mathrm{retro}} = \log_2 \Gamma$: the consistency factor that fixes forecast reliability and every payoff threshold is, in capacity units, the retrocausal communication strength of the induced act-to-forecast channel.

Section II constructs the consistency condition and opens the Bell contraction into a forward-time circuit. Section III then applies the filter to Newcomb's payoffs, and Secs. IV and V turn the resulting table into the Prisoner's-Dilemma–to–Stag-Hunt transition and its side-by-side/worldline reading. Section VI asks what remains quantum once the payoff statistics are known, and Sec. VII develops dephasing of the consistency setting and its consequences for coherence, classicality, and causal order. The payoff thresholds and coherence diagnostics hold for any realization of the forecast–act consistency constraint, and no causal reading of it, cyclic or ordinary, moves them; the Discussion separates what the laboratory statistics establish from the stronger time-machine reading. Fixed-prior generalizations, adaptive predictor response rules, and the longer causal and interpretive checks are left to the appendices.

II. P-CTC CONSTRUCTION OF THE CONSISTENCY CONDITION

Register Y stores the predictor's forecast and register X stores the act; their final computational-basis values are y and x. Agreement, $x = y$, is a correct forecast and mismatch, $x \neq y$, is a forecast error. We impose this relation with the postselected-teleportation representation of a P-CTC [22, 24]: a pair of loop registers L, L' is prepared in a Bell state $|\Phi\rangle_{LL'}$, L interacts with X and Y, and the pair is projected back onto the same Bell state; the contracted pair traces the loop, L the leg that meets the records and L' the return leg. For an interaction $\mathcal{U}_{L,XY}$, this Bell boundary, the paired preparation and final projection, induces the operator

$$C_{\mathcal{U}} = {}_{LL'}\langle\Phi| \left(\mathcal{U}_{L,XY} \otimes I_{L'}\right) |\Phi\rangle_{LL'} = \frac{1}{2} \operatorname{Tr}_L \mathcal{U}_{L,XY} \quad (2)$$

on XY. The selected branch is produced by the completely positive, trace-nonincreasing map $\rho \mapsto C_{\mathcal{U}}\rho C_{\mathcal{U}}^\dagger$, whose trace is the probability of the selected Bell outcome; conditional on that outcome, the P-CTC output is

$$\rho \longmapsto \frac{C_{\mathcal{U}}\rho C_{\mathcal{U}}^\dagger}{\operatorname{Tr}(C_{\mathcal{U}}\rho C_{\mathcal{U}}^\dagger)}. \quad (3)$$

The construction below chooses $\mathcal{U}$ so that the boundary favors agreement over mismatch, then derives an ordinary forward-time circuit with the same selected operator. Figure 1 fixes the geometry: only the Bell legs are contracted, while the record line remains open.

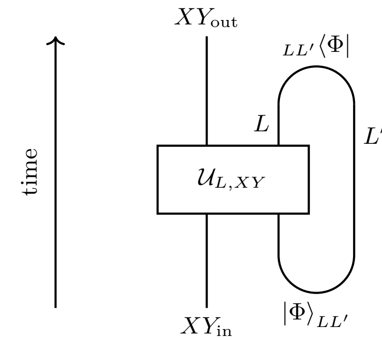


FIG. 1. Bell-boundary representation of the P-CTC contraction. The open line XY enters the interaction $\mathcal{U}_{L,XY}$ and continues to the output. The loop qubit L is entangled below with L' in $|\Phi\rangle_{LL'}$ and postselected above onto the same Bell state. Contracting the Bell legs induces $C_{\mathcal{U}} = {}_{LL'}\langle\Phi| (\mathcal{U}_{L,XY} \otimes I_{L'}) |\Phi\rangle_{LL'}$ on XY.

Assumption 1 (predictor access). *The predictor knows every classical fact about the setup, including the preparation of the act register. It has no access to the outcome produced by the final measurement.*

Quantum mechanics also quantifies the coherence cost of an earlier perfectly readable act record. Let $|C\rangle \equiv |0\rangle$ denote one-boxing and $|D\rangle \equiv |1\rangle$ denote two-boxing, and let a record $\mathcal{R}$ begin in $|0\rangle_{\mathcal{R}}$. A premeasurement, the correlating interaction of a measurement without its readout, records the act without changing it,

$$\begin{aligned} |C\rangle_X |0\rangle_{\mathcal{R}} &\mapsto |C\rangle_X |r_C\rangle_{\mathcal{R}}, \\ |D\rangle_X |0\rangle_{\mathcal{R}} &\mapsto |D\rangle_X |r_D\rangle_{\mathcal{R}}. \end{aligned}$$

Linearity then gives, for $c_C |C\rangle + c_D |D\rangle$,

$$\langle C|\rho'_X|D\rangle = c_C c_D^* \langle r_D|r_C\rangle, \quad (4)$$

where ρ'_X is the reduced act state after tracing out $\mathcal{R}$; its displayed off-diagonal element is the act's surviving superposition. A perfectly readable record has $\langle r_C|r_D\rangle = 0$, so it removes the C/D off-diagonal element while leaving the C/D probabilities unchanged. Preserving the magnitude of the act-register coherence requires $|\langle r_C|r_D\rangle| = 1$; the record states then coincide up to phase and reveal no act value. Section VI and Appendix H give the resulting distinguishability–visibility tradeoff [48–50].

A. Agreement/mismatch sectors and the coherent filter

The four possible basis records split into agreement and mismatch subspaces,

$$\begin{aligned} \Pi_= &:= |00\rangle\langle 00| + |11\rangle\langle 11|, \\ \Pi_{\neq} &:= |01\rangle\langle 01| + |10\rangle\langle 10|. \end{aligned} \quad (5)$$

Each sector is spanned by a pair of Bell states of the records, $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$ for agreement and $|\Psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$ for mismatch; we write Φ for the agreement sector and Ψ for the mismatch sector. These combinations share their form with the loop pair's Bell state but live on the record registers; only the loop pair is contracted. Perfect consistency selects only the agreement subspace, so the ideal selected operator is $\Pi_=$. Forecast errors require a nonzero selected weight for $\Pi_{\neq}$.

For concreteness, take

$$|\Phi\rangle_{LL'} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{6}$$

as the initial and final Bell state. The construction is parameterized by an angle θ, the consistency setting: the loop interaction below shifts the two sector angles together, $\theta/2$ on agreement and $\pi/2+\theta/2$ on mismatch, a right angle apart, so that the mismatch amplitude $\cos(\pi/2+\theta/2) = -\sin(\theta/2)$ vanishes at the ideal point $\theta = 0$ and is negative beyond it. Collecting the two angles into one Hermitian operator, define

$$\Theta_\theta = \frac{\theta}{2}\Pi_= + \left(\frac{\pi}{2} + \frac{\theta}{2}\right)\Pi_{\neq} \tag{7}$$

and the loop interaction, with σ_z^L the Pauli Z on the loop qubit,

$$\mathcal{U}_\theta = \exp(i\sigma_z^L \otimes \Theta_\theta). \tag{8}$$

On a sector whose angle in Eq. (7) is ϑ, the interaction applies the loop phase $e^{i\sigma_z^L\vartheta} = \cos\vartheta\, I_L + i\sin\vartheta\,\sigma_z^L$; the σ_z^L part moves the loop pair onto the orthogonal Bell state and is discarded by the final projection, so the sector survives with amplitude $\cos\vartheta$; the discarded amplitude, $i\sin\vartheta$, appears in the orthogonal Bell outcome. A decoupled loop, $\mathcal{U} = I_L \otimes U'$, contracts to the unitary U' and leaves every branch accepted. A traceless operator on the loop alone contracts to zero, so no branch is ever accepted; the in-loop NOT with which Lloyd et al. tested the grandfather paradox is of this kind, and its vanishing weight is how the P-CTC forbids the paradox [22]. Equation (2) therefore gives

$$\begin{aligned} C_{\mathcal{U}_\theta} &= \frac{1}{2}\left(e^{i\Theta_\theta} + e^{-i\Theta_\theta}\right) = \cos\Theta_\theta \\ &= \cos(\theta/2)\Pi_= - \sin(\theta/2)\Pi_{\neq} =: K(\theta). \end{aligned} \tag{9}$$

The coherent filter's selected map is $\Lambda_\theta : \rho \mapsto K(\theta)\rho K^\dagger(\theta)$, where $K(0) = \Pi_=$ is the ideal P-CTC consistency check. At nonzero θ, agreement and mismatch survive in one coherent selected branch, with probabilities $\cos^2(\theta/2)$ and $\sin^2(\theta/2)$; the relative minus sign remains observable on superpositions of the two sectors.

B. Forward-time opening

The Bell preparation and final Bell projection can be opened into an ordinary heralded circuit. Prepare an ancilla a in $|0\rangle$ and use two controlled-NOT (CNOT) gates to write the record parity into it, the ancilla ending in $|0\rangle$ on agreement and $|1\rangle$ on mismatch,

$$|x\rangle_X\,|y\rangle_Y\,|0\rangle_a \longmapsto |x\rangle_X\,|y\rangle_Y\,|x\oplus y\rangle_a\,. \tag{10}$$

Apply

$$R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \tag{11}$$

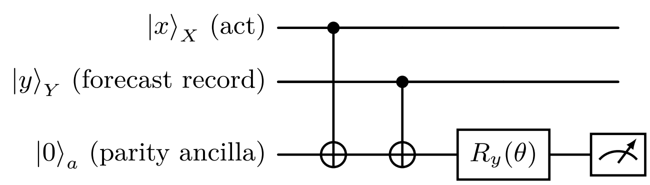


FIG. 2. Forward-time opening of the Bell contraction. The CNOT gates write the record parity $x\oplus y$ into the ancilla; after $R_y(\theta)$, the branch is retained only for the outcome $a = 0$. Agreement and mismatch then survive with weights $\cos^2(\theta/2)$ and $\sin^2(\theta/2)$, and the retained Kraus operator is exactly $K(\theta) = C_{\mathcal{U}_\theta}$.

and retain the outcome $a = 0$. The retained overlaps are

$$\begin{aligned} \langle 0|\,R_y(\theta)\,|0\rangle &= \cos(\theta/2), \\ \langle 0|\,R_y(\theta)\,|1\rangle &= -\sin(\theta/2), \end{aligned}$$

so the opened circuit acts with the same coefficient as $K(\theta)$ on every basis record. Figure 2 displays this forward-time realization. Hence

$$\begin{aligned} &\langle 0|_a\, R_y(\theta)\,\mathrm{CNOT}_{Y\to a}\,\mathrm{CNOT}_{X\to a}\,|0\rangle_a \\ &\quad = K(\theta) = C_{\mathcal{U}_\theta}. \end{aligned} \tag{12}$$

Equation (12) is the equivalence used throughout. In the laboratory, the ancilla outcome $a = 0$ heralds the selected branch. In the P-CTC representation, the same branch is selected by the final Bell projection. Because Eq. (12) is an operator identity, the two representations assign every input ρ the same selected branch $K(\theta)\rho K^\dagger(\theta)$, hence the same selection probability and conditional state on XY. From this point onward, A denotes the retained event in either representation: $a = 0$ in the opened circuit and the selected Bell outcome in the closed one; the distribution of (x, y) conditioned on A is the accepted ensemble.

If the ancilla outcome is ignored, the $a = 0$ branch $K(\theta)$ of Eq. (9) is summed with the complementary $a = 1$ branch $K_1(\theta) = \sin(\theta/2)\Pi_= + \cos(\theta/2)\Pi_{\neq}$, obtained from Eq. (11); their agreement–mismatch cross terms cancel, giving

$$\rho \longmapsto \Pi_=\rho\Pi_= + \Pi_{\neq}\rho\Pi_{\neq}.$$

Since $\Pi_=$ and $\Pi_{\neq}$ are complementary projectors, this is a trace-preserving channel. It removes coherence between the agreement and mismatch sectors without reweighting either one. The consistency bias appears only after conditioning on A.

C. Matched noisy implementation

Decoherence of the consistency setting turns the ideal filter into a noisy P-CTC. Keep the ideal parity interaction

$$\mathcal{U}_{\rm par} = I_L \otimes \Pi_= + \sigma_x^L \otimes \Pi_{\neq}, \qquad C_{\mathcal{U}_{\rm par}} = \Pi_=, \tag{13}$$

where σ_x^L is the Pauli X (NOT) gate on the loop qubit. A bit-flip channel of error probability ℓ on the loop, inserted between this interaction and the final Bell projection (the placement is immaterial, Appendix A), contracts to the *binary* noisy P-CTC, the incoherent two-branch selected map

$$\mathcal{N}_\ell^{\text{noise}}(\rho) = (1-\ell)\Pi_= \rho \Pi_= + \ell \Pi_{\neq} \rho \Pi_{\neq}. \quad (14)$$

The forecast errors are the loop-qubit flips: $p_{\text{match}} = 1-\ell$, $p_{\text{mismatch}} = \ell$, and $\Gamma = (1-\ell)/\ell$. The bit flip is a channel on L; its contraction, Eq. (14), takes XY states to unnormalized outputs, and normalizing them gives the nonlinear P-CTC update. Conditioning that update on success defines the accepted-record channel of Sec. II D, derived in Appendix J. The coherent filter and the binary noisy map are matched, with equal weights and equal Γ, at $\ell = \sin^2(\theta/2)$; throughout we take the agreement-favoring range $0 \le \ell \le 1/2$, equivalently $0 \le \theta \le \pi/2$ and $\Gamma \ge 1$. The two realizations therefore share the same agreement and mismatch probabilities while acting differently on superpositions of the two sectors. Section VII treats a distinct unrecorded distribution of the consistency setting. This places the construction in the noisy-P-CTC framework of Ji, Lloyd, and Wilde, in which the traversing system returns to its past through a noisy channel [38]; here the microscopic loop-noise channel is the bit-flip channel. The capacity calculation in Sec. II D concerns the accepted-record channel, from act X to forecast Y, under the retrocausal reading. The three implementations agree on every payoff-basis statistic and are separated in Sec. VI by local and joint complementary-basis measurements.

D. Consistency factor and accepted reliability

The selected maps constructed above, Λ_θ and $\mathcal{N}_\ell^{\text{noise}}$, are completely positive and trace-nonincreasing; for any such map Λ, define

$$\begin{aligned} p_{\text{sel}}(\rho) &:= \operatorname{Tr}[\Lambda(\rho)], \\ \rho_{\text{sel}}(\rho) &:= \frac{\Lambda(\rho)}{p_{\text{sel}}(\rho)}, \qquad p_{\text{sel}}(\rho) > 0. \end{aligned} \quad (15)$$

On a basis record,

$$\begin{aligned} p_{\text{match}} &:= \Pr(\mathsf{A} \mid x = y), \\ p_{\text{mismatch}} &:= \Pr(\mathsf{A} \mid x \neq y), \end{aligned} \quad (16)$$

recovering the consistency factor of Eq. (1), $\Gamma = p_{\text{match}}/p_{\text{mismatch}} = \cot^2(\theta/2)$ for the coherent realization. Each selected map fixes its acceptance probability through a single operator E, its selection effect, defined by $p_{\text{sel}}(\rho) = \operatorname{Tr}[E\rho]$; for a map with Kraus operators $\{K_r\}$ it is $E = \sum_r K_r^\dagger K_r$. The coherent filter and the matched noisy map share the selection effect

$$E = p_{\text{match}} \Pi_= + p_{\text{mismatch}} \Pi_{\neq}. \quad (17)$$

Thus Γ is the likelihood ratio that acceptance supplies between agreement and mismatch: the accepted record is no better than a coin flip at $\Gamma = 1$ and predicts the act with certainty as $\Gamma \to \infty$.

Suppose X and Y are prepared independently with $\Pr(X = 0) = \alpha$ and $\Pr(Y = 0) = \beta$. Before selection,

$$\begin{aligned} \Pr(x = y) &= \alpha\beta + (1-\alpha)(1-\beta), \\ \Pr(x \neq y) &= \alpha(1-\beta) + (1-\alpha)\beta. \end{aligned}$$

Bayes' rule gives[6]

$$\Pr(x = y \mid \mathsf{A}) = \frac{\Gamma \Pr(x = y)}{\Gamma \Pr(x = y) + \Pr(x \neq y)}, \quad (18)$$

or, equivalently,

$$\frac{\Pr(x = y \mid \mathsf{A})}{\Pr(x \neq y \mid \mathsf{A})} = \Gamma \frac{\Pr(x = y)}{\Pr(x \neq y)} : \quad (19)$$

Γ multiplies the prior odds of agreement. At the unbiased forecast prior $\beta = 1/2$, agreement and mismatch each have prior probability $1/2$ for every α, and the accepted-record accuracy is

$$q := \Pr(Y = X \mid \mathsf{A}) = \frac{\Gamma}{\Gamma + 1}. \quad (20)$$

The accepted record's distinguishability, the normalized advantage over chance for inferring X from Y, is

$$\mathcal{D} := 2q - 1 = \frac{\Gamma - 1}{\Gamma + 1}. \quad (21)$$

Section VI returns to this $\mathcal{D}$ as the which-way distinguishability paired with the transverse visibility.

The same accepted reliability also gives Γ an information-theoretic meaning. At $\beta = 1/2$, after conditioning on acceptance, the forecast Y is the output of a binary symmetric channel with input X, agreement probability $(1+\mathcal{D})/2$, and crossover probability $(1-\mathcal{D})/2$. Ji, Lloyd, and Wilde define retrocausal classical and quantum capacities for a noisy P-CTC channel, one whose output precedes its input, and express them through the max-information and the regularized Doeblin information of that channel, $C_{\text{retro}} = I_{\max} + I_{\text{doe}}^\infty = 2Q_{\text{retro}}$ [38]. Under the retrocausal-channel reading the accepted-record channel has that orientation, the later act X its input and the earlier forecast Y its output. Appendix J evaluates the two terms for it, $I_{\max} = \log_2(1+\mathcal{D})$ and $I_{\text{doe}}^\infty = -\log_2(1-\mathcal{D})$, hence

$$C_{\text{retro}} = 2Q_{\text{retro}} = \log_2 \frac{1+\mathcal{D}}{1-\mathcal{D}} = \log_2 \Gamma. \quad (22)$$

[6] Writing Bayes' rule treats A as an ordinary event to condition on, which is unproblematic in the open-circuit reading, whose causal structure is acyclic. In the closed P-CTC reading the loop makes it cyclic, so conditioning is no longer automatic; the operator identity of Sec. II B nonetheless gives the same posterior, and the cyclic quantum causal modelling framework of Ref. [63] obtains the cyclic probabilities from an associated acyclic model conditioned on successful postselection.

Thus the parameter that controls forecast reliability and the Newcomb payoff thresholds is, in capacity units, the information-carrying strength of the corresponding retrocausal channel: its asymptotic classical capacity is $\log_2 \Gamma$, and its quantum capacity is half as large. Its parameter $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$ is fixed by the accepted match/mismatch ratio and equals $1 - 2\ell$ for the binary noisy realization. The rate is per use of the retrocausal channel resource; the success probability of a forward-time heralded simulation depends on the full communication protocol (Appendix J).

E. Payoff-basis orderings depend only on Γ

The filters considered here preserve each computational-basis record and change only its selected weight,

$$\Lambda(|xy\rangle\langle xy|) = p_{xy}\,|xy\rangle\langle xy|\,,$$
$$p_{xy} = \begin{cases} p_{\mathrm{match}}, & x = y, \\ p_{\mathrm{mismatch}}, & x \neq y. \end{cases} \tag{23}$$

The coherent parity filter, the binary noisy P-CTC map, and the classical rejection filter, each diagonal in the record basis, all satisfy this condition.

Proposition (Γ-only orderings)**.** *For record-diagonal preparations* $\rho = \sum_{x,y} r_{xy}\,|xy\rangle\langle xy|$ *and payoff observables* $U = \sum_{x,y} u(x,y)\,|xy\rangle\langle xy|$*, the ordering of the selected-branch payoffs* $\mathrm{Tr}[U\Lambda(\rho)]$ *of any two preparations depends on a record-preserving filter only through* $\Gamma = p_{\mathrm{match}}/p_{\mathrm{mismatch}}$*. Provided both preparations have nonzero acceptance probability, the same holds for the accepted-history payoffs* $\mathrm{Tr}[U\rho_{\mathrm{sel}}(\rho)]$*, whose values themselves depend on the filter only through* Γ*.*

Proof. See Appendix B. □

The diagonal hypothesis is necessary: the transverse observables of Sec. VI are not diagonal in the record basis, so the proposition does not constrain them, and on the nondiagonal preparation used there their Γ-dependence separates the coherent construction from a classical rejection filter. Since only the ratio matters for orderings, a convenient normalization for $\Gamma \geq 1$ is the lossless one, $p_{\mathrm{match}} = 1$ and $p_{\mathrm{mismatch}} = 1/\Gamma$: agreement always passes, and discarding falls only on mismatch histories. The shared effect of Eq. (17) fixes the acceptance probabilities alone; the payoff read out after acceptance also requires the record-preserving property displayed above.

III. ACCEPTED-HISTORY PAYOFFS AND SELECTED PAYOFF YIELDS

A payoff observable $U = \sum_{x,y} u(x,y)\,|xy\rangle\langle xy|$, diagonal in the record basis, may be scored across the selected branch in two ways. The *accepted-history payoff* is the expected payoff after learning that acceptance occurred,

$$\bar{U}_{\mathrm{acc}}(\rho) = \mathrm{Tr}[U\rho_{\mathrm{sel}}(\rho)] = \frac{\mathrm{Tr}[U\Lambda(\rho)]}{p_{\mathrm{sel}}(\rho)}. \tag{24}$$

The *selected payoff yield* keeps the branch weight,

$$\bar{U}_{\mathrm{sel}}(\rho) = p_{\mathrm{sel}}(\rho)\bar{U}_{\mathrm{acc}}(\rho) = \mathrm{Tr}[U\Lambda(\rho)]. \tag{25}$$

The accepted-history payoff is what an agent inside the branch sees, since only accepted runs happen to that agent; the selected payoff yield is what an outside bookkeeper sees, counting every attempt and scoring a discarded run as no win. The two scorings can part only when $p_{\mathrm{sel}}(\rho)$ depends on the act preparation. At the unbiased forecast prior it does not (Theorem 1(i)), and the two give the same ordering of act preparations at every Γ.

For a deterministic record $|xy\rangle$, the selected map gives $\Lambda(|xy\rangle\langle xy|) = p_{xy}\,|xy\rangle\langle xy|$. The accepted-history payoff of the fixed record is therefore $u(x,y)$, while its selected-payoff entry is

$$u^{\mathrm{sel}}(x,y) = p_{xy}u(x,y). \tag{26}$$

The symmetric normal form of Secs. IV and V is a statement about this selected payoff yield. Appendix C gives the accepted-history calculation.

A. Base payoffs

Let m denote the transparent-box reward and M the possible opaque-box reward, with $M \gg m$ in the standard Newcomb setup [3]. Write $x \in \{C, D\}$ for the agent's action (one-boxing C versus two-boxing D), and $y \in \{C, D\}$ for the forecast variable (the predictor's setup); on the binary registers, $C \equiv 0$ and $D \equiv 1$. The choosing role receives the standard Newcomb payoffs, the payoff u of Sec. I,

$$\begin{aligned} u(C,C) &= M, & u(D,C) &= M + m, \\ u(C,D) &= 0, & u(D,D) &= m. \end{aligned} \tag{27}$$

We label Y so that $y = C$ means *predict* $X = C$ (opaque box filled) and $y = D$ means *predict* $X = D$ (opaque box empty). The diagonal records are correct forecasts, selected at p_{match}; the off-diagonal records are forecast errors: a two-boxer facing a filled opaque box receives $M + m$, and a one-boxer facing an empty one receives zero.

The reference case fixes an unbiased forecast record, $\beta := \Pr(Y = C) = 1/2$, independently of the act preparation; Appendix C gives both payoff scorings for general fixed β.

B. The invariant verdict at the unbiased forecast prior

At $\beta = 1/2$ the agent's preparation has no effect on the acceptance rate, and the two scoring conventions return the same verdict.

Theorem 1 (acceptance invariance and a normalization-independent verdict)**.** *Fix $M > m > 0$ and $1 \le \Gamma < \infty$, with the ideal case $p_{\text{mismatch}} = 0$ obtained as $\Gamma \to \infty$. Prepare the act and forecast registers independently, with* $\Pr(X = C) = \alpha$ *and* $\Pr(Y = C) = \beta$.[7]

(i) *At $\beta = 1/2$, the probability of acceptance is independent of α, so conditioning does not reweight the act marginal.*

(ii) *At that prior, the selected payoff yield and accepted-history payoff both assign a larger value to deterministic one-boxing than to deterministic two-boxing iff*

$$\Gamma > \frac{M+m}{M-m}.$$

(iii) *In terms of the accepted-record accuracy q of Eq. (20), the same boundary is*

$$q > \frac{M+m}{2M},$$

Proof sketch. At $\beta = 1/2$, agreement and mismatch each have prior probability $1/2$ for every α, so $\Pr(\mathsf{A}) = (p_{\text{match}} + p_{\text{mismatch}})/2$ is act-independent. The selected-yield comparison then gives $\Gamma(M - m) > M + m$, and the common acceptance probability makes the accepted-history ordering identical. Part (iii) substitutes $q = \Gamma/(\Gamma+1)$. Appendix C gives the fixed-prior formulas and the full proof.

The two normalizations are therefore two bookkeepers' ledgers. Both ledgers carry the accepted records with the Γ-weighted agreement odds of Eq. (19); only the outside one lists how often each act preparation is accepted, the information the normalization removes. At $\beta = 1/2$ the acceptance rate is the same for every act preparation, so the normalization cannot change the ordering. Away from that prior the two ledgers can part, a Simpson reversal [91] in which a selected-yield preference for one-boxing coexists with an accepted-history preference for two-boxing. The two scoring thresholds separate at every biased prior, in the direction fixed by the sign of $1 - 2\beta$; when both finite thresholds exist, the interval between them gives the reversal (Appendix C).

The boundary of Theorem 1 has a simple form in the accepted-record distinguishability $\mathcal{D}$ of Eq. (21). At $\beta = \frac{1}{2}$,

$$\textbf{one-box} \iff \mathcal{D} > \frac{m}{M}, \tag{28}$$

so the accepted forecast need only be biased toward the correct act by more than the reward ratio. The ratio m/M compares the reward in the observed transparent box with the reward in the unobserved opaque box: the former gives two-boxing its fixed m advantage, while the latter is the reward whose presence is correlated with the forecast. At the classic million-versus-thousand stakes ($m/M = 10^{-3}$) this threshold is $\mathcal{D} = 10^{-3}$, an accepted record one part in a thousand better than a coin flip ($q \approx 0.5005$). Lewis gives this cutoff as $(1 + r)/2$ in the reward ratio $r = m/M$, with the standard-stakes value 0.5005 [11, pp. 238–239]; the cutoff is standard in the decision-theoretic literature [60, 61]. For comparison, a distinguishability of 1.048596%[8] clears the threshold by more than a factor of ten.

The verdict is classical. By the proposition of Sec. II E, every quantity entering the comparison depends on the mechanism only through the two sector acceptance probabilities, and a classical rejection filter that keeps agreement profiles with probability p_{match} and disagreement profiles with probability p_{mismatch} reproduces all of them. The quantum content lives off the payoff basis. Section VI identifies a payoff-independent probe of whether the consistency condition was realized coherently.

C. Relation to the Wolpert–Benford resolution

Wolpert and Benford (WB) resolve Newcomb's paradox by distinguishing two probabilistic structures relating the choice x and the forecast y [10]: a Fearful decomposition $P(y \mid x)P(x)$ that treats forecast accuracy as given, and a Realist decomposition $P(x \mid y)P(y)$ with a y-independent strategy. The two cannot be merged (their Sec. 3.5). Li and Zenker criticize WB's expected-utility formula; the use made of WB here is confined to the two factorizations [61]. WB assume a perfect predictor and set the accuracy value aside as a red herring (their Sec. 3.1), so they never evaluate its threshold. Here the accepted reliability $q = \Gamma/(\Gamma + 1)$ of Sec. II D plays the accuracy parameter in their Fearful model. One-boxing is then optimal iff $q > (M + m)/(2M)$, the threshold of Theorem 1 and, in the symmetric game, the risk-dominance boundary of Sec. IV B. The Realist product distribution is recovered only at $\Gamma = 1$. Li and Zenker represent the one-boxing structure by an arrow from choice to forecast; after imposing the two-boxer condition $P(x \mid y) = P(x)$, their revised graph removes the choice–forecast edge [61]. Here the two graphs are endpoints of one physical family, joined by the accepted correlation at intermediate Γ, and

[7] Independence is of the prepared registers, $\Pr(X, Y) = \Pr(X)\Pr(Y)$. Under Assumption 1 the predictor may know the biases α and β, though not the realized act; the forecast–act correlation is absent from the preparation and appears only in the accepted ensemble, through the postselection. The theorem is a fixed-prior intervention: β remains fixed as α is varied. Responsive predictor protocols take $\beta = \beta(\alpha)$, as in Appendix F.

[8] Also known as the *Steins Gate* worldline divergence number, from *Steins;Gate* (2009). Here it labels a point between the one- and two-boxing attractors of the potential-game picture (Appendix E). For the standard million-versus-thousand Newcomb problem, $\mathcal{D} = 1.048596\%$ is approximately 10.5 times the one-boxing threshold $\mathcal{D} = m/M = 0.1\%$.

the underdetermination runs the other way: the accepted statistics fix the joint distribution at every Γ while still admitting the common-cause, collider, and retrocausal readings of Fig. 3 (Appendix J). The lower threshold $(M+m)/M$, where mutual one-boxing first becomes an equilibrium, is a property of the two-player table and has no single-agent counterpart; Sec. IV symmetrizes the payoffs and derives it, and the risk-dominance boundary above it, from the selected-payoff table.

Figure 3 separates the causal readings of the accepted correlation from the two Wolpert–Benford factorizations. Panel (c) gives the P-CTC reading its temporal content: under the stronger retrocausal-channel interpretation the accepted correlation is represented as an effective influence from the later act X toward the earlier forecast Y, running through the selected boundary A. This retrocausal representation stays distinct from the causal structure of the opened laboratory circuit, in which both records feed the herald, and the accepted statistics do not fix it: on the symmetric diagonal slice the asymmetric $X \to \mathsf{A} \to Y$ mechanism and the symmetric common-cause representation fit the same accepted statistics (Appendix J), so the asymmetry is just what the observational data cannot register. The reading is cyclic in spirit, a closed timelike curve, though panel (c) draws only the open leg $X \to \mathsf{A} \to Y$; the loop closes only when the final Bell projection identifies the boundary's two ends (Appendix A).

The construction does not merge the Fearful and Realist problems; $\Gamma \to \infty$ approaches the perfect-predictor limit. Wolpert and Benford's time-reversal result, that the paradox is unchanged if prediction occurs after the choice, leaves this distinction intact: it does not identify the two factorizations or the different quantities they hold fixed [10].

IV. THE INDUCED GAME: FROM PRISONER'S DILEMMA TO STAG HUNT

Section III assigns each record (x, y) the selected-payoff entry $p_{xy}u(x, y)$; every payoff in this section is that selected yield, an outside bookkeeper's scoring. Every result of this section is classical, reproduced at matched weights by a classical rejection filter, the orderings depending only on Γ (Sec. II E). The resulting normal form changes ordinal type as Γ grows: the game is a phase diagram in the two sector weights, with the thresholds of Sec. III reappearing as equilibrium boundaries. The parity filter is symmetric between the two registers, so the induced normal form is symmetric, and we use the standard 2×2 notation

$$\begin{array}{c|cc} & C & D \\ \hline C & (R,R) & (S,T) \\ D & (T,S) & (P,P) \end{array} \tag{29}$$

to classify the effective payoff ordering. Here R and P are the mutual-C and mutual-D payoffs, reward and punishment, while T and S are the off-diagonal temptation and sucker payoffs. The symmetric table is a bookkeeping device for the two-register payoff ordering: the register Y is still Ω's prior record, and equilibrium-selection language becomes literal only once Y is promoted to a second decision register (Sec. V).

The symmetrized payoff table has Prisoner's-Dilemma ordering [40] when

$$T > R > P > S,$$

so D strictly dominates C and (D, D) is the unique Nash equilibrium of the symmetric game. It has the Stag-Hunt structure [41] when

$$T < R > P > S,$$

so both (C, C) and (D, D) are Nash equilibria.

A. Acceptance-weighted Newcomb

To put the two roles on symmetric footing, write $u_1 := u$ for the choosing register's payoff and define the second by swapping arguments,

$$u_2(x, y) = u_1(y, x). \tag{30}$$

The swapped payoff u_2 arranges the match and mismatch sectors into a symmetric normal form. Under selected-yield scoring (26), each payoff entry of Eq. (27) is multiplied by the acceptance probability of its profile (x, y). Using Eq. (16), the row player's effective entries are

$$\begin{array}{c|cc} & C & D \\ \hline C & p_{\text{match}}M & p_{\text{mismatch}} \cdot 0 \\ D & p_{\text{mismatch}}(M+m) & p_{\text{match}}m \end{array} \tag{31}$$

with the column player's given by the transpose. Comparing with Eq. (29) identifies

$$\begin{aligned} R &= p_{\text{match}}M, & P &= p_{\text{match}}m, \\ T &= p_{\text{mismatch}}(M+m), & S &= 0. \end{aligned} \tag{32}$$

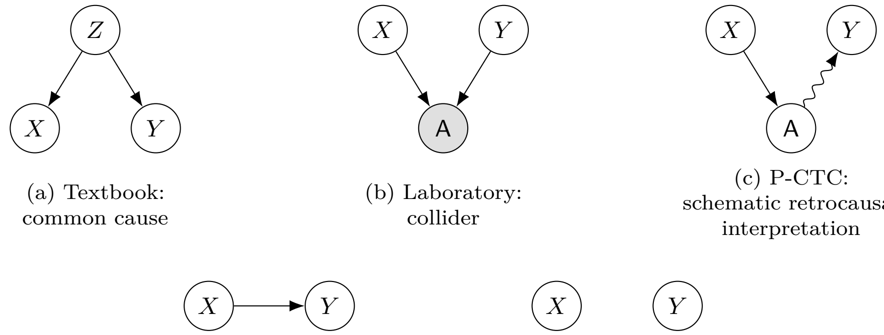
This zero is the Newcomb zero $u(C, D) = 0$ of Eq. (27); with every discarded run scored at zero, the payoff origin is part of the induced game's data. The temptation payoff $M + m$ lives off-diagonal in the Newcomb table: it is the reward for choosing D when the forecast variable is C. In the present mechanism that payoff is carried by a mismatch profile, hence inherits the acceptance probability p_{mismatch}.

The nontrivial condition $R > T$ reads

$$p_{\text{match}}M > p_{\text{mismatch}}(M+m), \tag{33}$$

equivalently

$$\Gamma > \frac{M+m}{M}. \tag{34}$$



(a) Textbook: common cause

(b) Laboratory: collider

(c) P-CTC: schematic retrocausal interpretation

(d) Fearful: $P(x,y) = P(y \mid x)P(x)$

(e) Realist: $P(x \mid y) = P(x)$ $P(x,y) = P(x)P(y)$

FIG. 3. Causal readings of the choice–forecast relation, followed by the two Wolpert–Benford decision models. In panel (a), a latent disposition Z is a common cause of act and forecast. In panel (b), X and Y are prepared independently and become associated only after conditioning on their common effect A; the shading marks A as the conditioned node. In panel (c), the P-CTC final boundary is given the stronger retrocausal-signaling interpretation: the later act affects the selected boundary, which acts back on the earlier forecast. The accepted statistics alone do not choose among these readings: A is an observed selection variable in panel (b), absorbed into the accepted distribution in panel (a), and treated as a fixed boundary condition in panel (c). Panels (a) and (c) are the two causal explanations admitted by Reichenbach's common cause principle, a common cause and a direct cause; panel (b) lies outside that dichotomy, an association induced by selection on a common effect [52, 53]. Panels (d) and (e) show the reduced choice–forecast relations of the Fearful and Realist probability structures: $P(y \mid x)P(x)$ for Fearful and, after imposing $P(x \mid y) = P(x)$, an independent pair for Realist [10, 61]. The row-1 arrows denote causal relations, whereas the row-2 arrows are probability factorizations that encode which conditional the agent holds fixed, and that Wolpert and Benford in turn read as a causal-net choice between act and forecast.

When this condition holds, a second diagonal Nash equilibrium appears: (C, C), the mutual one-boxing outcome. The threshold sits a distance m/M above $\Gamma = 1$: the excess consistency factor is $\Gamma - 1 = m/M$, or $\mathcal{D} = m/(2M + m)$ on the normalized distinguishability scale.

The normal form is therefore Prisoner's-Dilemma ordered below this boundary and a coordination game above it. Appendix D gives the finer ordinal subcases and the degenerate endpoints. In the strong-assurance regime above $\Gamma = (M + m)/m$, the temptation payoff falls below even mutual two-boxing.

B. Basin threshold and risk dominance

Once the Newcomb payoffs have been inserted, the condition $R > T$ marks the appearance of the one-boxing agreement equilibrium (C, C). The forecast prior β of Sec. III, read here as the setup prior of the symmetric game, then determines on which side of the Stag-Hunt basin boundary derived below the preparation lies. Choosing C has selected-yield expected payoff $U(C|\beta) = \beta R + (1 - \beta)S$, and choosing D has $U(D|\beta) = \beta T + (1 - \beta)P$; setting $U(C \mid \beta) = U(D \mid \beta)$ gives the indifference prior $\beta_{\text{ind}} = (P - S)/[(P - S) + (R - T)]$. For $\beta > \beta_{\text{ind}}$, the one-boxing branch has the larger selected payoff yield.

For the acceptance-weighted Newcomb matrix, $S = 0$. Substituting Eq. (32) gives

$$\beta_{\text{ind}} = \frac{P}{P + R - T} = \frac{\Gamma\, m}{(M + m)(\Gamma - 1)}. \tag{35}$$

Reading β as the probability the partner plays C, one-boxing is the better reply for $\beta > \beta_{\text{ind}}$ and two-boxing for $\beta < \beta_{\text{ind}}$. We call these two intervals of the setup prior the one-boxing and two-boxing basins; Eq. (35) is the boundary between them, and under the replicator adaptation of Appendix E they become basins of attraction in the dynamical sense. At $\beta_{\text{ind}} = 1$, C is a weak best reply to C because $R = T$. For $\Gamma > (M+m)/M$ the one-boxing basin has positive width; as Γ rises, β_{ind} falls and crosses $1/2$. Thus $\beta_{\text{ind}} = 1$ gives $\Gamma = (M + m)/M$, where the one-boxing diagonal first becomes a weak equilibrium, while $\beta_{\text{ind}} = 1/2$ gives $\Gamma = (M + m)/(M - m)$.

The latter condition is the Harsanyi–Selten risk-dominance criterion [42]: in a symmetric 2×2 coordination game the risk-dominant equilibrium is the unique best reply to a partner mixing uniformly over the two equilibrium strategies or, equivalently, the equilibrium whose basin exceeds half the prior interval, $\beta_{\text{ind}} < 1/2$. The agreement equilibrium (C, C) is risk-dominant when

$$R - T > P - S.$$

Using Eq. (32), this becomes

$$p_{\text{match}}(M - m) > p_{\text{mismatch}}(M + m), \tag{36}$$

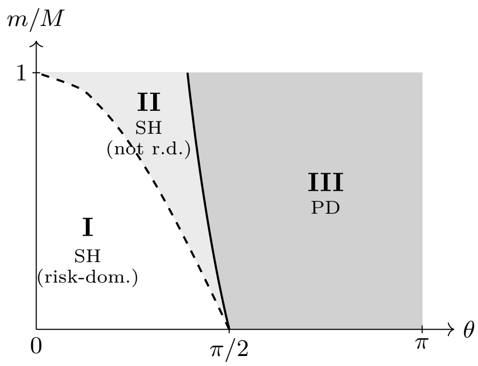


FIG. 4. Phase diagram of the selected-payoff table in the $(\theta,\, m/M)$ plane, with $\Gamma = \cot^2(\theta/2)$. **I** (white): Stag-Hunt payoff ordering with one-boxing (C, C) risk-dominant. **II** (light gray): Stag-Hunt payoff ordering, but two-boxing (D, D) is risk-dominant. **III** (dark gray): Prisoner's-Dilemma ordering, where D strictly dominates. The solid curve is the payoff-ordering boundary (34); the dashed curve is the risk-dominance boundary (36). Both curves meet at $\theta = \pi/2$ in the limit $m/M \to 0$. The agreement-favoring filter regime is $0 \le \theta \le \pi/2$; the continuation to $\theta > \pi/2$ favors the mismatch sector $\Pi_{\neq}$. The plotted range $0 < m/M < 1$ corresponds to $M > m > 0$.

equivalently

$$\Gamma > \frac{M+m}{M-m}.$$

At the unbiased setup prior, the risk-dominance comparison is a comparison of normal-form payoffs against a partner mixing uniformly: $U(C \mid \frac{1}{2}) = \frac{1}{2}(R+S)$ and $U(D \mid \frac{1}{2}) = \frac{1}{2}(T+P)$. The better reply against that mixture is the risk-dominant equilibrium. The single-agent comparison of Theorem 1 evaluates the same two numbers: at $\beta = 1/2$ the selected yields of deterministic one-boxing and two-boxing are $\frac{1}{2}(R+S)$ and $\frac{1}{2}(T+P)$; the unbiased forecast prior enters the yield exactly as the uniform partner mixture enters the normal form. Hence the two boundaries coincide. Appendix C shows that this coincidence is unchanged if every rejected run is assigned a common payoff other than zero, even though the threshold at which (C, C) first becomes an equilibrium then moves.

Figure 4 shows the two boundaries in the $(\theta,\, m/M)$ plane, with $\Gamma = \cot^2(\theta/2)$ throughout. The solid curve is the payoff-ordering boundary $R = T$: solving $\cot^2(\theta_{\mathrm{SH}}/2) = (M+m)/M$ gives

$$\theta_{\mathrm{SH}} = 2\arctan\Big[\big(1 + m/M\big)^{-1/2}\Big]. \tag{37}$$

The dashed curve is the risk-dominance boundary $R - T = P - S$: solving $\cot^2(\theta_{\mathrm{RD}}/2) = (M+m)/(M-m)$ gives

$$\theta_{\mathrm{RD}} = 2\arctan\Big[\Big(\frac{1 - m/M}{1 + m/M}\Big)^{1/2}\Big], \tag{38}$$

and it lies inside the Stag-Hunt region. For Gardner's values, the Γ thresholds are 1.001 and ≈ 1.002, with the Stag-Hunt and risk-dominance boundaries lying $0.0286°$ and $0.0573°$ below $\theta = \pi/2$, respectively. As θ increases from 0, the postselection filter becomes less selective: region I has Stag-Hunt payoff ordering and risk-dominant agreement, region II has Stag-Hunt payoff ordering with (D, D) risk-dominant, and region III has Prisoner's-Dilemma ordering, with D strictly dominant. For $M \gg m$ the two curves lie close to each other near $\theta = \pi/2$, their separation set by m/M. Reweighting a Prisoner's Dilemma with positive diagonal payoffs by a sufficiently large agreement-to-mismatch ratio yields a coordination game; what is specific here is that a single physical parameter $\Gamma = \cot^2(\theta/2)$ sets both thresholds and ties them to the reward ratio m/M.

Potential and basins. If the common one-boxing weight is adjusted toward the better-paying act across attempts, it ends at pure one-boxing or pure two-boxing, with β_{ind} dividing the two basins; the larger basin switches sides at the risk-dominance boundary. The adjustment is an added assumption; Appendix E gives an explicit replicator flow under which, in the coordination regime, β_{ind} is the unstable boundary between the two basins of attraction.

Adaptive prediction. Assumption 1 lets the forecast weight respond to the act preparation, and the verdict then depends on the predictor's response rule: an accuracy seeker makes pure one-boxing optimal already at the Stag-Hunt boundary, while an adversary concedes any one-boxing weight only above $\Gamma = (M+m)/m$, its prior pinned to β_{ind} (Appendix F).

V. LEWIS'S SIDE-BY-SIDE AND PARALLEL-WORLDLINE READINGS

Lewis's reading of the Prisoner's Dilemma as two Newcomb problems side by side (Sec. I), played as a single game, gives Sec. IV its equilibrium structure. This section takes a stronger reading: one agent playing against itself on parallel worldlines, the two tied at the selected boundary by the consistency condition, and the coding of the records, like the scoring, becomes a fact about the pair. Four facts about the pair support it: exchange symmetry makes X and Y interchangeable, coding inversion exchanges agreement and mismatch without changing the game, scoring only one register's yield destroys the coordination, and the Bell-component weight defined below belongs to the pair alone (Appendices G and H).

The parity-filter construction and the thresholds of Secs. II–IV do not depend on this one-agent/worldline reading. The symmetry holds for identically prepared registers, so the symmetric record model itself does not distinguish the one-agent and two-agent readings. Brams describes Prisoner's Dilemma as a symmetrized Newcomb problem, anticipating this move [60]. Badici reviews the epistemic-asymmetry and strategic–parametric objections to the side-by-side identification and argues that two juxtaposed Newcomb problems can be treated as a strategic game [59]. Promote Y from a forecast record to a second decision register: each act then forecasts the other, the parity filter supplying the correlation.

Each side retains the ordinary Newcomb dominance relation, two-boxing adding m against every fixed partner action, while at unbiased priors the partner's record forecasts one's own act with reliability $q = \Gamma/(\Gamma+1)$; Appendix G gives the formal side-by-side decomposition. Putting the two sides together gives the symmetric game of Sec. IV, so Lewis's Prisoner's-Dilemma case is the weak-correlation end of a continuous family that, as Γ grows, enters the Stag-Hunt regime of the selected-payoff table, where mutual one-boxing becomes an equilibrium before the single-agent expected-utility recommendation changes from two-boxing. The two thresholds, derived in Sec. IV, are

$$\Gamma_{\mathrm{SH}} = \frac{M+m}{M}, \qquad \Gamma_{\mathrm{RD}} = \frac{M+m}{M-m}, \tag{39}$$

and the decomposition keeps them distinct.

The change of game belongs to the selected-payoff table alone: under accepted-history scoring the correlation enters only through the accepted reliability q, no ordering changes at any Γ, and the single verdict switch falls at the risk-dominance threshold.

The mismatch weight that Secs. II–IV scored as forecast error admits a second reading here: each worldline brings its own coding of acts into bits, and the labels may come apart from the acts they encode. Let f_X and f_Y map each register's bit value to the physical alternative it encodes. The record-level Novikov self-consistency condition [71, 72] in this model is

$$f_X(x) = f_Y(y). \tag{40}$$

With aligned codings this is written $y = x$; with opposite codings the same physical alternative carries opposite bits,

$$y = x \quad \text{(aligned)}, \qquad y = x \oplus 1 \quad \text{(inverted)}. \tag{41}$$

Flipping both registers' labels at once changes no agreement or mismatch, so consistency is a relation between the two codings. When the codings differ, a bit-level mismatch is the consistent case: if one worldline reads 1 as alive and 0 as dead and the other the reverse, a record reading *mismatch, human is dead* decodes to a consistent history.[9] The parity filter relaxes the Novikov condition of Eq. (40) and then recovers it. A mismatch history, the predictor holding one act on record while the agent performs the other, is the self-contradicting case under aligned coding, kept at the reduced weight Γ^{-1} rather than forbidden; as $\Gamma \to \infty$ the weight vanishes and Novikov consistency is restored in full. At finite Γ the recovery takes two forms. Under the aligned coding the surviving mismatch is the bit-flip forecast error of the noisy P-CTC (Sec. II C): the inconsistency is realized physically as loop noise, and isolating the CTC interaction from that noise restores $y = x$. Under the inverted coding each mismatch history instead glues to a consistent one [Eq. (41)], and the ideal consistency check is the conjugated filter selecting the mismatch sector [Eq. (42) below], the same game read in a different coding.

To write inverted-coding consistency as an operator, relabel one worldline by the local bit flip: in a fixed laboratory basis, coding inversion is $U_{\mathrm{inv}} = I_X \otimes \sigma_x^{(Y)}$, which exchanges the two parity sectors. Conjugating the filter gives

$$\begin{aligned} K_{\mathrm{inv}}(\theta) &= U_{\mathrm{inv}} K(\theta) U_{\mathrm{inv}}^\dagger \\ &= \cos(\theta/2)\Pi_{\neq} - \sin(\theta/2)\Pi_{=}. \end{aligned} \tag{42}$$

At the ideal setting, $K(0) = \Pi_{=}$ and $K_{\mathrm{inv}}(0) = \Pi_{\neq}$. The binary noisy P-CTC of Eq. (14) can therefore be written

$$\mathcal{N}_\ell^{\mathrm{noise}}(\rho) = (1-\ell)K(0)\rho K(0)^\dagger + \ell K_{\mathrm{inv}}(0)\rho K_{\mathrm{inv}}(0)^\dagger. \tag{43}$$

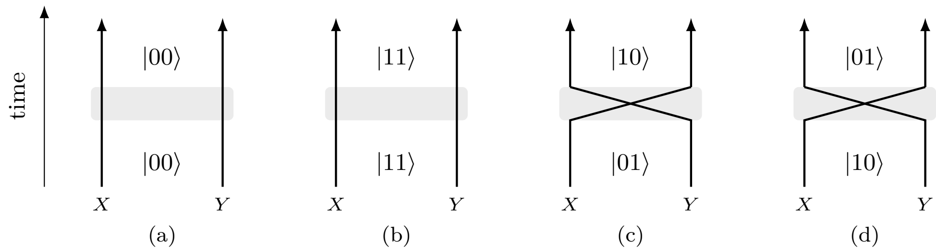
Its two branches are the ideal consistency checks for the aligned and inverted codings.

The parallel-worldline reading has a simple exchange symmetry. Let $\Pr(X = C) = \alpha_X$ and $\Pr(Y = C) = \alpha_Y$ before selection. Under the aligned coding, the consistent histories have weights $\alpha_X\alpha_Y$ and $(1-\alpha_X)(1-\alpha_Y)$; under the inverted coding, they have weights $\alpha_X(1-\alpha_Y)$ and $(1-\alpha_X)\alpha_Y$. Exchanging X and Y leaves each pair unchanged, for arbitrary preparation weights. The two worldline registers are therefore interchangeable within either coding class, the symmetry behind the single-agent reading.

On states, worldline exchange is the operator $\mathrm{SWAP}\,|x\rangle_X |y\rangle_Y = |y\rangle_X |x\rangle_Y$. It therefore preserves both the agreement and mismatch sectors. Coding inversion is different: Eq. (42) exchanges the two sectors, $\Phi^\pm \leftrightarrow \Psi^\pm$.

The parity-sector bias is the distinguishability $\mathcal{D}$ of Eq. (21), the number the game thresholds turn on. On the input $|+\rangle_X |+\rangle_Y$, with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, whose only Bell components are Φ^+ and Ψ^+, the accepted record state of the binary noisy P-CTC splits, for $\Gamma \geq 1$, as $\rho = \mathcal{D}\,|\Phi^+\rangle\langle\Phi^+| + (1-\mathcal{D})\rho_{\mathrm{sep}}$, with ρ_{sep} separable, and no decomposition can assign the separable part more weight (Appendix H). For $\Gamma < 1$, continuing the family below the agreement-favoring range of Sec. II C, the Bell component is Ψ^+, the consistent state of the inverted coding, the weight passing through zero at $\Gamma = 1$ as the component changes, the state varying continuously. The game thresholds of Sec. IV are conditions on the same

[9] One could imagine using a *PhoneWave* to send a banana to the past and getting a gellified banana in return rather than forbidding the CTC altogether: the same bit-level mismatch is inconsistent under aligned coding and Novikov-consistent under inverted coding. This is also our reading of the dead–alive swap in *Stag Hunt* [37]: decoded agreement is the film's both-cooperate outcome, the one-boxing outcome that isolating the CTC interaction from noise (Sec. VII) protects.



$\Phi^{\pm} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ $\quad$ $\Psi^{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$

FIG. 5. Parallel-worldline reading of the four basis records and their Bell combinations. Each panel follows the records X and Y of two versions of one agent, with the joint record written at entry and exit. In the parallel cases (a) and (b) the worldlines keep their sides, and the two cases combine into the agreement-sector Bell states $\Phi^{\pm}$. In the crossed cases (c) and (d) the exchange acts visibly, the crossing sending $|01\rangle \leftrightarrow |10\rangle$, and the two cases combine into the mismatch-sector Bell states $\Psi^{\pm}$. The crossing denotes the record-to-worldline identification; the shaded block denotes the diagonal filter $K(\theta)$. Exchanging the two registers is a symmetry of both classes: on the parallel records it acts invisibly, so panels (a) and (b) could equally be drawn crossed, and the parallel drawing is a choice of representative. The shaded block is the parity filter acting once on the joint record: diagonal in the record basis, it couples only the parity $x \oplus y$ to the loop and is retained on A, weighting the parallel pair by $\cos(\theta/2)$ and the crossed pair by $-\sin(\theta/2)$ [Eq. (9)]. Inverting the coding is a different operation [Eq. (42)]: it swaps the agreement and mismatch sectors, so the crossed records of (c) and (d) become the Novikov-consistent histories, written in the inverted coding [Eq. (41)]. The loop's own Bell state is equally conventional: every Bell state in the cup and cap induces the same selected map (Appendix A), so which sector decodes to agreement of the acts comes from the coding each worldline brings.

number, Stag-Hunt ordering at $\mathcal{D} > m/(2M+m)$ and risk dominance of mutual one-boxing at $\mathcal{D} > m/M$: one filter, measured in two experiments, returns one $\mathcal{D}$, as the diagnostic input's Bell weight and as the diagonal ensemble's accepted bias. The identity is a fact about the filter rather than a mechanism: the game is played on diagonal preparations whose accepted states carry no entanglement (Appendix H), and the classical rejection filter, unentangled, crosses the same thresholds.

The bit-level sectors Φ and Ψ are fixed subspaces of the records; the coding decides only which act each record encodes. The shared loop carries no side of that convention: every Bell state chosen for the loop preparation and projection induces the same selected map (Appendix A), and the flip that inverts one coding leaves each maximally entangled record's reduced states fixed, so on the maximally mixed marginals of unbiased priors neither worldline can decide from its own register which decoding holds; a biased preparation would mark the difference in its accepted marginals. Which acts are consistent is relative to the pair in two distinct ways: the parity is a fact of the pair, while the coding that maps the records to acts is the pair's shared convention.

VI. THE PREDICTOR'S PREDICAMENT: ACCURACY AGAINST VISIBILITY

Once the act is the computational-basis readout of a decision qubit, its two values play the role of the two paths of an interference experiment, and the forecast Y is a which-way record for them. The early access that Assumption 1 denies the predictor is then bounded by two-alternative complementarity: the accepted forecast's accuracy (the which-way distinguishability $\mathcal{D}$) and the act register's transverse visibility $\mathcal{V}$ are complementary quantities, constrained by an inequality the coherent filter meets with equality. This locates the construction's quantum content in the tradeoff between an early readable act record and the act register's surviving coherence, while the decision itself stays classical: the payoff verdicts read only the computational-basis records.

By Sec. II E, the coherent parity filter, the binary noisy P-CTC (Sec. II C), and a classical rejection filter matched on both sector weights agree on every basis-diagonal quantity and differ only off the diagonal. Two complementary-basis measurements read the difference and split it exactly: a local σ_x on one register flips the parity, so it reads only the local coherence, carried by the elements connecting the two sectors; the joint $\sigma_x \otimes \sigma_x$ preserves the parity, so it reads only the joint coherence, carried by the elements within each sector. At the unbiased forecast prior, the accepted reliability q of Eq. (20) gives the distinguishability $\mathcal{D} = 2q - 1$ of Eq. (21); define the transverse visibility

$$\mathcal{V} = \sqrt{\langle \sigma_x \rangle_X^2 + \langle \sigma_y \rangle_X^2}. \tag{44}$$

Here σ_y is the Pauli Y; the visibility is phase optimized, $\mathcal{V} = 0$ for a reduced act state diagonal in the payoff basis and $\mathcal{V} = 1$ for an equal coherent superposition. In the which-way setting, Englert's distinguishability–visibility relation [48], of the same form as the earlier predictability complementarity of Greenberger and Yasin [49, 50], gives

$$\mathcal{D}^2 + \mathcal{V}^2 \le 1. \tag{45}$$

Appendix H proves the general statement and records its equality conditions.

To evaluate the coherent filter, prepare the diagnostic input $|+\rangle_X |+\rangle_Y$, with $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. This run is a complementary calibration of the unmeasured filter.[10] Its computational-basis probabilities are those of the $\alpha = \beta = 1/2$ diagonal preparation, but it also carries phase coherence across all four records. The selected state, a coherent superposition of the agreement sector Φ and the mismatch sector Ψ (Appendix H carries out the postselection), is

$$|\psi(\theta)\rangle_{XY} = \cos(\theta/2)\left|\Phi^+\right\rangle - \sin(\theta/2)\left|\Psi^+\right\rangle. \tag{46}$$

The computational readout of Y is the optimal which-way measurement for the act. Tracing out either register gives

$$\rho_X(\theta) = \rho_Y(\theta) = \tfrac{1}{2}\big(I - \sin\theta\,\sigma_x\big). \tag{47}$$

The coincidence of the two reduced states is the exchange symmetry of the two records (Sec. V). The σ_x signal's magnitude is the visibility ($\langle\sigma_y\rangle_X = 0$ for the real-amplitude filter, its $K(\theta)$ coefficients real, Appendix H), and the accepted C/D statistics give the same distinguishability as the diagonal preparations (Sec. II D), so

$$\begin{aligned} \mathcal{D} &= \frac{\Gamma - 1}{\Gamma + 1} = \cos\theta, \\ \mathcal{V} &= |\langle\sigma_x\rangle_X| = \frac{2\sqrt{\Gamma}}{\Gamma + 1} = \sin\theta, \\ \mathcal{D}^2 + \mathcal{V}^2 &= 1. \end{aligned} \tag{48}$$

The coherent filter therefore saturates the bound. As $\Gamma \to \infty$, it reaches the Omega point $(\mathcal{D}, \mathcal{V}) = (1, 0)$, where every accepted forecast is errorless and Novikov-consistent under the aligned coding, $y = x$. Consistency is thus a joint constraint on X and Y: increasing forecast reliability removes the act register's complementary coherence. Conversely, at $\Gamma = 1$ the record has no predictive bias and the visibility is maximal (Sec. VI A).

The choice recommendation is carried by $\mathcal{D}$ alone. The visibility, the measured strength of the local coherence, lies off the payoff basis and leaves the strategic verdict untouched, but the two quantities lie on the same arc: every game boundary is at $\mathcal{D}$ of order m/M, so for $M \gg m$ the coordination transition occurs while the coherent implementation still has nearly full visibility. Figure 6 puts the decision threshold and the coherence tradeoff on the same plane.

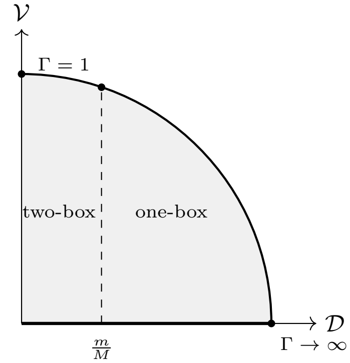


FIG. 6. Decision and coherence content in the $(\mathcal{D}, \mathcal{V})$ plane at the symmetric preparation. For the diagnostic family considered here, $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$; the coherent implementation traces the unit arc, while the binary noisy and classical maps lie on $\mathcal{V} = 0$ at every Γ, have zero local coherence (Appendix H), and are separated by the joint correlator, $\langle\sigma_x \otimes \sigma_x\rangle = 1$ against 0. Partially dephased settings fall strictly inside the disk (Sec. VII). The Newcomb verdict reads only the horizontal coordinate, one-boxing for $\mathcal{D} > m/M$; the dashed threshold is placed schematically, and its label gives the exact value. The visibility is maximal at $\Gamma = 1$ and vanishes in the perfect-prediction limit.

A. The σ_x coherence signal at the symmetric preparation

Evaluate the three matched filters on the same input $|+\rangle_X |+\rangle_Y$ and at the same consistency factor. Computational-basis one-boxing and two-boxing preparations carry neither local nor joint coherence, so this equal-superposition input isolates the two coherences, the part of the construction the payoff table cannot see.

Carrying the $|+\rangle_X |+\rangle_Y$ input through the two CNOTs, the ancilla rotation, and postselection on $|0\rangle_a$ reproduces the state of Eq. (46) directly in the opened circuit (Appendix H), with acceptance probability 1/2 independent of θ. The coherent output therefore has the local signal $\langle\sigma_x\rangle_X = -2\sqrt{\Gamma}/(\Gamma+1)$; the negative sign comes from the relative minus sign of the mismatch amplitude in $K(\theta)$. The binary noisy and classical outputs have $\langle\sigma_x\rangle_X = 0$ on the same input (Appendix H), so the local measurement separates the coherent filter from the other two.

The local σ_x signal is strongest at $\Gamma = 1$, where $\langle\sigma_x\rangle_X = -1$: neither sector is reweighted, the filter maps $|+\rangle|+\rangle$ to $|-\rangle|-\rangle$, and the coherence is carried through at full strength with its sign reversed. At the endpoints the pair retains joint coherence ($|\Phi^+\rangle$ as $\Gamma \to \infty$, $|\Psi^+\rangle$ as $\Gamma \to 0$) while the reduced state on X becomes $I/2$ and the local signal vanishes. By exchange symmetry $\langle\sigma_x\rangle_Y = \langle\sigma_x\rangle_X$.

On the same input and in the agreement-favoring range $\Gamma \geq 1$, all three maps have $\langle\sigma_z \otimes \sigma_z\rangle = \mathcal{D}$, while the binary noisy output and the real-amplitude coherent output both obey $\langle\sigma_x \otimes \sigma_x\rangle = 1$ and have concurrence $\mathcal{D}$. At $\Gamma = 1$ the binary noisy output is separable despite $\langle\sigma_x \otimes \sigma_x\rangle = 1$. The nonlinear witness

$$\mathcal{W}_{XZ} := |\langle\sigma_x \otimes \sigma_x\rangle| + |\langle\sigma_z \otimes \sigma_z\rangle| \tag{49}$$

therefore gives $1 + \mathcal{D}$ for both quantum outputs and $\mathcal{D}$ for the classical output; its separable bound is $\mathcal{W}_{XZ} \leq 1$, so $\mathcal{W}_{XZ} > 1$ certifies entanglement. Thus local coherence separates coherent from binary noisy enforcement, while joint coherence separates both quantum implementations

[10] $|+\rangle_X |+\rangle_Y$ can be thought of as the outside bookkeeper's view of the coherent symmetric game.

from the classical rejection filter; concurrence provides the corresponding separation for $\Gamma > 1$. Table I collects the three diagnostics at matched Γ. Appendix H gives the selected-event sampling cost.

As Γ grows the local signal $2\sqrt{\Gamma}/(\Gamma+1)$ falls toward zero and the sampling cost $O(\Gamma)$ climbs: the local-coherence diagnostic is hardest to resolve just where the accepted forecast is most reliable. The joint-coherence test has no such scaling: $\langle \sigma_x \otimes \sigma_x \rangle$ is 1 for the binary noisy map and 0 for the classical rejection filter at every Γ, a unit gap, so a fixed-confidence discrimination between the two needs only $O(1)$ accepted samples.

VII. NOISE, CLASSICALITY, AND CAUSAL COMPATIBILITY

The noisy P-CTC here arises from decoherence. In the opened circuit of Sec. II B, dephasing the consistency setting, an unrecorded distribution over θ, produces it directly; in the closed representation of Sec. II C the same noise is a channel on the loop qubit, the bit-flip channel giving the binary case of Eq. (14). Equation (8) factorizes as $\mathcal{U}_\theta = (e^{i\theta\sigma_z^L/2} \otimes I_{XY})\mathcal{U}_0$ with $\mathcal{U}_0 := \mathcal{U}_{\theta=0}$, so the setting distribution acts in the closed representation as a random z-axis-rotation channel on the loop qubit; a Pauli-Z channel of rate ℓ on $\mathcal{U}_0$ contracts to Eq. (14), as does a Pauli-X channel of the same rate on $\mathcal{U}_{\text{par}}$ (Appendix A). The expected dephasing and the binary map share the match and mismatch acceptance probabilities while acting differently on superpositions of the sectors; the binary map is what the dephasing of θ reduces to whenever its distribution is symmetric about the ideal setting, at any width. The setting distribution drags the consistency factor to an effective value set by the expected weights, fixing every payoff comparison, while a separate coefficient controls how much cross-sector coherence survives. This section fixes the ratio that controls the first and the coefficient that controls the second.

The accepted ensemble is the expectation of the unnormalized outputs $K(\theta)\rho K^\dagger(\theta)$, normalized once at the end; the expectation of the normalized fixed-θ states would give every setting the same weight regardless of its acceptance rate. The expected selected numerator, $\overline{\Lambda}(\rho) := \mathbb{E}[K(\theta)\rho K^\dagger(\theta)]$, is

$$\overline{\Lambda}(\rho) = \bar{p}_{\text{match}}\Pi_= \rho \Pi_= + \bar{p}_{\text{mismatch}}\Pi_{\neq}\rho\Pi_{\neq} - g\big(\Pi_=\rho\Pi_{\neq} + \Pi_{\neq}\rho\Pi_=\big), \tag{50}$$

where

$$\bar{p}_{\text{match}} := \mathbb{E}[\cos^2(\theta/2)], \qquad \bar{p}_{\text{mismatch}} := \mathbb{E}[\sin^2(\theta/2)], \tag{51}$$

and

$$g = \tfrac{1}{2}\mathbb{E}[\sin\theta]. \tag{52}$$

Here $\bar{p}_{\text{match}}$ and $\bar{p}_{\text{mismatch}}$ are the expected acceptance probabilities of agreement and mismatch, g multiplies the terms coupling the agreement and mismatch sectors, and $\bar{p}_{\text{match}} + \bar{p}_{\text{mismatch}} = 1$. The g terms preserve a phase relation already present between the parallel and crossed history classes of Fig. 5; they do not create either class or move weight between them (Appendix I). By the Cauchy–Schwarz inequality $g^2 \le \bar{p}_{\text{match}}\bar{p}_{\text{mismatch}}$, with equality only for a sharp θ (Appendix I).

The acceptance probability $\mathrm{Tr}[\overline{\Lambda}(\rho)]$ is linear in the map, so it depends only on the expected selection effect of Sec. II D,

$$E_{\text{eff}} = \bar{p}_{\text{match}}\Pi_= + \bar{p}_{\text{mismatch}}\Pi_{\neq}; \tag{53}$$

the cross terms are traceless, $\mathrm{Tr}[\Pi_=\rho\Pi_{\neq}] = \mathrm{Tr}[\rho\,\Pi_{\neq}\Pi_=] = 0$. Hence, for $\bar{p}_{\text{mismatch}} > 0$, every basis-diagonal threshold is controlled by

$$\Gamma_{\text{eff}} = \frac{\bar{p}_{\text{match}}}{\bar{p}_{\text{mismatch}}} = \frac{\mathbb{E}[\cos^2(\theta/2)]}{\mathbb{E}[\sin^2(\theta/2)]}, \tag{54}$$

with $\Gamma_{\text{eff}} = \infty$ at $\bar{p}_{\text{mismatch}} = 0$. The diagonal results follow by replacing Γ with Γ_{eff}, but the complementary-basis signal still depends on g. If $g = 0$, Eq. (50) reduces to the incoherent two-Kraus numerator of the binary noisy P-CTC with effective error probability $\ell_{\text{eff}} = \bar{p}_{\text{mismatch}}$; using Eq. (43), its two Kraus branches are the aligned- and inverted-coding ideal consistency checks of Sec. V. Thus $g = 0$ removes the coherence between these two consistency sectors. If $g \neq 0$, the same $\bar{p}_{\text{match}}$, $\bar{p}_{\text{mismatch}}$, and Γ_{eff} coexist with a nonzero cross-sector term.

The cross-sector term and the accepted parity bias are controlled separately by

$$2g = \mathbb{E}[\sin\theta], \qquad \bar{p}_{\text{match}} - \bar{p}_{\text{mismatch}} = \mathbb{E}[\cos\theta]. \tag{55}$$

The first measures the surviving coherence between the two coding-specific consistency sectors; the second controls their accepted weight imbalance and hence Γ_{eff}. Both are read off the diagnostic input of Sec. VI: its acceptance probability is $1/2$ at every θ (Sec. VI A), so selection does not reweight the spread and the accepted state is the expectation of the sharp-θ accepted states under the dephasing distribution itself, with visibility $\mathcal{V} = |\mathbb{E}[\sin\theta]| = 2|g|$ and distinguishability $\mathcal{D}_{\text{eff}} := (\Gamma_{\text{eff}} - 1)/(\Gamma_{\text{eff}} + 1) = \bar{p}_{\text{match}} - \bar{p}_{\text{mismatch}} = \mathbb{E}[\cos\theta]$. Together, $\mathcal{D}_{\text{eff}}^2 + \mathcal{V}^2 = (\mathbb{E}[\cos\theta])^2 + (\mathbb{E}[\sin\theta])^2 \le 1$. The sum is the squared length of the expected unit vector $(\cos\theta, \sin\theta)$, which reaches one only at a sharp θ: dephasing pulls the accepted state off the unit arc of Fig. 6 into the disk.

Substituting $\Gamma_{\text{eff}} = (1 - \ell_{\text{eff}})/\ell_{\text{eff}}$ into the Γ thresholds of Secs. III and IV and Appendix F 4, the diagonal boundaries become simple noise tolerances. Stag-Hunt ordering requires $\ell_{\text{eff}} < M/(2M + m)$; risk dominance of mutual one-boxing requires $\ell_{\text{eff}} < (M - m)/(2M)$; and the adversarial adaptive-preparation threshold requires $\ell_{\text{eff}} < m/(M + 2m)$. For $M \gg m$, the first two tolerate an error rate close to one half, whereas the adversarial bound is of order m/M.

at $\alpha = \beta = \frac{1}{2}$	coherent $R_y(\theta)$	binary noisy	classical
$\Pr(\mathsf{A})$	$1/2$	$1/2$	$1/2$
q	$\Gamma/(\Gamma+1)$	$\Gamma/(\Gamma+1)$	$\Gamma/(\Gamma+1)$
$\langle \sigma_x \rangle_X$	$-2\sqrt{\Gamma}/(\Gamma+1)$	0	0
$\langle \sigma_x \otimes \sigma_x \rangle$	1	1	0
$\mathcal{W}_{XZ}$	$1+\mathcal{D}$	$1+\mathcal{D}$	$\mathcal{D}$
concurrence	$\mathcal{D}$	$\mathcal{D}$	0

TABLE I. Coherent parity filter, binary noisy P-CTC, and classical rejection filter matched on the computational-basis sector weights. All entries are at $\Gamma \geq 1$, where the binary noisy concurrence is $\mathcal{D}$. The payoff-basis statistics agree. A local complementary-basis measurement separates the coherent filter from the other two; the nonlinear witness $\mathcal{W}_{XZ} > 1$ certifies entanglement for both quantum outputs when $\Gamma > 1$.

Because θ is a rotation angle, the dephasing distribution lives on a circle. The von Mises distribution is the circular analogue of a Gaussian, the maximum-entropy distribution on the circle at fixed $\mathbb{E}[\cos\theta]$, and is therefore the natural concentrated-noise model for this control parameter. Centered on the ideal setting $\theta = 0$, with density proportional to $e^{\kappa \cos\theta}$ and concentration parameter κ, it is symmetric and hence gives $g = 0$, with

$$\ell_{\text{eff}} = \tfrac{1}{2}\left(1 - \frac{I_1(\kappa)}{I_0(\kappa)}\right). \tag{56}$$

Here I_n is the modified Bessel function of the first kind, and $A(\kappa) := I_1(\kappa)/I_0(\kappa)$ is the length of the expected unit vector for this distribution, running from 0 for a uniform θ to 1 for a sharply concentrated one. Thus $\mathcal{D}_{\text{eff}}$ equals $A(\kappa)$: the concentration κ sets the sharpness of the consistency setting, the angular control precision of the forward circuit. For Gardner's values, the Stag-Hunt, risk-dominance, and adversarial boundaries are approximately $\kappa = 9.995 \times 10^{-4}$, 2.000×10^{-3}, and 250.751, respectively. Only the adversarial threshold lies in the large-κ regime; there the von Mises distribution approaches a wrapped Gaussian with variance $1/\kappa$, giving an angular jitter under 0.0632 rad, approximately 3.62°. Appendix I gives the exact $A(\kappa)$ conditions for these three thresholds.

Maximal-ignorance limit. Complete dephasing erases the accepted forecast–act correlation. Taking θ uniformly on $[-\pi, \pi)$ gives $\bar{p}_{\text{match}} = \bar{p}_{\text{mismatch}} = 1/2$, so $\Gamma_{\text{eff}} = 1$ and the diagonal consistency bias is gone; equivalently, the aligned- and inverted-coding consistency sectors of Eq. (43) enter with equal weight. The symmetry of the spread also gives $g = 0$, so the local transverse signal vanishes, though on the equal-superposition diagnostic input the joint coherence persists, keeping $\langle \sigma_x \otimes \sigma_x \rangle = 1$.

Under the uniform-circle ensemble, the settings for which the corresponding sharp filter has Stag-Hunt payoff ordering occupy $|\theta| < \theta_{\text{SH}}$, an interval of angular measure $2\theta_{\text{SH}}$. Hence

$$\Pr(\text{SH}) = \frac{\theta_{\text{SH}}}{\pi}, \qquad \Pr(\text{PD}) = 1 - \frac{\theta_{\text{SH}}}{\pi}. \tag{57}$$

Equivalently,

$$\Pr(\text{PD}) - \Pr(\text{SH}) = 1 - \frac{2\theta_{\text{SH}}}{\pi} = \frac{m}{\pi M} + O\left(\frac{m^2}{M^2}\right). \tag{58}$$

In the standard Newcomb regime $M \gg m$, the uniform ensemble therefore assigns almost equal measure to the Stag-Hunt and Prisoner's-Dilemma regions, with a Prisoner's-Dilemma excess of 3.18×10^{-4} for Gardner's values. Averaging over the same ensemble gives $\Gamma_{\text{eff}} = 1$: agreement and mismatch survive at equal weight, so the induced payoff ordering is Prisoner's Dilemma.

For the centered symmetric dephasing family used above, $g = 0$ while $\mathcal{D}_{\text{eff}} = \mathbb{E}[\cos\theta]$ decreases continuously as the spread is broadened; the uniform-circle limit gives $g = 0$ and $\mathcal{D}_{\text{eff}} = 0$ together. More generally, Γ_{eff} fixes only the diagonal parity bias and does not determine g; Appendix I gives an explicit $\Gamma_{\text{eff}} = 1$ ensemble with residual local coherence between the sectors. Removing the remaining joint coherence produces the classical rejection filter.

A. Dephasing and causal order

An experimenter can read a result and decide what to do next. Liu and Oreshkov require normalized probabilities for every such local choice rule, with no influence of later settings on earlier outcome probabilities within a laboratory [56]. They derive parity erasure without assuming a definite order between laboratories: for uniformly random later settings, the joint earlier outcomes of every nonempty subset of laboratories are independent of the parity of that subset's own later settings (Appendix J 3). Before conditioning on A, complete the retained branch by its complementary branch and let H record which occurred: $H = 1$ denotes A and $H = 0$ its complement. In the opened circuit these are the two ancilla outcomes; for the coherent Bell interaction they are the two nonzero Bell branches of the closed construction (Appendix A). Let s_X, s_Y be independently and uniformly chosen computational-basis settings, preparing $X = s_X$ and $Y = s_Y$. Their setting parity is therefore the record parity $X \oplus Y$.

In the opened circuit, X and Y precede H: their parity is written into the ancilla and the later measurement supplies the branch record. The alternative slicing gives each laboratory a copy of H as its earlier outcome, followed by its freely chosen setting s_X or s_Y. Its normalized distribution has

$$\Pr(H=1 \mid s_X, s_Y) = \begin{cases} \bar{p}_{\text{match}}, & s_X \oplus s_Y = 0, \\ \bar{p}_{\text{mismatch}}, & s_X \oplus s_Y = 1. \end{cases} \tag{59}$$

Hence the H-before-X, Y slicing satisfies parity erasure iff $\bar{p}_{\text{match}} = \bar{p}_{\text{mismatch}}$, equivalently $\Gamma_{\text{eff}} = 1$. For $\Gamma_{\text{eff}} > 1$, H is independent of either uniformly random setting separately, but distinguishes their two parity values with contrast

$$\Pr(H=1 \mid s_X \oplus s_Y = 0) - \Pr(H=1 \mid s_X \oplus s_Y = 1) = \mathcal{D}_{\text{eff}}. \tag{60}$$

For this three-record family, dephasing removes the parity dependence that obstructs this temporal ordering. At $\mathcal{D}_{\text{eff}} = 0$, equivalently $\Gamma_{\text{eff}} = 1$, H is an unbiased bit independent of the settings; it can be read before choosing them without changing the normalization. Parity erasure here is applied to $\Pr(H \mid s_X, s_Y)$; conditioning on $H = 1$ and renormalizing gives the selected basis-record distribution.

In Ferradini et al.'s approach to emergent causal order, the tensor network is initially undirected [62] (Appendix J 4). Contracting the internal Bell legs and tracing out the outgoing X, Y registers leaves one classical leg for each of the three records. For uniformly weighted s_X, s_Y, the normalized diagonal Choi tensor is

$$\chi_H = \tfrac{1}{8}\left(I - \mathcal{D}_{\text{eff}}\, \sigma_z \otimes \sigma_z \otimes \sigma_z\right). \tag{61}$$

Appendix J 4 derives this tensor directly from the two nonzero Bell branches. Assigning its three legs as inputs or outputs gives $2^3 = 8$ orientations: seven with at least one output and one with all three legs as inputs. An orientation is directly channel-normalized when tracing over its output legs gives the normalized identity on its inputs; if this condition fails, the generalized reorientation introduces a local self-cycle. For any of the seven assignments with an output, the required partial trace acts on at least one factor of σ_z. Since $\operatorname{Tr} \sigma_z = 0$, this removes the second term in Eq. (61), while the identity term gives the required normalized identity on the remaining inputs. These seven assignments therefore satisfy the normalization condition for every Γ_{eff}. In particular, the channel with X, Y as inputs and H as its output is directly normalized for every Γ_{eff} and requires no local self-cycle.

The all-input assignment is the only one that can fail direct normalization for this three-record tensor. Because it has no output leg, its normalization condition is applied directly to χ_H, which would have to equal $I/8$. Its deviation from that directly normalized form is

$$\chi_H - \tfrac{1}{8} I = -\frac{\mathcal{D}_{\text{eff}}}{8}\, \sigma_z \otimes \sigma_z \otimes \sigma_z. \tag{62}$$

For $\mathcal{D}_{\text{eff}} \neq 0$, the all-input orientation therefore fails the direct channel-normalization condition, so the generalized reorientation introduces a local self-cycle. As the symmetric spread is broadened, $|\mathcal{D}_{\text{eff}}|$ decreases and the explicit parity term in Eq. (62) shrinks. At $\mathcal{D}_{\text{eff}} = 0$, equivalently $\Gamma_{\text{eff}} = 1$, that term vanishes and $\chi_H = I/8$, so the all-input orientation is directly normalized without the self-cycle. Thus, for this three-record family, the same coefficient $\mathcal{D}_{\text{eff}}$ controls both the parity-erasure obstruction and the all-input normalization defect; both vanish at $\mathcal{D}_{\text{eff}} = 0$. Contracting the self-cycle introduced for $\mathcal{D}_{\text{eff}} \neq 0$ recovers χ_H up to an overall positive factor (Appendix J 4). Dephasing therefore restores compatibility with the assigned local direction from outcomes to subsequent choices. At $\chi_H = I/8$, every input/output assignment is normalized, so an ordered surrounding network can supply the tensor's time direction without additional feedback.

Neither causal result requires the accepted state to be diagonal in the payoff basis. For a consistency-angle distribution symmetric about $\theta = 0$, the local coherence between the agreement and mismatch sectors is already removed because $\mathbb{E}[\sin\theta] = 0$, even while $\Gamma_{\text{eff}} > 1$. At $\Gamma_{\text{eff}} = 1$, joint coherence within each parity sector can still survive; removing that remaining joint coherence gives the classical rejection filter.

VIII. DISCUSSION

Newcomb's problem reduces here to a parity bond of strength Γ tying the forecast to the act (Sec. II). Because the closed-loop contraction is operator-identical to the heralded circuit, every result is also an ordinary laboratory prediction conditioned on the herald; the closed representation changes no payoff verdict.

At the unbiased forecast prior, one-boxing has the larger expected payoff under both scorings once the accepted-record bias $\mathcal{D}$ exceeds m/M (Sec. III). Under selected-yield scoring, promoting the forecast to a second decision register produces a two-player game that moves from Prisoner's Dilemma to Stag Hunt, with mutual one-boxing becoming risk-dominant as Γ increases (Sec. IV).

Read as one agent on parallel worldlines, the same table is a game the agent plays against itself, the two sides tied at the selected boundary, with the coding of the records and the scoring facts about the pair (Sec. V).

A classical rejection filter with the same acceptance weights reproduces every payoff ordering (Sec. II E), so the strategic content is classical. The implementation-dependent quantum content lies instead in complementary-basis measurements: at finite Γ, the coherent filter saturates $\mathcal{D}^2 + \mathcal{V}^2 = 1$, both quantum outputs retain joint coherence, and only the coherent output retains the local coherence (Sec. VI).

Dephasing the consistency setting replaces Γ by Γ_{eff} in every payoff threshold. In the parallel-worldline reading, its incoherent limit mixes the two ideal consistency checks associated with the aligned and inverted codings of Sec. V. For the centered symmetric noise model, the local coherence vanishes and widening the spread lowers the accepted parity bias continuously to zero (Sec. VII). At equal agreement and mismatch weight, H loses the record parity, so the normalized three-record statistics become compatible with a slicing in which H precedes X and Y and parity erasure is satisfied, while the all-input orientation of the corresponding three-record tensor becomes directly normalized (Sec. VII A). At the same unbiased prior, the accepted act-to-forecast channel is binary symmetric and its retrocausal capacities obey $C_{\text{retro}} = 2Q_{\text{retro}} = \log_2 \Gamma$ (Eq. (22)), so the same consistency factor also measures the communication strength of the induced retrocausal channel [38].

A. What is quantum here, and what is left out?

Traditional quantum game theory places quantum structure in the strategies, replacing classical mixed moves with unitaries and using entanglement to alter the equilibrium set [92–94]. Earlier quantum treatments of Newcomb follow that route [95, 96]; Cavalcanti instead uses a Bell scenario to expose the decision-theoretic role of causal assumptions [97]. Other CTC work emphasizes computational or state-discrimination resources in Deutsch and postselected models [33, 98, 99]. Here the payoff strategies remain classical. The quantum operation lies in the selected consistency mechanism, and the change from Prisoner's Dilemma to Stag Hunt occurs with the payoff strategy class fixed. The accepted records form a pre- and post-selected ensemble [100]; a weak measurement before selection would probe the corresponding weak value [101], though the σ_x test of Sec. VI A is not of that kind, being an ordinary strong measurement of the accepted state after the selected outcome is known.

In the laboratory reading, the filter need not itself be the predictor: X and Y become associated through joint postselection even when no predictor has committed itself before the act. An actual predictor may still stand behind the preparation of Y, since the filter reads only the exposed forecast record and its agreement or mismatch with X (Appendix K). Wolpert and Benford's time-reversal result (Sec. III C) shows that the same probability structure arises when Ω observes the choice later [10]. The laboratory statistics therefore leave open how the record was produced; the cyclic reading supplies the forecast-before-act gloss conditional on a physical final boundary.

The same operator still leaves the causal story underdetermined. Read as a shared accepted state, the diagonal statistics have the ordinary common-cause embedding verified in Appendix J; read cyclically, the final boundary constrains the earlier record. The process-matrix check closes a modest question about the accepted slice; a generic postselected filter has no process-matrix representation when its normalization depends on the local operations. Time-symmetric ontologies raise the broader question of whether retrocausality is avoidable [74, 75]; in the divine-foreknowledge version of Newcomb's problem, Craig similarly distinguishes a forecast's counterfactual dependence on a later act from backward signaling [76]. Reversible dynamics by itself fixes no preferred temporal orientation [78, 79], and the selected operator is real and register-exchange symmetric, so it fixes no forecast/act ordering either. Newcomb's premise identifies Y as the earlier forecast and X as the later act; Appendix K gives its epistemic counterpart.

The parity filter also leaves the predictor's internal route unspecified. Appendix K makes this precise: different hidden predictor mechanisms can reduce to the same exposed forecast–act records and the same payoff-diagonal Γ, while off-diagonal measurements distinguish classical uncertainty from phase-sensitive nonclassical uncertainty encoded by retained coherence. It also separates the physical calculation from the additional choice of evidential, causal, or functional decision theory.

Dynamics and composition remain open: the potential landscape acquires dynamical meaning only once a replicator flow is specified, while adaptive prediction changes only the response rule applied to the same filter. A dynamical treatment of macroscopic worldline shifts and extensions beyond the single-link construction are left for future work.[11]

B. From a selected branch to a time machine

Everything used in the payoff and coherence results is contained in the selected map, and in the laboratory it is an ordinary heralded outcome [31, 32]: the opened circuit conditions on a recorded result, whereas the closed representation postulates the selected Bell boundary as part of the model. Turning that selected branch into a physical final boundary requires an additional hypothesis: the selection must be supplied by nature rather than by an experimenter who conditions on a herald. Call this conjecture Q_{post}. The calculations themselves do not require it [58].

The distinction is familiar from black-hole final-state proposals, where the final projection is supplied by nature rather than selected afterward by a laboratory experimenter [64–66]. If nature made such postselection an arbitrarily controllable resource for quantum computation, the computational class would extend from BQP to PostBQP = PP [30].

No exotic geometry is needed for the ingredients: the P-CTC can be represented, or conditionally simulated, by creating and projecting onto an entangled particle–antiparticle pair [21, 24], and Feynman's vacuum already draws worldlines that turn around in time, with pair

[11] Or past work, depending on perspective.

creation and annihilation as the turning points [23]. A boundary supplied by a genuine closed timelike curve requires a spacetime that contains one, together with a consistent physical identification around it [67, 69–73]. The present construction supplies the consistency mechanism that such a realization would have to enforce; it does not supply the spacetime geometry. General relativity itself admits geometries containing closed timelike curves; notably, the analytic extension of the Kerr black-hole solution contains regions with nontrivial causality violation [68]. Hawking's chronology-protection conjecture proposes that the laws of physics prevent closed timelike curves from forming, with quantum backreaction near chronology horizons providing a possible mechanism [77].

Once a physical final boundary is assumed, the side-by-side game of Sec. V admits a parallel-worldline reading, under which the dephasing calculation acquires a Novikov interpretation. The aligned and inverted codings define the two ideal consistency checks $K(0) = \Pi_=$ and $K_{\rm inv}(0) = \Pi_{\neq}$, while every sharp setting $K(\theta) = \cos(\theta/2)\Pi_= - \sin(\theta/2)\Pi_{\neq}$ combines those same two sectors coherently. For a sharp θ, the selected numerator $K(\theta)\rho K^\dagger(\theta)$ contains terms coupling the two sectors. Averaging over a symmetric spread of θ sets their coefficient $g = \frac{1}{2}\mathbb{E}[\sin\theta]$ to zero, leaving the incoherent binary noisy P-CTC of Eq. (43). Widening the spread then brings the weights of the two consistency sectors together, reaching equality at $\Gamma_{\rm eff} = 1$. The accepted forecast then becomes no better than chance, and H no longer distinguishes the parity of the two records. The temporal slicing with H earlier than both records therefore satisfies parity erasure [56]; at the same point, $\chi_H = I/8$, so the all-input orientation of the contracted three-record tensor is directly normalized without its local self-cycle. These results concern the causal ordering of the contracted X, Y, H statistics. They do not decide whether the Bell loop, or a spacetime loop implementing it, persists. As the dephasing spread increases, however, $\mathcal{D}_{\rm eff}$ decreases continuously: the forecast–act bias weakens, the parity information carried by H is reduced, and the parity-dependent tensor component proportional to $\mathcal{D}_{\rm eff}$ shrinks to zero. For the classic million-versus-thousand stakes, the strongest control requirement quantified in Sec. VII corresponds to angular jitter under 3.62°. On the P-CTC reading, $\mathcal{D}_{\rm eff}$ thus controls the observability of the consistency signal, while the persistence of the loop remains an independent physical question.

IX. CONCLUSION

Newcomb's problem is realized here by a postselected consistency condition on the forecast and act records. Opening the Bell contraction gives an ordinary heralded circuit with the same selected branch, so the payoff and coherence results do not require a physical closed timelike curve. On the payoff basis, one consistency strength fixes the predictor's reliability and every decision threshold. A classical rejection filter with the same weights reproduces every payoff ordering, while increasing the consistency strength carries the symmetric two-record game from Prisoner's Dilemma to Stag Hunt.

Away from the payoff basis, the coherent, noisy, and classical filters retain different combinations of local and joint coherence despite agreeing on the payoffs. A symmetric spread of the consistency setting removes the local coherence, while widening the spread gradually erases the accepted forecast–act bias. At complete parity erasure the forecast–act bias disappears and the selection outcome no longer carries the parity of the two records; the normalized three-record statistics then become compatible with an ordering in which that outcome precedes both records. Yet the accepted state need not become diagonal in the payoff basis.

The selected boundary may be understood as laboratory postselection, epistemic postselection, or a physical final boundary. On the time-machine reading, increasing this spread gradually erases the forecast–act bias and the parity information carried by the selection outcome. At the endpoint the previously parity-violating temporal slicing becomes compatible with ordinary locally sequential causality, while the calculation does not decide whether an underlying spacetime loop should still be regarded as present. On the P-CTC reading, decoherence may progressively erase what makes a time machine observable.

ACKNOWLEDGMENTS

The idea of reaching a Stag Hunt through Newcomb's problem and a P-CTC was inspired by the short film *Stag Hunt*, directed by Andrey Kezzyn and written by Michele Reilly. I thank Pablo Bueno for suggestions on presentation, and Hippolyte Dourdent for a brief exchange about process-matrix approaches to Newcomb's problem.

Appendix A: Bell-boundary geometry and conventions

In the tensor contraction of Sec. II, drawn in Fig. 1, time runs upward: the Bell cup prepares L, L' below the interaction, the Bell cap postselects the same pair above it, and only the L, L' indices are contracted, so the XY line is not part of the closed boundary. The projection tests the overlap of the evolved pair with the same Bell state fixed below. For any operator O on L, the Bell-state identity

$$(O \otimes I)\,|\Phi\rangle = (I \otimes O^T)\,|\Phi\rangle \tag{A1}$$

moves an operator across a Bell leg by transposition on the contracted index.

The loop pair need not be written specifically as Φ^+. If preparation and postselection use the same Bell state, all four choices give the same contraction:

$$\begin{aligned} {}_{LL'}\langle B|\,(\mathcal{U}_{L,XY} \otimes I_{L'})\,|B\rangle_{LL'} &= \frac{1}{2}\,\mathrm{Tr}_L\,\mathcal{U}_{L,XY}, \\ & B \in \{\Phi^+, \Phi^-, \Psi^+, \Psi^-\}. \end{aligned} \tag{A2}$$

Each Bell state is, up to a global phase, $(I \otimes \sigma)\,|\Phi^+\rangle$ for a Pauli σ on the spectator leg, where it commutes with $\mathcal{U}_{L,XY}$ and cancels between bra and ket. The same-state hypothesis is necessary: preparing $(I \otimes \sigma)\,|\Phi^+\rangle$ and postselecting $(I \otimes \sigma')\,|\Phi^+\rangle$ gives the twisted trace $\frac{1}{2}\,\mathrm{Tr}_L[(\sigma'^{\dagger}\sigma)^T \mathcal{U}_{L,XY}]$, with the transposition supplied by Eq. (A1).

The complementary ancilla outcome in the opened circuit gives

$$K_\perp(\theta) = \sin(\theta/2)\Pi_= + \cos(\theta/2)\Pi_{\neq}. \tag{A3}$$

Its sector weights are exchanged relative to $K(\theta)$, so on the record basis it implements inverted-label consistency. At nonzero θ it differs by a parity-sector phase from the inverted-coding filter $K_{\mathrm{inv}}(\theta)$ of Eq. (42), the parity filter with its two sectors swapped. The two operators obey

$$K_\perp(\theta) = Z_{\mathrm{par}} K_{\mathrm{inv}}(\theta), \qquad Z_{\mathrm{par}} := \Pi_{\neq} - \Pi_=. \tag{A4}$$

The two operators have the same selection effect and coincide at $\theta = 0$. Unitarity gives $K^\dagger K + K_\perp^\dagger K_\perp = I$. At the Bell boundary, evaluating the Bell contraction on each Bell projector, the Φ^+ outcome returns $K(\theta)$, the Φ^- outcome returns $iK_\perp(\theta)$ (the i a branch phase without physical effect), and the $\Psi^\pm$ outcomes vanish identically for $\mathcal{U}_\theta$, which is diagonal on L in the computational basis. The two nonzero Bell branches therefore reproduce the two ancilla outcomes. Bell states of the external records X, Y instead encode the aligned or inverted relations of Sec. V; the auxiliary and external Bell structures have different physical roles.

Where a noise channel is inserted on the loop is likewise immaterial for the controlled interactions used here. Both $\mathcal{U}_{\mathrm{par}}$ and $\mathcal{U}_\theta$ have the block form $\sum_{s\in\{=,\neq\}} u_s \otimes \Pi_s$ with u_s acting on L, so a loop channel with Kraus operators $\{b_r\}$, inserted after the interaction, gives the selected Kraus operators

$$\begin{aligned} N_r &= \frac{1}{2}\,\mathrm{Tr}_L(b_r \mathcal{U}) \\ &= \frac{1}{2}\,\mathrm{Tr}(b_r u_=)\Pi_= + \frac{1}{2}\,\mathrm{Tr}(b_r u_{\neq})\Pi_{\neq}. \end{aligned} \tag{A5}$$

A channel can sit in three places: on L after the interaction, on L before it, or on the spectator leg L'. Placing it before replaces $\mathrm{Tr}(b_r u_s)$ by $\mathrm{Tr}(u_s b_r)$, the same number by cyclicity of the trace. Placing it on L' replaces each b_r by its transpose on L, by Eq. (A1); Writing $C_b(\mathcal{U}) := \frac{1}{2}\,\mathrm{Tr}_L(b\mathcal{U})$, the contraction gives $C_I(\mathcal{U}_\theta) = K(\theta)$, $C_{\sigma_z^L}(\mathcal{U}_\theta) = iK_\perp(\theta)$, and $C_{\sigma_x^L}(\mathcal{U}_\theta) = 0$, while for the parity interaction $C_I(\mathcal{U}_{\mathrm{par}}) = \Pi_=$, $C_{\sigma_x^L}(\mathcal{U}_{\mathrm{par}}) = \Pi_{\neq}$, and $C_{\sigma_z^L}(\mathcal{U}_{\mathrm{par}}) = 0$. The σ_z^L branch of $\mathcal{U}_\theta$ is therefore the complementary Bell outcome. At $\theta = 0$ a Pauli-Z channel of rate ℓ on $\mathcal{U}_0$ and a Pauli-X channel of the same rate on $\mathcal{U}_{\mathrm{par}}$ both induce Eq. (14). Finally, the loop operators used here, I, σ_x^L, σ_z^L, and $e^{i\sigma_z^L \vartheta}$, are symmetric matrices in the computational basis, so transposition changes nothing. All three placements give the same selected operators.

Appendix B: Reduction to the two acceptance probabilities

This appendix proves the proposition of Sec. II E for a general trace-nonincreasing completely positive map Λ obeying the record-preserving condition stated there.

The condition first fixes the Kraus structure. Choose Kraus operators $\{K_r\}$ for Λ and, for each basis record, write $|v_{r,xy}\rangle := K_r\,|xy\rangle$. Record preservation gives $\sum_r |v_{r,xy}\rangle\langle v_{r,xy}| = p_{xy}\,|xy\rangle\langle xy|$; every term on the left is positive, while the right-hand side has support only on $|xy\rangle$, so $|v_{r,xy}\rangle = c_{r,xy}\,|xy\rangle$ for every r. Each K_r is therefore diagonal in the record basis, with $p_{xy} = \sum_r |c_{r,xy}|^2$, and the selection effect is

$$\begin{aligned} E_\Lambda &= \sum_r K_r^\dagger K_r \\ &= \sum_{xy} p_{xy}\,|xy\rangle\langle xy| \\ &= p_{\mathrm{match}}\Pi_= + p_{\mathrm{mismatch}}\Pi_{\neq}. \end{aligned} \tag{B1}$$

Proof of the proposition of Sec. II E. For $\rho = \sum_{x,y} r_{xy}\,|xy\rangle\langle xy|$ and $U = \sum_{x,y} u(x,y)\,|xy\rangle\langle xy|$,

$$\begin{aligned} \mathrm{Tr}[U\Lambda(\rho)] = p_{\mathrm{match}} &\sum_{x=y} r_{xy} u(x,y) \\ &+ p_{\mathrm{mismatch}} \sum_{x\neq y} r_{xy} u(x,y), \end{aligned} \tag{B2}$$

linear in the two weights; the acceptance probability $p_{\mathrm{sel}}(\rho) = \mathrm{Tr}[E_\Lambda \rho]$ is the case $u \equiv 1$. The two rates are fixed only up to rescaling the filter, which multiplies both by a common positive factor, leaving the normalized accepted state and every payoff ordering unchanged while rescaling the absolute acceptance probability and selected yield, so only their ratio Γ carries decision content. Separating that ratio from the overall scale, write $(p_{\text{match}}, p_{\text{mismatch}}) = p_{\text{tot}}\big(\frac{\Gamma}{1+\Gamma}, \frac{1}{1+\Gamma}\big)$ with $p_{\text{tot}} = p_{\text{match}} + p_{\text{mismatch}} > 0$ the overall normalization. Every selected payoff yield is then p_{tot} times a function of Γ alone, so the scale multiplies all preparations equally and drops from their ordering. Every accepted-history payoff is a ratio of two such yields, well defined when the acceptance probability is nonzero, and there p_{tot} cancels outright:

$$\bar{U}_{\text{acc}} = \frac{\Gamma \sum_{x=y} r_{xy} u(x,y) + \sum_{x\neq y} r_{xy} u(x,y)}{\Gamma \sum_{x=y} r_{xy} + \sum_{x\neq y} r_{xy}}. \tag{B3}$$

The value depends on the filter only through Γ. At $p_{\text{mismatch}} = 0$ only matching records survive selection, the endpoint $\Gamma \to \infty$; the conditioned state is then defined for preparations with nonzero weight in the surviving sector. □

For the coherent parity filter, the binary noisy channel, and the matched classical rejection filter the two rates already sum to one, $p_{\text{tot}} = 1$, so the weights are themselves $\Gamma/(1+\Gamma)$ and $1/(1+\Gamma)$, functions of Γ alone. The payoff reduction requires only the two acceptance probabilities, not a particular coherent implementation. By the diagonality of every K_r, a completely positive map escapes the reduction only by rewriting a basis record after acceptance, and then the effect still fixes the acceptance probabilities while the payoff read out afterward is no longer a function of the two sector weights.

Appendix C: Fixed-prior payoff comparisons and proof of Theorem 1

This appendix works out the two scorings of Sec. III at a fixed forecast prior β: first the selected payoff yield, then the accepted-history utility. It also records how a nonzero payoff on rejection changes the induced game.

1. Selected payoff yield

Let $\beta := \Pr(Y = C)$ be the forecast prior before the filter acts, fixed independently of the preparation of X. The consistency factor Γ then multiplies the prior odds of forecast–act agreement. Write $\bar{u}_{\text{sel}}(x)$ for the selected payoff yield $\bar{U}_{\text{sel}}$ of the deterministic preparation $X = x$. Then

$$\bar{u}_{\text{sel}}(C) = \beta p_{\text{match}} M, \tag{C1}$$
$$\bar{u}_{\text{sel}}(D) = \beta p_{\text{mismatch}}(M+m) + (1-\beta) p_{\text{match}} m. \tag{C2}$$

Using $p_{\text{match}} = \Gamma p_{\text{mismatch}}$, one-boxing has the larger selected payoff yield iff

$$\Gamma > \frac{\beta(M+m)}{\beta M - (1-\beta) m}, \tag{C3}$$

provided $\beta M > (1-\beta)m$; if $\beta M \le (1-\beta)m$, increasing Γ can never make one-boxing preferable. At $\beta = 1/2$, Eq. (C3) becomes the boundary of Theorem 1.

The same calculation gives the acceptance invariance used there. If $\alpha := \Pr(X = C)$, then

$$\begin{aligned}\Pr(\mathsf{A} \mid \alpha, \beta) = {}& p_{\text{match}}[\alpha\beta + (1-\alpha)(1-\beta)] \\ &+ p_{\text{mismatch}}[\alpha(1-\beta) + (1-\alpha)\beta],\end{aligned} \tag{C4}$$

and

$$\frac{\partial \Pr(\mathsf{A} \mid \alpha, \beta)}{\partial \alpha} = (p_{\text{match}} - p_{\text{mismatch}})(2\beta - 1).$$

Thus at $\beta = 1/2$ every act preparation has the same acceptance probability.

Rejection payoff. If every rejected run is assigned a constant payoff r instead of zero, the four selected-yield entries become

$$\begin{aligned}
R &= p_{\text{match}} M + (1-p_{\text{match}}) r,\\
P &= p_{\text{match}} m + (1-p_{\text{match}}) r,\\
T &= p_{\text{mismatch}}(M+m) + (1-p_{\text{mismatch}}) r,\\
S &= (1-p_{\text{mismatch}}) r.
\end{aligned}$$

The equilibrium condition $R > T$ becomes

$$\Gamma > \frac{M+m-r}{M-r} \qquad (r < M),$$

while the risk-dominance comparison $R - T > P - S$ remains $\Gamma > (M+m)/(M-m)$ because the r terms cancel. At $\beta = 1/2$ the same cancellation leaves the single-agent verdict unchanged. The differences $P - S$ and $R - T$ themselves shift by the common amount $r(p_{\text{mismatch}} - p_{\text{match}})$, so the indifference prior β_{ind} does depend on r, and the basin structure of Appendix E is stated in the $r = 0$ convention.

2. Accepted-history utilities

Accepted-history scoring conditions on the selected branch and retains accepted forecast errors with their ordinary mismatch payoffs. It also supports two comparisons that must be distinguished. With the forecast value $Y = y$ held fixed, an accepted record receives the original Newcomb payoff. Comparing the deterministic preparations $X = C$ and $X = D$ instead averages over Y, and each preparation can induce a different forecast distribution after conditioning on A.

Concretely, let $\Pr(x, y)$ denote the joint distribution over profiles before the ancilla is measured. Conditioning on A produces a new distribution over the accepted runs,

$$\Pr(x, y \mid \mathsf{A}) = \frac{p_{xy}\,\Pr(x, y)}{\sum_{x',y'} p_{x'y'}\,\Pr(x', y')}, \tag{C5}$$

and the conditional expected utility is

$$\mathbb{E}\big[u \,\big|\, \mathsf{A}\big] = \sum_{x,y} \Pr(x, y \mid \mathsf{A})\, u(x, y). \tag{C6}$$

For the Newcomb payoffs (27), conditional scoring restores dominance entrywise:

$$\begin{aligned} u(D, C) - u(C, C) &= (M + m) - M = m, \\ u(D, D) - u(C, D) &= m - 0 = m. \end{aligned} \tag{C7}$$

Thus D strictly dominates C entrywise, independently of the consistency factor. This is not yet the preparation comparison because the accepted mixture of forecast values need not be the same under the two acts.

Let $\mathrm{U}_{\mathsf{A}}(x) := \mathbb{E}[u \mid \mathsf{A}, X = x]$. If the prior forecast distribution is $\Pr(Y = y)$, then

$$\Pr(Y = y \mid \mathsf{A}, X = x) = \frac{p_{xy}\,\Pr(Y = y)}{\sum_{y'} p_{xy'}\,\Pr(Y = y')}. \tag{C8}$$

At the unbiased forecast prior $\beta = 1/2$, a one-boxer sees $Y = C$ with accepted probability $\Gamma/(\Gamma + 1)$, whereas a two-boxer sees it with probability $1/(\Gamma + 1)$. Hence $\mathrm{U}_{\mathsf{A}}(C) = M\Gamma/(\Gamma+1)$ and $\mathrm{U}_{\mathsf{A}}(D) = (M+m+m\Gamma)/(\Gamma+1)$, so

$$\mathrm{U}_{\mathsf{A}}(D) - \mathrm{U}_{\mathsf{A}}(C) = \frac{(M + m) - \Gamma(M - m)}{\Gamma + 1}. \tag{C9}$$

The preparation comparison therefore favors one-boxing for $\Gamma > (M + m)/(M - m)$, the risk-dominance boundary of Sec. IV. Entrywise dominance and the preparation comparison use different postselected distributions of Y.

For general $\beta \in (0, 1)$,

$$\begin{aligned} \mathrm{U}_{\mathsf{A}}(C) &= \frac{\beta\Gamma M}{\beta\Gamma + 1 - \beta}, \\ \mathrm{U}_{\mathsf{A}}(D) &= \frac{\beta(M + m) + (1 - \beta)\Gamma m}{\beta + (1 - \beta)\Gamma}. \end{aligned} \tag{C10}$$

Since both denominators are positive, clearing them and collecting powers of Γ gives $\mathrm{U}_{\mathsf{A}}(C) > \mathrm{U}_{\mathsf{A}}(D)$ iff

$$\begin{aligned} \beta(1 - \beta)(M - m)\Gamma^2 + m\big(2\beta(1 - \beta) - 1\big)\Gamma& \\ - \beta(1 - \beta)(M + m) > 0.& \end{aligned} \tag{C11}$$

At $\beta = 1/2$, the left-hand side is $(\Gamma+1)[\Gamma(M-m)-(M+m)]/4$, recovering the boundary above. For fixed finite Γ, both $\beta \to 0$ and $\beta \to 1$ favor two-boxing; the latter limit gives the fixed-record comparison M versus $M + m$.

Entrywise dominance at a fixed forecast and the accepted-history comparison between deterministic act preparations are therefore different questions. The first always favors two-boxing; the second is prior-dependent because the selection event changes the forecast mixture differently under $X = C$ and $X = D$.

3. Conditioning and the Simpson reversal

Away from the unbiased forecast prior, the two payoff scorings can disagree. Selected-yield scoring asks what the branch is worth before conditioning on its occurrence; accepted-history scoring asks what to do once the successful branch is the only history under consideration. A selected-yield one-boxing advantage can then coexist with accepted-history two-boxing dominance. The boundaries of the preceding subsections locate the window: substituting the selected-yield boundary (C3) into the quadratic above, with $\beta M - (1-\beta)m = \beta(M + m) - m$, the accepted-history condition evaluates to $Mm\,\beta^2(M + m)(1 - 2\beta)/[\beta(M + m) - m]^2$, so whenever $\beta M > (1 - \beta)m$, and Eq. (C3) therefore sets a finite selected-yield threshold, the two boundaries coincide at $\beta = 1/2$ and separate at every other prior, with every Γ strictly between them reversing the comparison. For $0 < \beta \le m/(M + m)$ the selected-yield threshold disappears: selected-yield scoring favors two-boxing at every Γ, while accepted-history scoring turns to one-boxing at sufficiently large Γ. The acceptance event is a common effect of X and Y, and conditioning on it changes the forecast mixture against which the acts are compared [91].

The reversal is absent at $\beta = 1/2$ because the acceptance rate is independent of the act preparation. This scoring question is separate from the causal representation of the same correlation: collider, common-cause, and retrocausal descriptions concern how the accepted association is represented, while the two scorings concern which normalization is retained.

Proof of Theorem 1. Part (i) is the differentiation displayed above, the derivative of $\Pr(\mathsf{A} \mid \alpha, \beta)$ in α vanishing at $\beta = 1/2$. For (ii), at $\beta = 1/2$ the selected payoff yields are $\bar{u}_{\mathrm{sel}}(C) = \frac{1}{2}p_{\mathrm{match}}M$ and $\bar{u}_{\mathrm{sel}}(D) = \frac{1}{2}[p_{\mathrm{mismatch}}(M + m) + p_{\mathrm{match}}m]$, so one-boxing wins iff $\Gamma > (M + m)/(M - m)$. Equation (C9) shows that the accepted-history difference changes sign at the same value. Part (iii) substitutes $q = \Gamma/(\Gamma + 1)$, giving $q > (M + m)/(2M)$. ∎

Appendix D: Payoff-ordering classification

Section IV identifies when mutual one-boxing appears and when it becomes risk-dominant. Here we classify the full payoff ordering. For the entries in Eq. (32), $M > m > 0$ and positive sector weights imply $R > P > S$ and $T > S$ throughout. Only R versus T and T versus P can change.

Proposition (Payoff-ordering boundaries). *Assume $M > m > 0$ and $p_{\mathrm{match}}, p_{\mathrm{mismatch}} > 0$, and set $\Gamma = p_{\mathrm{match}}/p_{\mathrm{mismatch}}$. For the selected-payoff table, the entries in Eq. (32) satisfy:*

(i) *For $\Gamma < (M + m)/M$, the strict ordering is $T > R > P > S$, so the game is a Prisoner's Dilemma.*

(ii) *For $\Gamma > (M+m)/M$, one has $R > T$, $R > P$, and $P > S$. The game is then a coordination game with the two strict Nash equilibria (C, C) and (D, D).*

(iii) *Within that region, $R > T > P > S$ iff $(M + m)/M < \Gamma < (M + m)/m$, whereas $R > P > T > S$, the strong-assurance ordering, holds iff $\Gamma > (M + m)/m$. In the latter case the equilibria are unchanged, but mutual two-boxing has a larger selected payoff yield than unilateral two-boxing.*

(iv) *At $\Gamma = (M+m)/M$, $R = T$, so D weakly dominates C and (C, C) is a weak Nash equilibrium. At $\Gamma = (M + m)/m$, $T = P$ and the ordering is $R > P = T > S$.*

(v) *Within the coordination region, (C, C) is risk-dominant iff $R - T > P - S$, equivalently $\Gamma > (M + m)/(M - m)$.*

At the excluded endpoints, $p_{\rm match} = 0$ gives $T > R = P = S = 0$, while $p_{\rm mismatch} = 0$ gives $R > P > T = S = 0$.

Proof. The comparison $R > T$ is equivalent to $p_{\rm match}M > p_{\rm mismatch}(M + m)$, hence to $\Gamma > (M + m)/M$. The other variable comparison is $T > P$, equivalent to $p_{\rm mismatch}(M+m) > p_{\rm match}m$ and therefore to $\Gamma < (M+m)/m$. Together with the fixed inequalities $R > P > S$ and $T > S$, these give parts (i)–(iv). Finally, $R - T > P - S$ is equivalent to $p_{\rm match}(M - m) > p_{\rm mismatch}(M + m)$, hence to $\Gamma > (M + m)/(M - m)$. □

The risk-dominance boundary always lies above the equilibrium threshold. It lies below the $T = P$ boundary iff $M > 2m$; thus, for the standard Newcomb regime $M \gg m$, the thresholds occur in the order $(M+m)/M < (M + m)/(M - m) < (M + m)/m$.

Appendix E: Potential landscape and basins

This appendix gives one explicit dynamical completion of the basin picture in Sec. IV. Let $\nu \in [0, 1]$ be the common one-boxing weight of the two preparations, $\alpha = \beta = \nu$; the dynamics below specifies how this weight changes across attempts and is additional to the P-CTC consistency condition. This is a mixed preparation across many attempts, with each realized act still C or D. Since ν is also each player's probability of meeting C, the indifference point below is the same $\beta_{\rm ind}$ found in Eq. (35). In the original single-agent reading, β continues to denote the forecast prior. The dynamics introduced here applies to the symmetrized game rather than to the selected branch itself.

The symmetric 2×2 game is an exact potential game in the sense of Monderer and Shapley [43]: a single function Ξ on the four profiles reproduces every player's incentive, in that changing one player's action against a fixed partner moves Ξ by exactly that player's own payoff change,

$$\Xi(a_i', a_j) - \Xi(a_i, a_j) = u_i(a_i', a_j) - u_i(a_i, a_j), \tag{E1}$$

for each player i and payoff u_i against a fixed partner action a_j. Symmetry is what lets such a Ξ exist: each player's cross-difference of payoffs equals the same quantity $R-T-S+P$, and that agreement is the integrability condition for Eq. (E1) to be solvable. The relation fixes Ξ up to an additive constant. Anchoring at $\Xi(D, D) = 0$ and reading R, S, T, P from the selected-yield table (Eq. (32)), a one-player move to one-boxing against a two-boxing partner gives $\Xi(C, D) = \Xi(D, C) = S-P$, and moving the second player as well gives $\Xi(C, C) = (R - T) + (S - P)$:

$$\begin{aligned} \Xi(D, D) &= 0, \\ \Xi(C, D) = \Xi(D, C) &= S - P, \\ \Xi(C, C) &= R - T + S - P. \end{aligned} \tag{E2}$$

Let $\Delta := R - S - T + P$ and $V(\nu) := (P - S)\nu - \frac{1}{2}\Delta\nu^2$. If both players independently choose C with probability ν, then averaging Eq. (E2) over that mixture gives

$$\overline{\Xi}(\nu) = \Delta\nu^2 - 2(P - S)\nu = -2V(\nu). \tag{E3}$$

Suppose the common weight ν is adjusted across many attempts, rising when one-boxing's payoff against the current mixture, $U(C \mid \nu)$, beats the mixture average and falling otherwise. In the continuous-time limit of small adjustments this reinforcement rule is the two-strategy replicator equation [44, 45]; the factor $\nu(1 - \nu)$ below makes the pure strategies $\nu = 0, 1$ invariant, keeping the flow in $[0, 1]$. With $h(\nu)$ denoting the payoff advantage of one-boxing,

$$h(\nu) := U(C \mid \nu) - U(D \mid \nu) = \Delta\nu - (P - S), \tag{E4}$$

the adjustment is the replicator flow

$$\dot{\nu} = f(\nu) := \nu(1 - \nu)h(\nu). \tag{E5}$$

The quadratic

$$V(\nu) = (P - S)\nu - \tfrac{1}{2}\Delta\nu^2 \tag{E6}$$

decreases along the flow, $\dot{V} = -\nu(1 - \nu)h(\nu)^2 \le 0$: the replicator flow moves downhill in V, equivalently uphill in the expected exact potential $\overline{\Xi} = -2V$.

In the Prisoner's-Dilemma region, $h(\nu) < 0$ throughout $0 < \nu < 1$, so every interior state flows toward $\nu = 0$, mutual two-boxing. In the coordination region $R > T$, the interior fixed point is the zero of h: with $\nu(1-\nu) \neq 0$ there, $\dot{\nu} = 0$ forces $h(\nu^*) = \Delta\nu^* - (P - S) = 0$, and since $\Delta = (R-T)+(P-S)$ this recovers the indifference prior $\beta_{\rm ind}$ of Eq. (35),

$$\nu^* = \frac{P - S}{\Delta} = \beta_{\rm ind}. \tag{E7}$$

Its stability and that of the endpoints follow directly from

$$\begin{aligned} f'(0) &= S - P < 0, \\ f'(1) &= T - R < 0, \\ f'(\nu^*) &= \nu^*(1 - \nu^*)\Delta > 0. \end{aligned} \tag{E8}$$

Hence $\nu = 0$ and $\nu = 1$ are attractors, while ν^* is an unstable basin boundary: initial conditions with $\nu < \nu^*$ flow to two-boxing and those with $\nu > \nu^*$ flow to one-boxing. The basin lengths are therefore

$$|\mathcal{B}_D| = \nu^*, \qquad |\mathcal{B}_C| = 1 - \nu^*. \tag{E9}$$

At $R = T$, $\nu^* = 1$ and $f'(1) = 0$. Extending the interior fixed-point branch through the boundary, it crosses the permanent branch $\nu = 1$ and exchanges stability with it: this is a boundary transcritical bifurcation of the replicator flow [46].

For $\Delta > 0$, V is a concave quadratic on the closed interval $[0, 1]$. In the coordination region its stable endpoints are two constrained boundary wells, separated by the maximum at ν^*. The Lyapunov barriers measured from the two-boxing and one-boxing endpoints are

$$\begin{aligned} B_D &:= V(\nu^*) - V(0) = \frac{(P - S)^2}{2\Delta}, \\ B_C &:= V(\nu^*) - V(1) = \frac{(R - T)^2}{2\Delta}. \end{aligned} \tag{E10}$$

For the selected-payoff Newcomb entries of Eq. (32), these become

$$\begin{aligned} B_D &= \frac{p_{\text{match}}^2 m^2}{2(M + m)(p_{\text{match}} - p_{\text{mismatch}})}, \\ B_C &= \frac{[p_{\text{match}} M - p_{\text{mismatch}}(M + m)]^2}{2(M + m)(p_{\text{match}} - p_{\text{mismatch}})}. \end{aligned} \tag{E11}$$

They are Lyapunov barriers, features of the deterministic landscape: under the flow alone a trajectory never leaves its basin, so on their own they set no rate. Only once a stochastic perturbation is specified can these barriers be related to switching rates between the two basins.

Risk dominance is read directly from either the basins or the barriers. Mutual one-boxing has the larger basin iff

$$\nu^* < \frac{1}{2} \quad \Longleftrightarrow \quad R - T > P - S \quad \Longleftrightarrow \quad \Gamma > \frac{M + m}{M - m}. \tag{E12}$$

Within the coordination region the same inequality is equivalent to $B_C > B_D$: the risk-dominant endpoint has the larger basin and the larger barrier separating it from the unstable boundary. It also lies lower in V: the difference

$$V(1) - V(0) = \tfrac{1}{2}\left[p_{\text{mismatch}}(M + m) - p_{\text{match}}(M - m)\right] \tag{E13}$$

changes sign at $\Gamma = (M + m)/(M - m)$, so on each side the risk-dominant endpoint is the one lying lower in V. At the risk-dominance boundary, $\nu^* = 1/2$, $V(1) = V(0)$, and $B_C = B_D$.

The three regions of Fig. 4 therefore have a direct basin interpretation: in region III only two-boxing attracts; in region II both endpoints attract but the two-boxing basin is larger; in region I both attract but the one-boxing basin is larger. As $\Gamma \to \infty$, the separating point approaches $\nu^* \to m/(M + m)$, so the two-boxing attractor persists while its basin shrinks.

The two thresholds admit a Landau-style reading of this Lyapunov landscape [47]. Throughout the coordination region the curvature is constant, $V''(\nu) = -\Delta$ for every ν, so the second derivative vanishes nowhere on the interval. Crossing $\Gamma = (M + m)/M$, the boundary transcritical bifurcation brings the separating maximum ν^* into the open interval and turns the one-boxing endpoint into a local boundary well. A second locally stable boundary well appears together with the unstable separator, the onset of metastability as at a spinodal, though the bifurcation is transcritical rather than a fold: they enter at the end of the interval, where $\nu = 1$ is a permanent fixed point, and the constant curvature leaves no interior pair of extrema behind. At $\Gamma = (M + m)/(M - m)$ no fixed point changes stability. The two endpoint wells reach equal depth, $V(1) = V(0)$, with equal barriers $B_C = B_D$ and separating maximum at $\nu^* = 1/2$, which plays the role of a coexistence point. Both readings are statements about the constrained Lyapunov function V of the replicator flow.

The filter defines the selected histories and the induced equilibria, but no dynamics for the preparation weight ν; the replicator flow is an additional model of adaptation, outside the P-CTC consistency condition.

Appendix F: Adaptive preparation: predictor response rules

The fixed-prior comparison treats the preparation of Y as external. Assumption 1 allows Ω to know the act preparation, so the forecast weight β may respond to the act's one-boxing weight α. The resulting verdict depends not only on Γ but also on the response rule assigned to the predictor. Over an ensemble of dispositions α, a responding forecast weight $\beta(\alpha)$ correlates the registers before any selection, the common-cause route of Fig. 3(a); the analysis below works at fixed α, where the preparation remains a product. Holding the parity filter fixed, we compare three predictor-response prescriptions as a sensitivity analysis: an accuracy-seeking forecaster, a calibration rule in two variants, and an adversarial lower envelope.

Prepare

$$\begin{aligned} |\psi(\alpha)\rangle_X &= \sqrt{\alpha}\,|C\rangle + \sqrt{1 - \alpha}\,|D\rangle, \\ \rho_Y(\beta) &= \beta\,|C\rangle\langle C| + (1 - \beta)\,|D\rangle\langle D|. \end{aligned}$$

The final computational-basis measurement gives one-boxing with Born weight α; the realized act is not read out before postselection. A relative phase in $|\psi(\alpha)\rangle_X$ is irrelevant here because the effect of the filter, the state of Y, and the payoff observable are all diagonal in the $\{C, D\}$ basis. The agent's selected payoff yield averages the four selected-payoff entries of Eq. (32) over the independently prepared weights α and β: for the selected-payoff table

(R, S, T, P),

$$W(\alpha,\beta) = \alpha\beta R + \alpha(1-\beta)S + (1-\alpha)\beta T + (1-\alpha)(1-\beta)P, \tag{F1}$$

or

$$\begin{aligned} W &= P + \alpha(S-P) + \beta(T-P) + \alpha\beta\Delta, \\ \Delta &= R - S - T + P. \end{aligned} \tag{F2}$$

For the Newcomb entries, $\Delta = (M+m)(p_{\text{match}} - p_{\text{mismatch}})$, so $\Delta > 0$ for $\Gamma > 1$ and $\Delta = 0$ at $\Gamma = 1$. Because W is affine in either variable when the other is fixed, the extrema below occur at corners or, in the adversarial problem, where two endpoint responses tie.

1. Accuracy-seeking forecaster

Under Assumption 1, an accuracy-seeking forecaster predicts the more likely act,

$$\beta_{\text{acc}}(\alpha) = \mathbf{1}[\alpha > \tfrac{1}{2}], \tag{F3}$$

with the tie assigned to $\beta = 0$. The payoff has two branches: $W_0 = P + \alpha(S-P)$ for $0 \le \alpha \le \frac{1}{2}$, and $W_1 = T + \alpha(R-T)$ for $\frac{1}{2} < \alpha \le 1$. Since $P > S$, W_0 decreases and has maximum P at $\alpha = 0$.

If $R > T$, W_1 increases to R at $\alpha = 1$; because $R > P$, this is the unique global optimum. If $R = T$, then $W_1 = R > P$, so every $\alpha \in (\frac{1}{2}, 1]$ is optimal. If $R < T$, W_1 is decreasing in α and rises toward its supremum $(R+T)/2$ as $\alpha \downarrow \frac{1}{2}$. The Prisoner's-Dilemma ordering $T > R > P > S$ ensures that this exceeds the lower branch's maximum P, but no preparation attains it: at $\alpha = \frac{1}{2}$ the tie rule selects $\beta = 0$ and gives $(S+P)/2 < P$. Thus $\alpha = \frac{1}{2} + \varepsilon$ approaches the best value while remaining just likely enough to be forecast as a one-boxer.

For the Newcomb entries, $R > T$ iff $\Gamma > (M+m)/M$. The result is therefore a unique pure one-boxing optimum above the Stag-Hunt boundary, a continuum of optima at the boundary, and only a supremum below it. The last non-attainment is a tie-breaking effect: choosing $\beta = 1$ at equality would make $\alpha = \frac{1}{2}$ attain the same value.

2. Direct calibration

The simplest marginal-calibration rule sets $\beta = \alpha$. Then $W(\alpha, \alpha)$ is convex for $\Gamma > 1$, so its maximum occurs at an endpoint. Since $R > P$, the preferred endpoint is pure one-boxing,

$$\alpha^*_{\text{cal}} = 1 \qquad (\Gamma \ge 1).$$

This is a calibration rule rather than an accuracy-seeking response rule; its purpose is to show how strongly the answer can depend on the response assigned to Ω.

3. Accepted-frequency calibration

A self-calibrated predictor sets its forecast prior β equal to the long-run accepted one-boxing probability $\Pr(X = C \mid \mathsf{A})$ that same prior generates, rather than to the preselection weight α; because acceptance filters the two acts unequally, these differ. The condition is a population-level fixed point, not a finite-sample frequency. This subsection solves that self-consistency condition for the calibration curve $\alpha(\beta)$, then maximizes the agent's selected payoff yield W along it and finds the same pure one-boxing optimum as the direct rule $\beta = \alpha$ above.

Take Γ finite, so $p_- := p_{\text{mismatch}} > 0$, and write $p_+ := p_{\text{match}}$ and $\delta := p_+ - p_- \ge 0$, the selection bias by which postselection favors a match over a mismatch ($\delta > 0$ iff $\Gamma > 1$). A one-boxing branch and a two-boxing branch survive postselection at different rates, so the accepted-run one-boxing frequency is α reweighted by those rates. Averaging over the forecast at prior β, an $X = C$ branch matches (weight p_+) with probability β and mismatches (weight p_-) with probability $1-\beta$, and symmetrically for $X = D$, giving the per-act acceptance weights $w_C := p_- + \delta\beta = \beta p_+ + (1-\beta)p_-$ and $w_D := p_+ - \delta\beta = \beta p_- + (1-\beta)p_+$. Self-calibration sets the assumed prior equal to the accepted-run frequency it produces, the Bayes reweighting of α by these weights, $\beta = \Pr(X = C \mid \mathsf{A}) = \alpha w_C / [\alpha w_C + (1-\alpha) w_D]$, which inverts to

$$\begin{aligned} \alpha(\beta) &= \frac{\beta w_D}{Z}, \\ Z &= p_- + 2\delta\beta(1-\beta), \\ \frac{\mathrm{d}\alpha}{\mathrm{d}\beta} &= \frac{p_-[p_+ - 2\delta\beta(1-\beta)]}{Z^2} > 0. \end{aligned} \tag{F4}$$

The bracket is at least $(p_+ + p_-)/2$, so this is a unique increasing response curve from $(0,0)$ to $(1,1)$ for every finite $\Gamma \ge 1$. It is the calibration relation; at $\Gamma = 1$ it reduces to $\beta = \alpha$.

Writing $W(\beta) := W(\alpha(\beta), \beta)$ for the payoff along the calibration curve, substitution into Eq. (F1) gives

$$R - W(\beta) = \frac{(1-\beta)P_{\text{cal}}(\beta)}{Z}, \tag{F5}$$

where

$$\begin{aligned} P_{\text{cal}}(\beta) = {} & p_+ p_- (M - m) \\ & + \delta\big[3Mp_- + (2M - m)\delta\big]\beta \\ & + \delta\big[m\delta - M(p_+ + p_-)\big]\beta^2. \end{aligned} \tag{F6}$$

Its quadratic coefficient is $\delta[m\delta - M(p_+ + p_-)] \le 0$, while $P_{\text{cal}}(0) = p_+ p_-(M - m) > 0$ and $P_{\text{cal}}(1) = p_+(Mp_+ - mp_-) > 0$. Hence P_{cal} is concave and positive on $[0,1]$, so $W(\beta) < R$ for $\beta < 1$ and $W(1) = R$. Because the response curve reaches $\beta = 1$ only at $\alpha = 1$, the unique payoff optimum is $(\alpha^*, \beta^*) = (1,1)$. At the perfect-consistency endpoint $p_- = 0$ the calibration relation becomes set-valued, its calibrated pairs with nonzero acceptance being $\beta = 0$ with $\alpha < 1$, $\beta = 1$ with $\alpha > 0$, and $\alpha = \frac{1}{2}$ with $0 < \beta < 1$; every such pair obeys $W = \alpha\beta R + (1-\alpha)(1-\beta)P \leq R$, with equality only at $(1,1)$. Thus pure one-boxing is optimal in the zero-noise limit; accepted-run calibration and $\beta = \alpha$ differ only away from the optimum.

4. Adversarial lower envelope

For a worst-case bound, after observing the one-boxing weight α, let the predictor choose $\beta \in [0,1]$ to minimize the selected payoff yield:

$$V_{\text{adv}}(\alpha) := \inf_{\beta\in[0,1]} W(\alpha, \beta).$$

Because W is affine in β, only the endpoint priors matter. From Eq. (F2),

$$W_0(\alpha) := W(\alpha, 0) = P + \alpha(S - P),$$
$$W_1(\alpha) := W(\alpha, 1) = T + \alpha(R - T),$$

and $V_{\text{adv}} = \min\{W_0, W_1\}$, with $W_1 - W_0 = T - P + \alpha\Delta$. For the Newcomb entries, $\Delta \geq 0$ when $\Gamma \geq 1$, while $P > S$ and $R > P$.

If $T \geq P$, then $W_1 \geq W_0$ throughout. Since W_0 decreases, the unique worst-case optimum is pure two-boxing, $\alpha^* = 0$. This includes $T = P$: the endpoint priors tie at $\alpha = 0$, but every $\alpha > 0$ has a smaller worst-case payoff.

If $T < P$, then $W_1 < W_0$ at $\alpha = 0$ and $W_1 > W_0$ at $\alpha = 1$. Moreover $R > T$, so W_1 rises toward their unique crossing while W_0 falls away from it. The crossing is therefore the unique maximum of the lower envelope:

$$\alpha^* = \frac{P - T}{\Delta}, \qquad V^*_{\text{adv}} = \frac{RP - ST}{\Delta}, \tag{F7}$$

$$\beta^* = \frac{P - S}{\Delta}. \tag{F8}$$

Here β^* makes the agent indifferent between its preparations. At α^* the payoff is independent of β, and at β^* it is independent of α, so (α^*, β^*) is a saddle point.

For the Newcomb table, $T < P$ is equivalent to

$$\Gamma > \frac{M + m}{m}. \tag{F9}$$

Above this threshold,

$$\alpha^* = \frac{\Gamma m - (M + m)}{(M + m)(\Gamma - 1)}, \quad \beta^* = \frac{\Gamma m}{(M + m)(\Gamma - 1)}. \tag{F10}$$

Thus a positive one-boxing weight appears only when $\mathcal{D} > M/(M+2m)$ on the distinguishability scale of Fig. 7. The dual prior β^* equals the fixed-prior basin boundary β_{ind} in Eq. (35), and

$$\beta^* - \alpha^* = \frac{1}{\Gamma - 1}.$$

Both approach $m/(M + m)$ as $\Gamma \to \infty$. At perfect consistency mismatch records never survive, so in the lossless normalization ($p_{\text{match}} = 1$, $p_{\text{mismatch}} \to 0$) the two endpoint yields collapse to $W_1(\alpha) = \alpha M$ and $W_0(\alpha) = (1 - \alpha)m$. The adversary always returns the smaller of the two, so the agent's best guarantee equalizes them, $\alpha M = (1 - \alpha)m$, giving $\alpha^* = m/(M + m)$ and the guaranteed value $Mm/(M + m)$. The optimum stays close to pure two-boxing ($\alpha^* \ll 1$ for $M \gg m$) because the predictor, free to reset the prior after seeing α, can undercut any larger one-boxing weight. Table II summarizes the three predictor response rules.

Figure 7 places these questions on the same distinguishability axis. The equilibrium and unbiased-verdict thresholds are both of order m/M, while a positive adversarial one-boxing weight appears only near perfect distinguishability, at $\mathcal{D} > M/(M + 2m)$.

Appendix G: Side-by-side decomposition, pairwise scoring, and worldline symmetry

1. Side-by-side decomposition

Lewis identified the Prisoner's Dilemma with two Newcomb problems, one faced by each player [11]. This appendix separates the two objects that identification couples. A *component* is a single-agent Newcomb decision problem: one register is the act, the other its forecast, and only that player's payoff is scored. The *composite* is the one game the two components make together, here the symmetric 2×2 game $G(\Gamma)$: each player chooses C or D, the row player receives the selected yields R, S, T, P of Eq. (32) at the profiles $(C, C), (C, D), (D, C), (D, D)$, and the column player the transpose. The result proved below is that the component and the composite separate at unbiased priors: the composite acquires the mutual one-boxing equilibrium at $\Gamma = (M + m)/M$, before either component's verdict changes at $\Gamma = (M + m)/(M - m)$.

Each component retains the ordinary Newcomb dominance relation. Holding player j's action fixed gives

$$u_i(D, x_j) - u_i(C, x_j) = m,$$

so two-boxing adds the transparent reward within every fixed profile. Under the symmetric accepted distribution obtained from $\alpha = \beta = 1/2$, $\Pr(X_j = x_i \mid X_i = x_i, \mathsf{A}) = q = \Gamma/(\Gamma + 1)$ for either $x_i \in \{C, D\}$. A one-boxer collects the opaque reward M when the forecast matches its act (probability q) and nothing otherwise; a two-boxer always collects the transparent m and additionally the opaque M when the forecast mismatches (probability $1 - q$). Either component's two acts therefore carry the accepted expected payoffs

$$V_C(q) = Mq, \qquad V_D(q) = m + M(1 - q) = M + m - Mq. \tag{G1}$$

The single-agent recommendation changes to one-boxing at $\Gamma = (M + m)/(M - m)$.

Response rule	β	optimum	threshold
Accuracy-seeking	$\beta_{\rm acc} = \mathbf{1}[\alpha > \frac{1}{2}]$	$\alpha^* = 1$	$\Gamma > \dfrac{M+m}{M}$
Calibrated	$\beta = \alpha$	$\alpha^* = 1$	all $\Gamma \geq 1$
Adversarial	$\arg\min_\beta W$	$\alpha^* = 0$ below, rising toward $\frac{m}{M+m}$ above	$\Gamma > \dfrac{M+m}{m}$

TABLE II. Optimal act preparation under three predictor response rules, with the same parity filter throughout. Among these rules, the accuracy-seeking row is the closest to Newcomb's predictor under Assumption 1. The calibrated result also holds for accepted-frequency calibration; the adversarial row is a worst-case bound rather than a forecast model.

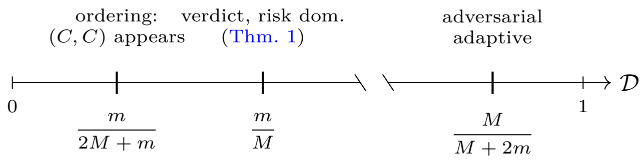


FIG. 7. Threshold ladder on the distinguishability axis $\mathcal{D} = (\Gamma-1)/(\Gamma+1)$. The one-boxing equilibrium is weak at $m/(2M+m)$ and strict above it; the single-agent verdict and risk dominance switch at m/M; a nonzero adversarial one-boxing weight appears only above $M/(M+2m)$. For Gardner's values these are approximately 5.0×10^{-4}, 10^{-3}, and 0.998. The axis break separates the two near-zero thresholds from the adversarial threshold near one; positions within each segment are schematic, while the labels give the exact values.

Proposition (Side-by-side decomposition)**.** *Let $M > m > 0$ and $\Gamma \geq 1$. Each component has the ordinary Newcomb payoffs: holding the forecast y fixed gives $u_i(D,y) - u_i(C,y) = m > 0$. At unbiased priors, the accepted forecast has reliability $q = \Gamma/(\Gamma+1)$, so Eq. (G1) favors one-boxing iff $\Gamma > (M+m)/(M-m)$. The composite $G(\Gamma)$ has Prisoner's-Dilemma ordering for $\Gamma < (M+m)/M$; (C,C) is a weak Nash equilibrium at equality and the table has Stag-Hunt ordering above it. Moreover, in the coordination region (C,C) is risk-dominant iff $\Gamma > (M+m)/(M-m)$. Thus the component verdict switches at the threshold where (C,C) becomes risk-dominant, after the equilibrium first appears weakly at the lower boundary.*

Proof. Equation (27) gives the entrywise difference m. At unbiased priors, agreement and disagreement have equal prior weight, and the filter changes their accepted odds by Γ, giving $q = \Gamma/(\Gamma+1)$. Equation (G1) then gives $V_C > V_D$ iff $q > (M+m)/(2M)$, or $\Gamma > (M+m)/(M-m)$; the common acceptance probability makes accepted-history and selected-yield component orderings identical. For the composite, Eq. (32) gives $R > T$ iff $\Gamma > (M+m)/M$, while Appendix D gives risk dominance iff $R-T > P-S$, or $\Gamma > (M+m)/(M-m)$. □

Hence, for $(M+m)/M < \Gamma < (M+m)/(M-m)$, mutual one-boxing is already a Nash equilibrium of the composite, but either component still favors two-boxing at unbiased priors and (D,D) has the larger basin. This is the precise sense in which $G(\Gamma)$ is two Newcomb problems side by side while their composite changes from Prisoner's Dilemma to Stag Hunt. The broader identification criticized in Ref. [13] is not required.

2. Mixed-observer scoring

Under accepted-history scoring, which conditions on acceptance and averages within the accepted runs, each fixed accepted record carries the bare Newcomb payoff at every Γ: the acceptance probability divides out, so each accepted record (x,y) carries its plain payoff $u(x,y)$ (Eq. (27)). That normalization keeps the Γ-dependent correlation in the accepted ensemble but leaves it unrewarded: at unbiased priors the accepted records agree at odds Γ to one, the bias residing in their joint parity $x \oplus y$ while each one-register marginal stays uniform. The selected yield keeps it, weighting each history by its acceptance probability, $p_{xy}u(x,y)$ (Eq. (26)).

The mutual-one-boxing pure equilibrium does not survive at finite Γ if only one side uses selected-yield scoring.[12] Give the row player the entries (R,S,T,P) and let the column player condition on acceptance. Against a pure opponent, the conditional player's acceptance factor cancels, leaving the raw Newcomb table; two-boxing then gains m against either fixed act. Hence no pure profile with a one-boxing conditional player is an equilibrium, and mutual two-boxing is the unique pure equilibrium. Away from pure opponents, the conditional payoff is a ratio rather than an affine mixed-strategy extension, so the hybrid object is not the standard multilinear extension of a 2×2 normal-form game. At the uniform mixture, Theorem 1 makes the two comparisons share a sign, but the full coordination structure still requires the same selected-yield convention on both sides. In this precise sense the

[12] Adopting this scoring is a normative choice: it weights each history's utility by its consistency weight.

coordination is a property of the pairwise ledger: the yield cannot be assigned independently to one worldline while the other conditions on acceptance.

3. Exchange, coding, and Bell representatives

The exchange symmetry of Sec. V respects the coding classes: exchange preserves the aligned-coding weight pair $\alpha_X\alpha_Y$, $(1-\alpha_X)(1-\alpha_Y)$ and the inverted-coding pair $\alpha_X(1-\alpha_Y)$, $(1-\alpha_X)\alpha_Y$ separately, so the two worldline registers are interchangeable within either coding class.

Coding inversion is the distinct operation $U_{\mathrm{inv}} = I_X \otimes \sigma_x^{(Y)}$. It exchanges $\Pi_=$ and $\Pi_{\neq}$ and maps $\Phi^{\pm} \leftrightarrow \Psi^{\pm}$. More generally, if U_X, U_Y encode the same physical label $s \in \{0,1\}$ into the two record qubits, the physical-consistency projector is

$$\Pi_{\mathrm{phys}}(U_X, U_Y) = \sum_{s=0}^{1} U_X \left|s\right\rangle\!\left\langle s\right| U_X^{\dagger} \otimes U_Y \left|s\right\rangle\!\left\langle s\right| U_Y^{\dagger}. \quad \text{(G2)}$$

It equals $\Pi_=$ for aligned codings and $\Pi_{\neq}$ for opposite codings. A coherent switch between the codings would require an additional coding register and lies outside the present model.

Within a fixed coding, the equal-superposition input gives the coherent and noisy states in Eqs. (46) and (H11). The former superposes the two history classes; the latter is their incoherent mixture. Appendix H proves the Bell-plus-separable decomposition (H15), the separability witness, and the concurrence identity.

Appendix H: Complementarity and the σ_x coherence measurement

This appendix proves the distinguishability–visibility bound of Sec. VI and gives the symmetric-preparation state algebra behind $\langle\sigma_x\rangle_X$, the nonlinear witness, and the equality case.

Proposition (distinguishability–visibility complementarity)**.** *For any process that creates the record Y before the final act readout, Eq. (45) holds: $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$. Equality requires that the designated readout of Y be optimal and that access to the purifying environment F not increase the optimal guessing bias beyond what is already achievable from Y.* For unequal act priors, $\mathcal{D}$ includes predictability arising from the prior imbalance as well as information carried by Y; on the equal-prior diagnostic input of Sec. VI, it is purely the record's which-alternative distinguishability.

Proof. Purify the joint state of X and Y by adjoining an environment F, and write the purification in the act basis as

$$\left|\psi\right\rangle_{XYF} = \left|C\right\rangle_X \left|u\right\rangle_{YF} + \left|D\right\rangle_X \left|v\right\rangle_{YF}, \quad \text{(H1)}$$

where $\langle u|u\rangle + \langle v|v\rangle = 1$. Tracing out YF gives $\langle C|\rho_X|D\rangle = \langle v|u\rangle$, so

$$\mathcal{V} = 2|\langle v|u\rangle|. \quad \text{(H2)}$$

Here $\langle C|\rho_X|D\rangle = (\langle\sigma_x\rangle_X - i\langle\sigma_y\rangle_X)/2$, so $2|\langle C|\rho_X|D\rangle|$ is the transverse Bloch length, the phase-optimized visibility of Eq. (44). If the full system YF were available, the optimal guessing bias would be the Helstrom value $\|p_C\rho_C - p_D\rho_D\|_1$, where ρ_C, ρ_D are the normalized conditional YF states given the act, which with the act priors $p_C = \langle u|u\rangle$ and $p_D = \langle v|v\rangle$ absorbed into the subnormalized states is the trace norm of $|u\rangle\langle u| - |v\rangle\langle v|$. Restricted to $\mathrm{span}\{u, v\}$, its two possibly nonzero eigenvalues have sum $\langle u|u\rangle - \langle v|v\rangle$ and product $|\langle v|u\rangle|^2 - \langle u|u\rangle\langle v|v\rangle$, nonpositive by Cauchy–Schwarz. They therefore have opposite signs or one is zero, and the trace norm $|\lambda_+| + |\lambda_-| = \sqrt{(\lambda_+ + \lambda_-)^2 - 4\lambda_+\lambda_-}$ gives

$$\mathcal{D}_{YF} = \big\| \left|u\right\rangle\!\left\langle u\right| - \left|v\right\rangle\!\left\langle v\right| \big\|_1 \quad \text{(H3)}$$

$$= \sqrt{\big(\langle u|u\rangle + \langle v|v\rangle\big)^2 - 4|\langle v|u\rangle|^2} \quad \text{(H4)}$$

$$= \sqrt{1 - 4|\langle v|u\rangle|^2} = \sqrt{1 - \mathcal{V}^2}. \quad \text{(H5)}$$

The chosen readout of Y cannot outperform the optimal measurement on Y, and access to Y alone cannot outperform access to YF. Hence, writing $\mathcal{D}_Y$ for the optimal guessing bias from Y alone, $\mathcal{D} \leq \mathcal{D}_Y \leq \mathcal{D}_{YF}$, which gives Eq. (45). Equality holds when both inequalities are equalities. □

Corollary (a perfectly readable record removes local act coherence)**.** $\Pr(y = x) = 1$ with a designated computational-basis readout of Y means the joint distribution of the X-readout and the Y-readout is supported on the diagonal; the vanishing populations remove, by positivity, every matrix element involving an off-diagonal record. The joint state may still carry coherences inside that diagonal subspace, but it has the form

$$\rho_{XY} = \sum_{y,y'} r_{yy'} \left|y\right\rangle_X \left|y\right\rangle_Y \left\langle y'\right|_X \left\langle y'\right|_Y$$

for some matrix $r_{yy'}$. Tracing over Y removes the $y \neq y'$ terms, so

$$\rho_X = \sum_y r_{yy} \left|y\right\rangle\!\left\langle y\right|$$

is diagonal and $\mathcal{V} = 0$; equivalently, the proposition gives $\mathcal{V} = 0$ directly from $\mathcal{D} = 1$. A perfect classical readout of X is created by a $\{C, D\}$-basis measurement, which leaves the diagonal Born weights untouched and removes only the off-diagonal coherence; the accepted weights equal the original preparation weights when the record-creation process is nondemolition in that basis, as for an ordinary controlled copy of the act value, and then $p_C = \alpha$. The endpoint $\Gamma \to \infty$ of (48) realizes this, with $\mathcal{D} \to 1$, $\mathcal{V} \to 0$. □

The σ_x state algebra. The two CNOTs from X and Y into the ancilla map $|+\rangle_X \otimes |+\rangle_Y \otimes |0\rangle_a$ to

$$\tfrac{1}{2}\big(|00\rangle\,|0\rangle + |01\rangle\,|1\rangle + |10\rangle\,|1\rangle + |11\rangle\,|0\rangle\big)_{XYa}, \tag{H6}$$

with the ancilla carrying the parity $x \oplus y$. Write $c := \cos(\theta/2)$ and $s := \sin(\theta/2)$. The rotation $R_y(\theta)$ on the ancilla followed by postselection on $|0\rangle_a$ yields the normalized joint state

$$|\psi(\theta)\rangle_{XY} = \tfrac{1}{\sqrt{2}}\big(c\,|00\rangle - s\,|01\rangle - s\,|10\rangle + c\,|11\rangle\big). \tag{H7}$$

At the symmetric preparation each parity sector has prior weight $1/2$, so the acceptance probability is $(c^2+s^2)/2 = 1/2$, independent of θ. Within the accepted ensemble the match and mismatch probabilities are c^2 and s^2, recovering $\Gamma = \cot^2(\theta/2)$.

Tracing out Y sums the act coherence over the two forecast values, each branch carrying the mismatch sign, $\langle C|\rho_X|D\rangle = \sum_y \langle Cy|\psi\rangle\langle\psi|Dy\rangle = -cs = -\frac{1}{2}\sin\theta$, so the reduced state on X is

$$\rho_X(\theta) = \tfrac{1}{2}\big(I - \sin\theta\,\sigma_x\big) = \tfrac{1}{2}\left(I - \frac{2\sqrt{\Gamma}}{\Gamma+1}\sigma_x\right), \tag{H8}$$

so $\langle\sigma_x\rangle_X = -2\sqrt{\Gamma}/(\Gamma+1)$. At $\Gamma = 1$, one has $c = s$ and Eq. (H7) factorizes as $|-\rangle_X\,|-\rangle_Y$, so the local signal reaches -1 even though the record has no predictive bias. In the perfect-consistency limit $\Gamma \to \infty$, the joint state tends to $|\Phi^+\rangle$, retaining joint coherence while the reduced state of X becomes maximally mixed and the local signal vanishes.

The distinguishability comes from the same state. Grouped by the act value, $|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}\big(|C\rangle_X\,|\phi_C\rangle_Y + |D\rangle_X\,|\phi_D\rangle_Y\big)$ with conditional records $|\phi_C\rangle = c\,|0\rangle - s\,|1\rangle$ and $|\phi_D\rangle = -s\,|0\rangle + c\,|1\rangle$. Their density-matrix difference is diagonal,

$$|\phi_C\rangle\langle\phi_C| - |\phi_D\rangle\langle\phi_D| = (c^2 - s^2)\,\sigma_z = \cos\theta\,\sigma_z, \tag{H9}$$

so the computational-basis readout of Y already attains the optimal which-way measurement, with $\mathcal{D} = \frac{1}{2}\|\cos\theta\,\sigma_z\|_1 = \cos\theta$. With $\mathcal{V} = \sin\theta$ from Eq. (H8), the coherent filter saturates the bound, $\mathcal{D}^2 + \mathcal{V}^2 = \cos^2\theta + \sin^2\theta = 1$, recovering Eq. (48). This is the equality case deferred in the proof of (45): the accepted state is pure and its record measurement optimal.

Phase and implementation dependence. A phase-generalized coherent filter has $K_{\theta,\varphi} = c\Pi_= - e^{i\varphi}s\Pi_{\neq}$. On the same input it gives

$$\rho_X(\theta,\varphi) = \tfrac{1}{2}\big(I - \sin\theta\cos\varphi\,\sigma_x\big), \qquad \langle\sigma_y\rangle_X = 0. \tag{H10}$$

The two cross terms carry $e^{+i\varphi}$ and $e^{-i\varphi}$ and add to the real element $\langle C|\,\rho_X\,|D\rangle = -\frac{1}{2}\sin\theta\cos\varphi$, which is why the phase attenuates the σ_x signal rather than rotating it into σ_y. Thus the real-amplitude circuit maximizes the local signal, and the σ_x measurement witnesses that representative. At $\varphi = \pi/2$ the selected XY state is still pure while the reduced act state carries no transverse coherence: the phase moves the signal into joint correlations, and a computational-basis readout of Y ceases to be optimal.

At the same Γ, set $\ell = s^2$ in the binary noisy P-CTC selected map of Eq. (14). Its normalized accepted state is

$$\rho^{\text{noise}}_{XY} = (1-\ell)\,\big|\Phi^+\big\rangle\big\langle\Phi^+\big| + \ell\,\big|\Psi^+\big\rangle\big\langle\Psi^+\big|, \qquad \rho^{\text{noise}}_X = \tfrac{1}{2}I. \tag{H11}$$

The two environment-recorded branches project onto complementary sectors, so the flip record in the loop environment fixes which sector survived: tracing it out removes the local coherence between agreement and mismatch, the local signal vanishes, and the joint coherence within each sector remains. By contrast, a computational-basis classical rejection filter produces

$$\rho^{\text{cl}}_{XY} = \frac{1-\ell}{2}\big(|00\rangle\langle 00| + |11\rangle\langle 11|\big) + \frac{\ell}{2}\big(|01\rangle\langle 01| + |10\rangle\langle 10|\big), \tag{H12}$$

which has no off-diagonal terms at all. Since both $|\Phi^+\rangle$ and $|\Psi^+\rangle$ are $+1$ eigenstates of $\sigma_x \otimes \sigma_x$, any state on their span gives $\langle\sigma_x \otimes \sigma_x\rangle = 1$ at every Γ, mixture or superposition alike, whereas $\sigma_x \otimes I$ maps Φ^+ to Ψ^+, so the local signal lives on the cross term alone. The binary noisy P-CTC therefore has $\langle\sigma_x \otimes \sigma_x\rangle = 1$, while the classical state, diagonal in a basis where $\sigma_x \otimes \sigma_x$ is purely off-diagonal, has zero. The real-amplitude coherent state, itself a $+1$ eigenstate, gives the same value, and its nonzero local expectation is what separates it from the noisy state. Since $\sigma_z \otimes \sigma_z$ is $+1$ on the agreement sector and -1 on the mismatch sector, all three maps give $\langle\sigma_z \otimes \sigma_z\rangle = (1-\ell) - \ell = \mathcal{D}$ on this input.

The $\sigma_x \otimes \sigma_x$ value alone is not an entanglement witness because separable states can attain it. For a product state, Cauchy–Schwarz and the one-qubit Bloch bound give

$$|\langle\sigma_x\rangle_1\langle\sigma_x\rangle_2| + |\langle\sigma_z\rangle_1\langle\sigma_z\rangle_2| \le 1,$$

and convexity extends the inequality to every separable state. Hence the nonlinear witness of Eq. (49) obeys

$$\mathcal{W}_{XZ} \le 1 \tag{H13}$$

for separable two-qubit states. The two quantum outputs give $\mathcal{W}_{XZ} = 1 + \mathcal{D}$, while the classical rejection filter gives $\mathcal{W}_{XZ} = \mathcal{D}$. Thus $\Gamma > 1$ yields an entanglement certificate from the two correlators together.

Read jointly, the two records form a two-qubit repetition code: at perfect consistency either coding encodes one logical qubit in the two records, and within a sector $\sigma_x \otimes \sigma_x$ acts as that code's logical σ_x, so the joint measurement reads the coherence of the encoded qubit.

The witness certifies entanglement; the concurrence quantifies it. Concurrence is an entanglement monotone, zero iff a two-qubit state is separable and one iff it is maximally entangled [90]. The noisy state is Bell diagonal with largest weight $1-\ell$, and for Bell-diagonal states the concurrence is $\max(0, 2p_{\max} - 1)$ in the largest weight $p_{\max}$, so for $\ell \le 1/2$,

$$\mathcal{C}\big(\rho_{XY}^{\text{noise}}\big) = 1 - 2\ell = \frac{\Gamma - 1}{\Gamma + 1} = \mathcal{D}. \tag{H14}$$

The pair is entangled iff the predictor is better than chance, and by that amount: the concurrence is the distinguishability. The coherent state of Eq. (H7) is pure, with amplitudes $(a_{00}, a_{01}, a_{10}, a_{11}) = (c, -s, -s, c)/\sqrt{2}$ on the basis records $|xy\rangle$; for a pure two-qubit state the concurrence is $2|a_{00}a_{11} - a_{01}a_{10}|$, here $|c^2 - s^2| = \mathcal{D}$ as well, while the classical rejection filter is separable, concurrence zero. This value is specific to the real-amplitude filter: the phase-generalized filter $K_{\theta,\varphi}$ gives concurrence $\sqrt{\mathcal{D}^2 + (1 - \mathcal{D}^2)\sin^2\varphi}$, equal to $\mathcal{D}$ only at $\varphi = 0$ and reaching 1 at $\varphi = \pi/2$. For $\Gamma > 1$, the local signal, the nonlinear witness, and concurrence therefore distinguish the coherent, binary noisy, and classical implementations at equal Γ. At $\Gamma = 1$, concurrence and the witness excess $\mathcal{W}_{XZ} - 1$ vanish for the real-amplitude representative, although the local and joint correlators still separate the three implementations.

Optimal Bell-plus-separable decomposition. For $\Gamma \ge 1$, substitute $1 - \ell = (1 + \mathcal{D})/2$ and $\ell = (1 - \mathcal{D})/2$ and regroup the equal-weight part of the Bell mixture, $(1 - \ell)\,|\Phi^+\rangle\langle\Phi^+| + \ell\,|\Psi^+\rangle\langle\Psi^+| = \mathcal{D}\,|\Phi^+\rangle\langle\Phi^+| + \frac{1-\mathcal{D}}{2}\big(|\Phi^+\rangle\langle\Phi^+| + |\Psi^+\rangle\langle\Psi^+|\big)$. This gives the optimal Bell-plus-separable split

$$\rho_{XY}^{\text{noise}} = \mathcal{D}\,\big|\Phi^+\big\rangle\big\langle\Phi^+\big| + (1 - \mathcal{D})\rho_{\text{sep}}, \tag{H15}$$

with

$$\rho_{\text{sep}} = \tfrac{1}{4}(I + \sigma_x \otimes \sigma_x) = \tfrac{1}{2}(|{+}{+}\rangle\langle{+}{+}| + |{-}{-}\rangle\langle{-}{-}|),$$

which is manifestly separable. If a decomposition writes the same state as $(1-w)\sigma_{\text{sep}} + w\sigma_{\text{ent}}$, then convexity gives $\mathcal{C}(\rho) \le w$, because the separable part contributes zero and a two-qubit state has concurrence at most one. The Bell-diagonal concurrence, Eq. (H14), is $\mathcal{D}$, so the split with Bell weight $w = \mathcal{D}$ saturates this bound and is optimal. For $\Gamma < 1$, the same argument uses Ψ^+ and $|\mathcal{D}|$. The Bell weight should not be confused with the fully entangled fraction $\max(1-\ell, \ell) = (1+|\mathcal{D}|)/2$, the maximum overlap with a maximally entangled state, which is larger.

Range of the identity. The split (H15) was derived at the equal-superposition input; how far it extends is a question about preparations. For the product preparation with one-boxing amplitudes $\sqrt{\alpha}, \sqrt{\beta}$, the noisy map returns a state whose only coherences are the two within-sector cross terms, and the two sectors share one geometric mean: the agreement amplitudes give $\mu\bar{\mu}$ with $\mu = \sqrt{\alpha\beta}$ and $\bar{\mu} = \sqrt{(1 - \alpha)(1 - \beta)}$, the mismatch amplitudes give $\sqrt{\alpha(1 - \beta)}\,\sqrt{(1 - \alpha)\beta}$, and the two products are equal. The unnormalized agreement and mismatch cross terms are then $(1 - \ell)\mu\bar{\mu}$ and $\ell\mu\bar{\mu}$, and Wootters' formula for states of this form gives

$$\mathcal{C} = \frac{2|\mathcal{D}|\,\mu\bar{\mu}}{p_{\text{sel}}}, \tag{H16}$$

with p_{sel} the preparation's acceptance probability, reducing to $|\mathcal{D}|$ at $\alpha = \beta = 1/2$ and vanishing whenever either register holds a definite value; diagonal preparations, mixtures of definite records, give diagonal accepted states, which are separable. Equality with a maximally entangled component is a property of balanced sectors: subtracting weight $\mathcal{C}$ of Φ^+ leaves a state of the same cross-term pattern whose positivity and partial-transpose conditions reduce to the single requirement $2\mu\bar{\mu} \ge \mu^2 + \bar{\mu}^2$, which holds iff $\mu = \bar{\mu}$. Among interior preparations with nonzero concurrence, the split of Eq. (H15) therefore extends for $\Gamma > 1$ to the balanced family $\beta = 1 - \alpha$, and only there; for $\Gamma < 1$ the mirror condition is $\beta = \alpha$, with Ψ^+ as the component.

Sampling cost. Let $v_\Gamma = 2\sqrt{\Gamma}/(\Gamma + 1)$ be the absolute σ_x expectation of the real-amplitude circuit. Since a σ_x outcome is bounded by ± 1, resolving v_Γ from zero at fixed significance and power requires a number of accepted samples

$$N_{\text{acc}} \gtrsim C_{\text{conf}}\, v_\Gamma^{-2} = C_{\text{conf}} \frac{(\Gamma + 1)^2}{4\Gamma} = O(\Gamma), \tag{H17}$$

where C_{conf} depends only on the chosen statistical criterion. The acceptance probability of this input is 1/2, so circuit attempts and accepted samples differ only by a factor of two.

Appendix I: Noise map, tolerances, and complete symmetric dephasing

The noise model of Sec. VII averages over an unrecorded spread of the consistency angle. Its effects are captured by two quantities: the averaged agreement and mismatch weights, which determine Γ_{eff} and every basis-diagonal payoff threshold, and the coefficient g, which controls coherence between the two parity sectors.

1. Full averaged numerator

For a fixed θ, $K(\theta) = c_\theta \Pi_= - s_\theta \Pi_{\neq}$ with $c_\theta = \cos(\theta/2)$ and $s_\theta = \sin(\theta/2)$, so expanding $\overline{\Lambda}(\rho) = \mathbb{E}[K(\theta)\rho K^\dagger(\theta)]$ term by term gives $\mathbb{E}[c_\theta^2]\,\Pi_=\rho\Pi_= + \mathbb{E}[s_\theta^2]\,\Pi_{\neq}\rho\Pi_{\neq} - \mathbb{E}[c_\theta s_\theta](\Pi_=\rho\Pi_{\neq} + \Pi_{\neq}\rho\Pi_=)$, which is Eq. (50). Write $\rho_{rs} = \Pi_r\rho\Pi_s$ for $r, s \in \{=, \neq\}$; relative to $\mathcal{H}_= \oplus \mathcal{H}_{\neq}$,

$$\overline{\Lambda}(\rho) = \begin{pmatrix} \bar{p}_{\text{match}}\rho_{==} & -g\rho_{=\neq} \\ -g\rho_{\neq=} & \bar{p}_{\text{mismatch}}\rho_{\neq\neq} \end{pmatrix}. \tag{I1}$$

The off-diagonal blocks preserve a phase relation already present between the two history classes. They do not transfer population:

$$\Pi_{\neq}\overline{\Lambda}(\Pi_=\rho\Pi_=)\Pi_{\neq} = 0, \qquad \Pi_=\overline{\Lambda}(\Pi_{\neq}\rho\Pi_{\neq})\Pi_= = 0. \tag{I2}$$

Thus $\bar{p}_{\text{match}} = \mathbb{E}[c_\theta^2]$, $\bar{p}_{\text{mismatch}} = \mathbb{E}[s_\theta^2]$, and $g = \mathbb{E}[c_\theta s_\theta] = \frac{1}{2}\mathbb{E}[\sin\theta]$. By the Cauchy–Schwarz inequality $g^2 \le \bar{p}_{\text{match}}\bar{p}_{\text{mismatch}}$, with equality only for a sharp θ, since equality requires $s_\theta \propto c_\theta$ wherever the distribution has weight, a constant $\tan(\theta/2)$; the magnitude of this coefficient is therefore bounded by the two sector weights.

2. Thresholds and the von Mises spread

Writing the matched binary-noise weight as $\ell_{\rm eff} = \bar{p}_{\rm mismatch}$, one has $\Gamma_{\rm eff} = (1-\ell_{\rm eff})/\ell_{\rm eff}$; the payoff-ordering, risk-dominance, and adversarial-adaptive conditions then take the $\ell_{\rm eff}$ form displayed in Sec. VII. For a von Mises distribution centered on the ideal setting, $\ell_{\rm eff} = \mathbb{E}[\sin^2(\theta/2)] = \frac{1}{2}\big(1-A(\kappa)\big)$, which is Eq. (56). Each tolerance $\ell_{\rm eff} < \ell^*$ of Sec. VII converts exactly to $A(\kappa) > 1-2\ell^*$: Stag-Hunt ordering requires $A(\kappa) > m/(2M+m)$, risk dominance of mutual one-boxing $A(\kappa) > m/M$, and the adversarial condition $A(\kappa) > M/(M+2m)$.

3. Complete symmetric dephasing

For θ uniform on the full phase circle, $\bar{p}_{\rm match} = \bar{p}_{\rm mismatch} = 1/2$ and $g = 0$: complete symmetric dephasing.

Equal sector weights alone do not imply the loss of local coherence between the sectors: the half-circle ensemble, θ uniform on $[0,\pi]$, shares $\bar{p}_{\rm match} = \bar{p}_{\rm mismatch} = 1/2$ and $\Gamma_{\rm eff} = 1$ but has $g = 1/\pi$, retaining interference between the two classes. The two conditions $g = 0$ and $\Gamma_{\rm eff} = 1$ are logically independent.

Appendix J: Accepted records, capacity, and causal representations

This appendix separates four causal objects associated with the selected construction. The first subsection studies the accepted-record channel and its retrocausal capacity; the second embeds the post-heralding accepted slice as a causally separable common-cause process. The final two retain the branch record before conditioning and study the resulting normalized three-record statistics through parity erasure and causal tensor reorientation. None of these constructions represents the full nonlinear P-CTC update for arbitrary local instruments.

1. Retrocausal capacity of the accepted-record channel

For the capacity calculation we apply Ref. [38] to the induced future-to-past record channel $\mathcal{T}_{\mathcal{D}} : X \to Y$. Initialize Y in $I_Y/2$ and let $p_{\rm tot} := p_{\rm match} + p_{\rm mismatch} > 0$. Every record-preserving selected map gives $\mathrm{Tr}_X[\Lambda(\rho_X \otimes I_Y/2)] = \frac{1}{2}\sum_{x,y} p_{xy} \langle x|\,\rho_X\,|x\rangle\,|y\rangle\langle y|$, whose trace is $p_{\rm tot}/2$ independently of ρ_X. Conditioning therefore gives the binary channel

$$\mathcal{T}_{\mathcal{D}}(\rho) = \sum_{x,y} B_{\mathcal{D}}(y \mid x)\,\langle x|\rho|x\rangle\; |y\rangle\langle y|\,, \tag{J1}$$

with transition probabilities

$$B_{\mathcal{D}}(y \mid x) = \begin{cases} (1+\mathcal{D})/2, & y = x, \\ (1-\mathcal{D})/2, & y \neq x. \end{cases} \tag{J2}$$

The agreement weight is the accepted reliability, $(1+\mathcal{D})/2 = q$ by Eq. (21). The orientation matches the retrocausal setting of Ji, Lloyd, and Wilde [38]: the output, the forecast record, precedes the input, the act, in time. They express the asymptotic retrocausal classical capacity as $C_{\rm retro} = I_{\max} + I^{\infty}_{\rm doe}$ and the quantum capacity as $Q_{\rm retro} = C_{\rm retro}/2$ [38]. Both information measures are defined by dominations of the Choi state χ, the channel acting on one half of a maximally entangled pair of the input with a reference copy, the same Bell state as Eq. (6): the max-information $I_{\max}$ is the smallest $\log_2\lambda$ admitting a state σ with $\chi \le \lambda\,(I/2)\otimes\sigma$, the Doeblin information $I_{\rm doe}$ is the smallest $\log_2\lambda$ admitting a unit-trace Hermitian τ with $(I/2)\otimes\tau \le \lambda\chi$, and $I^{\infty}_{\rm doe} := \lim_{n\to\infty} I_{\rm doe}(\mathcal{T}_{\mathcal{D}}^{\otimes n})/n$ is its regularization. The Choi state of the diagonal channel (J1) is itself diagonal, $\chi = \frac{1}{2}\sum_{x,y} B_{\mathcal{D}}(y \mid x)\,|xy\rangle\langle xy|$, so both dominations reduce to scalar inequalities on the transition probabilities, saturated by diagonal σ and τ with entries proportional to $\max_x B_{\mathcal{D}}(y \mid x)$ and $\min_x B_{\mathcal{D}}(y \mid x)$:

$$\begin{aligned} 2^{I_{\max}(\mathcal{T}_{\mathcal{D}})} &= \sum_y \max_x B_{\mathcal{D}}(y \mid x) = 1+\mathcal{D}, \\ 2^{I_{\rm doe}(\mathcal{T}_{\mathcal{D}})} &= \left[\sum_y \min_x B_{\mathcal{D}}(y \mid x)\right]^{-1} = \frac{1}{1-\mathcal{D}}. \end{aligned} \tag{J3}$$

For product channels the minima factorize, $\sum_{y_1 y_2} \min_{x_1 x_2} B_{\mathcal{D}}(y_1 \mid x_1)\, B_{\mathcal{D}}(y_2 \mid x_2) = \big[\sum_y \min_x B_{\mathcal{D}}(y \mid x)\big]^2$, so the Doeblin information is additive for this channel and equals its regularization. Hence

$$C_{\rm retro}(\mathcal{T}_{\mathcal{D}}) = 2Q_{\rm retro}(\mathcal{T}_{\mathcal{D}}) = \log_2\frac{1+\mathcal{D}}{1-\mathcal{D}} = \log_2\Gamma, \tag{J4}$$

which proves Eq. (22). This is the retrocausal capacity of $\mathcal{T}_{\mathcal{D}}$ in the sense of Ji, Lloyd, and Wilde. It differs from the ordinary forward Shannon capacity of the same binary symmetric channel, $1 - h_2((1-\mathcal{D})/2)$, where h_2 is the binary entropy: the retrocausal capacity can exceed one bit and diverges as $\Gamma \to \infty$. The distinction is operational. $C_{\rm retro}$ treats $\mathcal{T}_{\mathcal{D}}$ as the communication resource, with noiseless forward-in-time quantum memory supplied free, and quotes the asymptotic rate per use. A chronology-respecting heralded simulation must account for the success probability of the full encoding and decoding protocol. The protocol's forward memory can correlate the later input with earlier records, so the independent-preparation acceptance probability $p_{\mathrm{tot}}/2$ does not supply a rate conversion. Although $\mathcal{T}_{\mathcal{D}}$ is entanglement-breaking as an ordinary forward channel, its retrocausal quantum capacity is positive for $\mathcal{D} > 0$; the protocol of Ref. [38] allows a noiseless forward-in-time quantum memory to close the causal loop. Thus, when $\mathcal{T}_{\mathcal{D}}$ is taken as the future-to-past channel resource of Ref. [38], Eq. (22) makes Γ its capacity parameter, independently of the payoff interpretation.

2. Accepted-slice process matrix

Scope of the check. This subsection first embeds the accepted, basis-diagonal slice at the symmetric preparation $\alpha = \beta = 1/2$, and then notes that the same common-cause embedding extends, at fixed preparation, to the full accepted state including its local and joint coherences. Neither result is a claim that the complete postselected mechanism itself admits a process-matrix representation.

Process matrices assign probabilities to local quantum operations without presupposing a fixed global causal order [102, 103]. A bipartite process matrix is called causally separable when it can be written as a probabilistic mixture of process matrices with definite causal order. Their linear relation to a special class of postselected-CTC computations is discussed in Ref. [104]. The closest classical consistency-based comparison is the Baumeler–Wolf framework of logically consistent classical processes [105]. Here we ask only whether the accepted Newcomb correlation itself requires causal nonseparability.

For the present filter, the trace-decreasing numerator $\rho \mapsto K\rho K^\dagger$ is linear, while the normalized postselected update $\rho \mapsto K\rho K^\dagger/\operatorname{Tr}(K\rho K^\dagger)$ is linear only on preparation families for which the acceptance probability is constant. By Theorem 1(i), this constancy holds on the diagonal preparation family used for the payoff comparison when $\beta = 1/2$. The process matrix below implements the narrower check described above: it embeds the accepted diagonal slice after success has been conditioned on. The calculation shows that this payoff-level correlation, considered as a diagonal distribution, has a causally separable bipartite embedding.

The uniform-marginal member of this slice is the symmetric preparation $\alpha = \beta = 1/2$. Before conditioning, the four branches (x, y) have equal weight; the heralding event then distinguishes only the agreement and disagreement sectors. The accepted ensemble therefore has total agreement probability equal to the accepted-record accuracy of Eq. (20),

$$\begin{aligned} q &= \Pr(X = Y \mid \mathsf{A}) \\ &= \frac{p_{\mathrm{match}}}{p_{\mathrm{match}} + p_{\mathrm{mismatch}}} = \frac{\Gamma}{\Gamma + 1}, \\ \mathcal{D} &= 2q - 1 = \frac{\Gamma - 1}{\Gamma + 1}. \end{aligned} \tag{J5}$$

The two agreement branches share the total weight q equally, and the two disagreement branches share the total weight $1 - q$ equally. With $z_C = +1$ and $z_D = -1$, this four-cell distribution is equivalently

$$\begin{aligned} \Pr(x, y \mid \mathsf{A}) &= \tfrac{1}{4}\big(1 + \mathcal{D} z_x z_y\big), \\ z_C = +1, &\qquad z_D = -1. \end{aligned} \tag{J6}$$

The absence of terms proportional to z_x or z_y records the symmetric preparation: after conditioning, both one-register marginals remain uniform, and the only diagonal statistic is the forecast–act correlation. For $\beta = 1/2$ but $\alpha \neq 1/2$, the accepted distribution is still diagonal and causally separable, but its one-register marginals are not uniform; Eq. (J6) is the symmetric point used to test whether the Newcomb correlation itself carries causal-order structure.

Embed Eq. (J6) as a bipartite process with qubit input spaces A_I, B_I and unused qubit output spaces A_O, B_O:

$$\begin{aligned} W_{\mathrm{acc}} &= \tfrac{1}{4}\big(I + \mathcal{D}\, \sigma_z^{A_I} \otimes \sigma_z^{B_I}\big) \otimes I^{A_O B_O}, \\ \mathcal{D} &= \frac{\Gamma - 1}{\Gamma + 1}. \end{aligned} \tag{J7}$$

In this reduced embedding the output factors are spectators, included to use the standard process-matrix normalization. Equation (J7) stores the accepted-record distribution. For computational-basis readouts take the local instrument elements

$$M_A^x = \Pi_x^{A_I} \otimes \frac{I^{A_O}}{2}, \qquad M_B^y = \Pi_y^{B_I} \otimes \frac{I^{B_O}}{2}, \tag{J8}$$

with $\Pi_x = (I + z_x \sigma_z)/2$. The process-matrix Born rule gives

$$\operatorname{Tr}[W_{\mathrm{acc}}(M_A^x \otimes M_B^y)] = \tfrac{1}{4}\big(1 + \mathcal{D} z_x z_y\big), \tag{J9}$$

and therefore

$$\begin{aligned} \Pr(X = Y) &= \sum_x \operatorname{Tr}[W_{\mathrm{acc}}(M_A^x \otimes M_B^x)] \\ &= \frac{1 + \mathcal{D}}{2} = \frac{\Gamma}{\Gamma + 1} = q. \end{aligned} \tag{J10}$$

Equation (J7) is a valid process matrix. It is positive, from $|\mathcal{D}| \leq 1$, and its contraction with every pair of local CPTP Choi operators M^A, M^B is normalized to one: $\operatorname{Tr}_{A_O} M^A = I_{A_I}$ and $\operatorname{Tr}_{B_O} M^B = I_{B_I}$ reduce the contraction to the trace of the shared state in Eq. (J7), which is one. Being that shared state tensored with $I^{A_O B_O}$, W_{acc} is non-signalling and compatible with both fixed orders $A \prec B$ and $B \prec A$, hence causally separable: a common-cause embedding of the accepted diagonal statistics, with no remaining dependence on either party's output. This statement concerns the post-heralding slice. The physical route producing that slice is the trace-decreasing filter itself: in the laboratory reading the association is generated by conditioning on A, while in the cyclic P-CTC reading the same structure is read as final-boundary influence on the earlier forecast register. The Pauli support on the input spaces is only $I \otimes I$ and $\sigma_z \otimes \sigma_z$, the diagonal correlation of a common cause. Within this reduced embedding, the Newcomb payoff comparison uses only that common-cause record correlation.

Quantum coherence of the accepted state does not by itself imply causal nonseparability. At any fixed preparation for which the accepted state ρ_{acc} is defined, the full post-heralding state, including its local and joint coherences, has the common-cause process embedding

$$W_{\rho_{\mathrm{acc}}} = \rho_{\mathrm{acc}}^{A_I B_I} \otimes I^{A_O B_O}.$$

Because the output spaces are spectators, this process is non-signalling and compatible with both fixed orders $A \prec B$ and $B \prec A$, hence causally separable. Equation (J7) is the basis-diagonal member used for the Newcomb payoff statistics. The obstruction arises instead for the complete normalized postselected transformation when its success probability depends on the local operations.

Equation (J7) is compatible with the recent parity-erasure characterization of valid process matrices [56]. The parity erased there is the collective parity of the later settings, whereas the record parity $x \oplus y$ of Sec. II is the agreement variable selected by the heralding event. In the reduced W above, the output systems are spectators, so no later settings act through them; the record parity remains as an ordinary input–input correlation in the accepted records. This post-heralding shared-state process is valid and causally separable for every Γ.

3. Parity erasure and temporal order

Liu and Oreshkov formulate the constraint that local causality places on multipartite correlations, ordered or not, as a principle they name *parity erasure* [56]. Each of N laboratories applies an earlier instrument, with its own setting and outcome, and then a later one; the sequence is causal within each laboratory, so the earlier outcome may inform the choice of the later setting while the later setting cannot influence the earlier outcome. Their first theorem states that, for such locally sequential statistics with the later settings uniformly random bits, the joint earlier outcomes of every nonempty subset of laboratories are independent of the parity of that subset's own later settings, for every choice of the earlier settings. A single later setting may still show in another laboratory's earlier outcome, since a process matrix may route one laboratory's later output into another's earlier input; the parity over a subset includes each member's own later setting, and that combination is what the theorem forbids. Their second theorem states that an N-partite channel corresponds to a process matrix exactly when its local-process-tomography representation satisfies parity erasure.

To put the three-record construction of Sec. VII A into the locally sequential notation, let $s_X, s_Y \in \{0,1\}$ be independently and uniformly chosen computational-basis settings, preparing $X = s_X$ and $Y = s_Y$. Before conditioning on the retained event, let $H = 1$ denote A and $H = 0$ its complementary branch. To match the two-laboratory theorem, take each laboratory's earlier outcome to be a copy of the common branch record H, with no nontrivial earlier setting; the later settings are s_X and s_Y. Parity erasure is applied to the normalized behavior $\Pr(H \mid s_X, s_Y)$. This is distinct from the accepted-record channel $\mathcal{T}_{\mathcal{D}} : X \to Y$ of Appendix J 1 and the accepted common-cause process of Appendix J 2.

Completing the two acceptance probabilities with their rejection complements gives the normalized conditional distribution

$$\begin{aligned} \Pr(H = 1 \mid s_X, s_Y) &= \begin{cases} p_{\mathrm{match}}, & s_X \oplus s_Y = 0, \\ p_{\mathrm{mismatch}}, & s_X \oplus s_Y = 1, \end{cases} \\ \Pr(H = 0 \mid s_X, s_Y) &= 1 - \Pr(H = 1 \mid s_X, s_Y). \end{aligned} \tag{J11}$$

The calculation below uses only the necessary direction supplied by their first theorem, and only for the classical diagonal distribution of Eq. (J11). Parity erasure is applied to this normalized distribution. The trace-decreasing map $\rho \mapsto K\rho K^\dagger$ and conditioning on $H = 1$ are treated separately.

For either $u \in \{0,1\}$, uniformity of the other setting gives

$$\begin{aligned} \Pr(H = 1 \mid s_X = u) &= \frac{p_{\mathrm{match}} + p_{\mathrm{mismatch}}}{2}, \\ \Pr(H = 1 \mid s_Y = u) &= \frac{p_{\mathrm{match}} + p_{\mathrm{mismatch}}}{2}. \end{aligned} \tag{J12}$$

Each single-laboratory parity-erasure condition therefore holds for every $p_{\mathrm{match}}, p_{\mathrm{mismatch}}$: H carries no information about either uniformly random setting separately. The nontrivial condition is their collective parity. For the two-laboratory subset,

$$\begin{aligned} \Pr(H = 1 \mid s_X \oplus s_Y = 0) &= p_{\mathrm{match}}, \\ \Pr(H = 1 \mid s_X \oplus s_Y = 1) &= p_{\mathrm{mismatch}}. \end{aligned} \tag{J13}$$

Because both laboratories receive the same value of H, their joint earlier outcome is independent of the collective later parity if and only if $p_{\mathrm{match}} = p_{\mathrm{mismatch}}$. The complementary probabilities show the same equivalence for $H = 0$. For the averaged filter, set $p_{\mathrm{match}} = \bar{p}_{\mathrm{match}}$ and $p_{\mathrm{mismatch}} = \bar{p}_{\mathrm{mismatch}}$. Its condition is

$$\bar{p}_{\mathrm{match}} = \bar{p}_{\mathrm{mismatch}} \iff \Gamma_{\mathrm{eff}} = 1. \tag{J14}$$

This independence is the two-laboratory parity-erasure condition of the first theorem above. When $\Gamma_{\mathrm{eff}} \neq 1$, Eq. (J11) makes the probability of the assigned earlier outcome H depend on the collective parity of the two later settings. It therefore violates the necessary condition and lies outside the locally sequential behaviors the theorem covers. When $\Gamma_{\mathrm{eff}} = 1$, the normalized distribution admits a causally ordered realization in which H is sampled before the two later settings.

These statements compare temporal slicings of the same three-record statistics. With the herald placed after the decisions, the ordinary forward circuit is a valid process matrix at every Γ. With H assigned earlier than both later settings, Eq. (J11) violates parity erasure whenever $p_{\text{match}} \neq p_{\text{mismatch}}$. Conditioning on acceptance is a separate step: it produces a nonlinear P-CTC update when the acceptance probability depends on the input. Such an update cannot be represented by a process matrix valid for arbitrary local preparations [102, 104]. At $\mathcal{D}_{\text{eff}} = 0$, $E_{\text{eff}} = I/2$, and normalizing Eq. (50) gives an ordinary quantum channel. The operator identity of Sec. II is what lets these temporal orderings be discussed for the same selected construction. Parity erasure determines whether its normalized three-record statistics are compatible with the chosen locally sequential ordering. Under the Q_{post} hypothesis of Sec. I, where the selected final boundary is supplied by nature, the normalized earlier-outcome assignment at $\Gamma_{\text{eff}} > 1$ would describe a resource outside the process-matrix class, and $\mathcal{D}_{\text{eff}} = (\Gamma_{\text{eff}} - 1)/(\Gamma_{\text{eff}} + 1)$ is the normalized contrast between the herald probabilities for the two setting parities. Ref. [57] shows that, without postselection and under its closed-laboratory assumptions, protocols in a classical acyclic spacetime are behaviourally equivalent to quantum-controlled causal-order circuits. A nature-supplied parity-biasing postselection of the Q_{post} hypothesis is therefore not contained in the regime characterized by that theorem.

This conclusion concerns the normalized three-record statistics. The reduced accepted slice in Eq. (J7) remains a valid common-cause process for every Γ. A direction of time is not fixed by the selected operator: being real and diagonal, it is unchanged by transpose or adjoint, with register exchange a further symmetry (Sec. V). The forecast-before-act order is supplied by Newcomb's premise and the chronology-respecting laboratory. At $\Gamma_{\text{eff}} = 1$, the H-before-X, Y slicing becomes compatible with the locally sequential criterion.

4. Causal reorientations of the herald tensor

The tensor-network-to-causal-model mapping of Ferradini et al. [62] turns an undirected tensor network into a causal model by assigning a direction to each edge and reading a vertex as a channel from its input legs to its output legs. For that vertex to represent a normalized channel, the total output probability must equal one for every input. In the normalized Choi representation, this is equivalent to tracing over the output legs and obtaining the normalized identity on the input legs. If that condition fails, the generalized mapping adds a self-cycle, an edge from the vertex back to itself that represents local feedback.

The tensor reoriented here is obtained from the Bell-boundary instrument before conditioning on A. Appendix A shows that, for $\mathcal{U}_\theta$, the two nonzero Bell branches give $K(\theta)$ and $iK_\perp(\theta)$, with

$$K^\dagger K + K_\perp^\dagger K_\perp = I. \tag{J15}$$

Writing $H = 1$ for the retained branch and $H = 0$ for the complementary branch therefore gives a normalized binary branch record. For the averaged filter, $\mathcal{D}_{\text{eff}} = \bar{p}_{\text{match}} - \bar{p}_{\text{mismatch}}$, and

$$\Pr(H = h \mid s_X, s_Y) = \tfrac{1}{2}\big[1 - \mathcal{D}_{\text{eff}}\,(-1)^{s_X + s_Y + h}\big]. \tag{J16}$$

For uniformly weighted computational-basis settings, the normalized diagonal Choi tensor left after contracting the internal Bell legs and tracing out the outgoing X, Y registers is

$$\chi_H = \tfrac{1}{8}\big(I - \mathcal{D}_{\text{eff}}\,\sigma_z \otimes \sigma_z \otimes \sigma_z\big). \tag{J17}$$

Here the equal weighting of the four (s_X, s_Y) input basis states is the weighting used to form the normalized Choi tensor; the parity-erasure argument above separately assumes uniformly sampled later settings. On a basis state $|h, s_X, s_Y\rangle$, the operator $\sigma_z \otimes \sigma_z \otimes \sigma_z$ has eigenvalue $(-1)^{h+s_X+s_Y}$, so the second term distinguishes the even and odd joint parity of the three bits. Its coefficient $\mathcal{D}_{\text{eff}}$ is the corresponding parity bias. The coherence coefficient g of Eq. (52) is absent: this herald tensor is fixed by the two acceptance probabilities alone.

The tensor networks in this mapping are closed, with no open external legs. To place Eq. (J17) in that setting, put it at the center of a three-edge star and attach a maximally mixed tensor $I/2$ to each leaf. Each leaf satisfies the normalization condition with its edge directed either way, so the nontrivial check is at the central tensor.

For this contracted three-leg tensor, it is useful to count the eight assignments by the number of input legs at the center. There is one with no inputs, three with one input, three with two inputs, and one with all three inputs. The first seven all have at least one output. For each of them, the channel-normalization test traces over at least one factor of σ_z in Eq. (J17). Since $\operatorname{Tr}\sigma_z = 0$, that partial trace sets the parity-dependent second term to zero, while the identity term gives the required normalized identity on the remaining inputs. The zero-input case simply prepares the state χ_H, whose trace is one. Representatives of the two-input and one-input channel classes are

$$\begin{aligned} \mathcal{E}_{H|XY}(\rho) &= \tfrac{1}{2}\big[I - \mathcal{D}_{\text{eff}}\operatorname{Tr}(\rho\,\sigma_z \otimes \sigma_z)\,\sigma_z\big], \\ \mathcal{E}_{XY|H}(\rho) &= \tfrac{1}{4}\big[I - \mathcal{D}_{\text{eff}}\operatorname{Tr}(\rho\,\sigma_z)\,\sigma_z \otimes \sigma_z\big]. \end{aligned} \tag{J18}$$

The first line takes X, Y as inputs and H as the output: it is the parity-herald map used in Sec. VII A. Each reorientation changes which records are freely supplied as inputs and which are generated as outputs.[13] The other two assignments in this two-input/one-output class are obtained by choosing X or Y as the output instead. The second line takes H as the input and X, Y as outputs; choosing X or Y as the sole input gives the other two

[13] As per the physicist's wager of Sec. I, we shall not discuss free will in this paper.

members of the one-input/two-output class. In that class each single-output marginal is $I/2$, while the joint output parity depends on the input.

The only assignment left is the all-input case. It has no output leg, so no partial trace is taken. The channel-normalization condition is applied directly to χ_H and requires $\chi_H = I/8$, which by Eq. (J17) holds exactly when

$$\mathcal{D}_{\text{eff}} = 0 \quad \Longleftrightarrow \quad \Gamma_{\text{eff}} = 1. \tag{J19}$$

For $\mathcal{D}_{\text{eff}} \neq 0$, this is the only assignment of the contracted H, X, Y tensor for which the generalized mapping introduces a local feedback loop. The reason is explicit from Eq. (J17):

$$\chi_H - \frac{1}{8}I = -\frac{\mathcal{D}_{\text{eff}}}{8}\,\sigma_z \otimes \sigma_z \otimes \sigma_z. \tag{J20}$$

At $\mathcal{D}_{\text{eff}} = 0$, this parity-dependent term vanishes and the required equality $\chi_H = I/8$ holds, so the all-input orientation is directly normalized without the local self-cycle. The other seven orientations are directly normalized for every Γ_{eff}.

Conditioning χ_H on $H = 1$ and renormalizing gives the selected basis-record distribution. The full quantum update is obtained by normalizing Eq. (50). For $\mathcal{D}_{\text{eff}} \neq 0$, the all-input reorientation introduces a self-cycle under the generalized mapping. Contracting that added cycle recovers χ_H up to a positive factor [62]. After normalization, the cyclic model therefore reproduces the same three-record tensor. This self-cycle is introduced after the internal Bell legs have been contracted. At $\mathcal{D}_{\text{eff}} = 0$, $\chi_H = I/8$, the all-input orientation is directly normalized, and no self-cycle is introduced.

Appendix K: Epistemic coarse-graining, decision theory, and relational readings

The same formalism admits an operational epistemic reading in which postselection summarizes a consistency update over possible causal routes, a coarse-graining onto the two public records rather than a ψ-epistemic account of the state. Such epistemic "time travel" can be done on a computer or in imagination; laboratory postselection realizes it with a herald, while the P-CTC reading has nature supply the final boundary. The predictor's internal model may contain hidden variables, memories, simulations, and self-referential hypotheses, while the experiment exposes only the binary forecast $Y \in \{C, D\}$ and later act $X \in \{C, D\}$. If λ labels the joint internal routes of a run and $\pi(\lambda) = (x, y)$ coarse-grains them to these records, the payoff-diagonal update keeps only their total agreement and mismatch weights and reweights them by $\Gamma = p_{\text{match}}/p_{\text{mismatch}}$. This ratio fixes the accepted reliability at the unbiased prior without describing the routes. Different predictor mechanisms with the same record preparation and Γ are therefore indistinguishable on the normalized payoff diagonal, although their absolute acceptance rates may differ. This reduction requires the record-preserving class of Sec. II E; outside it, a map may distinguish records within a sector or rewrite a basis record after acceptance, and the payoff need no longer reduce to Γ.

In an epistemic reading, coherence should not be identified simply with ignorance about a hidden classical act. More cautiously, it can be read as the phase-sensitive part of a model in which the act has not yet been externalized as a public fact. A diagonal mixture over C and D describes uncertainty about an already classical alternative; a coherent superposition describes an unresolved act whose alternatives remain phase-related before externalization. Thus even a predictor with perfect knowledge of all settled classical facts need not have a determinate act-value to read off before the act is made public.

A decision theory is an additional input. Ahmed defends the evidential use of an act's information about the world, whereas Mackie makes the direction of causation central to the Newcomb dispute [4, 5]. Causal decision theory evaluates the consequences of intervening on the act and retains the familiar dominance motivation for two-boxing [7]. Functional decision theory instead treats the decision as the output of a mathematical procedure whose correlated instances are varied together and is designed to one-box in Newcomb's problem [8]; predictor-responsive cases such as Death in Damascus show why merely choosing between evidential and causal conditioning may be insufficient [9]. The physical calculation takes no side in that dispute: given a selected map and a scoring rule, it computes the reliability at which one-boxing overtakes two-boxing.

The postselected representation of Newcomb's problem keeps the dominance argument entrywise, two-boxing adding m against every fixed forecast, and keeps the predictive correlation as a property of the accepted ensemble, $q = \Gamma/(\Gamma + 1)$ at the unbiased prior. It does not keep the predictor's act of inference, which selection replaces. A causal decision theorist can therefore object that the agent chooses a preparation whose value is compared across postselected ensembles, a choice between protocols rather than between acts inside one fixed problem. Under Q_{post} the postselection would be supplied by nature rather than chosen by an experimenter; Sec. VIII B states the additional physical hypothesis required for that reading.

If facts are taken relationally, one may instead regard Y as a fact relative to the predictor and X as a later fact relative to the agent, with the parity filter constraining only the relation between their exposed records [80–82]. The construction does not require an observer-relative interpretation, impose cross-perspective links, or address the Wigner-friend disputes surrounding them [83–89].

[1] R. Nozick, "Newcomb's problem and two principles of choice," in *Essays in Honor of Carl G. Hempel*, edited by N. Rescher (D. Reidel, 1969), pp. 114–146.

[2] M. Gardner, "Free will revisited, with a mind-bending prediction paradox by William Newcomb," Mathematical Games, *Scientific American* **229**(1), 104 (July 1973); https://doi.org/10.1038/scientificamerican0773-104.

[3] M. Gardner, "Reflections on Newcomb's problem," Mathematical Games, *Scientific American* **230**(3), 102 (March 1974); https://doi.org/10.1038/scientificamerican0374-102.

[4] A. Ahmed, *Evidence, Decision and Causality* (Cambridge University Press, 2014); https://doi.org/10.1017/CBO9781139107990.

[5] J. L. Mackie, "Newcomb's paradox and the direction of causation," *Canadian Journal of Philosophy* **7**(2), 213–225 (1977); https://doi.org/10.1080/00455091.1977.10717014.

[6] A. Gibbard and W. L. Harper, "Counterfactuals and two kinds of expected utility," in *Foundations and Applications of Decision Theory*, Vol. 1, edited by C. A. Hooker, J. J. Leach, and E. F. McClennen (D. Reidel, 1978), pp. 125–162.

[7] P. Weirich, "Causal Decision Theory," in *The Stanford Encyclopedia of Philosophy* (Winter 2024 edition), edited by E. N. Zalta and U. Nodelman; https://plato.stanford.edu/archives/win2024/entries/decision-causal/.

[8] E. Yudkowsky and N. Soares, "Functional Decision Theory," arXiv:1710.05060 (2017).

[9] B. A. Levinstein and N. Soares, "Cheating Death in Damascus," *The Journal of Philosophy* **117**(5), 237–266 (2020).

[10] D. H. Wolpert and G. Benford, "The lesson of Newcomb's paradox," *Synthese* **190**, 1637–1646 (2013); https://doi.org/10.1007/s11229-011-9899-3.

[11] D. K. Lewis, "Prisoners' Dilemma Is a Newcomb Problem," *Philosophy & Public Affairs* **8**(3), 235–240 (1979).

[12] J. H. Sobel, "Some versions of Newcomb's problem are prisoners' dilemmas," *Synthese* **86**(2), 197–208 (1991); https://doi.org/10.1007/BF00485808.

[13] J. L. Bermúdez, "Prisoner's dilemma and Newcomb's problem: why Lewis's argument fails," *Analysis* **73**(3), 423–429 (2013); https://doi.org/10.1093/analys/ant034.

[14] J. H. Schmidt, "Newcomb's paradox realized with backward causation," *Br. J. Philos. Sci.* **49**, 67–85 (1998); https://doi.org/10.1093/bjps/49.1.67.

[15] P. Slezak, "Demons, deceivers and liars: Newcomb's *malin génie*," *Theory Decis.* **61**, 277–303 (2006); https://doi.org/10.1007/s11238-006-9002-6.

[16] C. Schmidt-Petri, "Newcomb's problem and repeated prisoners' dilemmas," *Philosophy of Science* **72**(5), 1160–1173 (2005); https://doi.org/10.1086/508962.

[17] G. A. Benford, D. L. Book, and W. A. Newcomb, "The tachyonic antitelephone," *Phys. Rev. D* **2**, 263–265 (1970); https://doi.org/10.1103/PhysRevD.2.263.

[18] D. Deutsch, "Quantum mechanics near closed timelike lines," *Phys. Rev. D* **44**, 3197 (1991); https://doi.org/10.1103/PhysRevD.44.3197.

[19] C. H. Bennett *et al.*, "Can closed timelike curves or nonlinear quantum mechanics improve quantum state discrimination or help solve hard problems?" *Phys. Rev. Lett.* **103**, 170502 (2009); https://doi.org/10.1103/PhysRevLett.103.170502.

[20] C. H. Bennett and B. Schumacher, "Simulated time travel, teleportation and quantum computers," unpublished lectures (2002 and 2005).

[21] G. Svetlichny, "Time travel: Deutsch vs. teleportation," *Int. J. Theor. Phys.* **50**, 3903–3914 (2011); arXiv:0902.4898; https://doi.org/10.1007/s10773-011-0973-x.

[22] S. Lloyd *et al.*, "Closed timelike curves via postselection: theory and experimental test of consistency," *Phys. Rev. Lett.* **106**, 040403 (2011); arXiv:1005.2219; https://doi.org/10.1103/PhysRevLett.106.040403.

[23] R. P. Feynman, "The theory of positrons," *Phys. Rev.* **76**, 749–759 (1949); https://doi.org/10.1103/PhysRev.76.749.

[24] S. Lloyd, L. Maccone, R. Garcia-Patron, V. Giovannetti, and Y. Shikano, "Quantum mechanics of time travel through post-selected teleportation," *Phys. Rev. D* **84**, 025007 (2011); arXiv:1007.2615; https://doi.org/10.1103/PhysRevD.84.025007.

[25] K. Singh, "US acting spy chief announces new round of staff reductions," *Reuters*, 11 July 2026, updated 13 July 2026, online.

[26] S. Kleeman, "Donald Trump reviews Tesla Model S: 'Everything's computer'," *Business Insider*, 11 March 2025, online.

[27] S. Lloyd, "The universe as quantum computer," in *A Computable Universe: Understanding and Exploring Nature as Computation*, edited by H. Zenil (World Scientific, Singapore, 2012); arXiv:1312.4455.

[28] D. Deutsch, "Quantum theory, the Church–Turing principle and the universal quantum computer," *Proc. R. Soc. Lond. A* **400**, 97–117 (1985); https://doi.org/10.1098/rspa.1985.0070.

[29] N. Bostrom, "Are you living in a computer simulation?" *Philosophical Quarterly* **53**(211), 243–255 (2003); https://doi.org/10.1111/1467-9213.00309.

[30] S. Aaronson, "Quantum computing, postselection, and probabilistic polynomial-time," *Proc. R. Soc. A* **461**, 3473–3482 (2005); arXiv:quant-ph/0412187; https://doi.org/10.1098/rspa.2005.1546.

[31] J.-M. A. Allen, "Treating time travel quantum mechanically," *Phys. Rev. A* **90**, 042107 (2014); https://doi.org/10.1103/PhysRevA.90.042107.

[32] T. A. Brun and M. M. Wilde, "Simulations of closed timelike curves," *Found. Phys.* **47**, 375–391 (2017); https://doi.org/10.1007/s10701-017-0066-7.

[33] T. A. Brun and M. M. Wilde, "Perfect state distinguishability and computational speedups with postselected closed timelike curves," *Found. Phys.* **42**, 341–361 (2012); https://doi.org/10.1007/s10701-011-9601-0.

[34] J. Gill, "Computational complexity of probabilistic Turing machines," *SIAM J. Comput.* **6**, 675–695 (1977); https://doi.org/10.1137/0206049.

[35] E. Bernstein and U. Vazirani, "Quantum complexity theory," *SIAM J. Comput.* **26**, 1411–1473 (1997); https://doi.org/10.1137/S0097539796300921.

[36] L. M. Adleman, J. DeMarrais, and M.-D. A. Huang, “Quantum computability,” *SIAM J. Comput.* **26**, 1524–1540 (1997); https://doi.org/10.1137/S0097539795293639.

[37] A. Kezzyn (dir.), *Stag Hunt*, short film featuring Michele Reilly and Seth Lloyd (2023); video.

[38] K. Ji, S. Lloyd, and M. M. Wilde, “Retrocausal capacity of a quantum channel: Communicating through noisy closed timelike curves,” *Phys. Rev. Lett.* **136**, 230801 (2026); arXiv:2509.08965; https://doi.org/10.1103/znyd-npk5.

[39] J. F. Nash, “Equilibrium points in n-person games,” *Proc. Natl. Acad. Sci. USA* **36**, 48–49 (1950); https://doi.org/10.1073/pnas.36.1.48.

[40] A. Rapoport and A. M. Chammah, *Prisoner’s Dilemma: A Study in Conflict and Cooperation* (University of Michigan Press, Ann Arbor, 1965).

[41] B. Skyrms, *The Stag Hunt and the Evolution of Social Structure* (Cambridge University Press, 2004).

[42] J. C. Harsanyi and R. Selten, *A General Theory of Equilibrium Selection in Games* (MIT Press, 1988).

[43] D. Monderer and L. S. Shapley, “Potential games,” *Games Econ. Behav.* **14**, 124–143 (1996); https://doi.org/10.1006/game.1996.0044.

[44] P. D. Taylor and L. B. Jonker, “Evolutionary stable strategies and game dynamics,” *Math. Biosci.* **40**, 145–156 (1978); https://doi.org/10.1016/0025-5564(78)90077-9.

[45] T. Börgers and R. Sarin, “Learning through reinforcement and replicator dynamics,” *J. Econ. Theory* **77**, 1–14 (1997); https://doi.org/10.1006/jeth.1997.2319.

[46] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed. (Westview Press, Boulder, 2015).

[47] N. Goldenfeld, *Lectures on Phase Transitions and the Renormalization Group* (Addison–Wesley, Reading, MA, 1992).

[48] B.-G. Englert, “Fringe visibility and which-way information: an inequality,” *Phys. Rev. Lett.* **77**, 2154–2157 (1996); https://doi.org/10.1103/PhysRevLett.77.2154.

[49] D. M. Greenberger and A. YaSin, “Simultaneous wave and particle knowledge in a neutron interferometer,” *Phys. Lett. A* **128**, 391–394 (1988); https://doi.org/10.1016/0375-9601(88)90114-4.

[50] G. Jaeger, A. Shimony, and L. Vaidman, “Two interferometric complementarities,” *Phys. Rev. A* **51**, 54–67 (1995); https://doi.org/10.1103/PhysRevA.51.54.

[51] D. Genkina, G. Chiribella, and L. Hardy, “Optimal probabilistic simulation of quantum channels from the future to the past,” *Phys. Rev. A* **85**, 022330 (2012); https://doi.org/10.1103/PhysRevA.85.022330.

[52] H. Reichenbach, *The Direction of Time* (University of California Press, Berkeley, 1956).

[53] J.-M. A. Allen, J. Barrett, D. C. Horsman, C. M. Lee, and R. W. Spekkens, “Quantum common causes and quantum causal models,” *Phys. Rev. X* **7**, 031021 (2017); arXiv:1609.09487; https://doi.org/10.1103/PhysRevX.7.031021.

[54] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Rev. Mod. Phys.* **75**, 715–775 (2003); arXiv:quant-ph/0105127; https://doi.org/10.1103/RevModPhys.75.715.

[55] W. H. Zurek, “Quantum Darwinism,” *Nat. Phys.* **5**, 181–188 (2009); arXiv:0903.5082; https://doi.org/10.1038/nphys1202.

[56] Z. Liu and O. Oreshkov, “Parity erasure: a foundational principle for indefinite causal order,” arXiv:2512.08635v3 [quant-ph] (2025).

[57] M. Salzger and V. Vilasini, “Higher-order quantum processes respecting closed labs in a spacetime have quantum controlled causal order,” arXiv:2605.08351 (2026).

[58] V. Vilasini and R. Colbeck, “General framework for cyclic and fine-tuned causal models and their compatibility with space-time,” *Phys. Rev. A* **106**, 032204 (2022); arXiv:2109.12128; https://doi.org/10.1103/PhysRevA.106.032204.

[59] E. Badici, “Prisoner’s Dilemma and Newcomb’s Problem: Two Problems or One?,” *Philosophia* **51**, 2543–2557 (2023); https://doi.org/10.1007/s11406-023-00694-1.

[60] S. J. Brams, “Newcomb’s Problem and Prisoners’ Dilemma,” *Journal of Conflict Resolution* **19**, 596–612 (1975); https://doi.org/10.1177/002200277501900402.

[61] Z. Li and F. Zenker, “Newcomb’s problem isn’t a choice dilemma,” *Synthese* **199**, 5125–5143 (2021); https://doi.org/10.1007/s11229-020-03018-y.

[62] C. Ferradini, G. Mazzola, and V. Vilasini, “Emergent causal order and time direction: bridging causal models and tensor networks,” arXiv:2603.12283v1 [quant-ph] (2026).

[63] C. Ferradini, V. Gitton, and V. Vilasini, “Cyclic quantum causal modelling with a graph separation theorem,” arXiv:2502.04168v2 (2025).

[64] G. T. Horowitz and J. Maldacena, “The black hole final state,” *JHEP* **02** (2004) 008; arXiv:hep-th/0310281; https://doi.org/10.1088/1126-6708/2004/02/008.

[65] S. Lloyd and J. Preskill, “Unitarity of black hole evaporation in final-state projection models,” *JHEP* **08** (2014) 126; arXiv:1308.4209; https://doi.org/10.1007/JHEP08(2014)126.

[66] D. Gottesman and J. Preskill, “Comment on ‘The black hole final state’,” *JHEP* **03** (2004) 026; arXiv:hep-th/0311269; https://doi.org/10.1088/1126-6708/2004/03/026.

[67] K. Gödel, “An example of a new type of cosmological solutions of Einstein’s field equations of gravitation,” *Rev. Mod. Phys.* **21**, 447–450 (1949); https://doi.org/10.1103/RevModPhys.21.447.

[68] B. Carter, “Global Structure of the Kerr Family of Gravitational Fields,” *Phys. Rev.* **174**, 1559–1571 (1968); https://doi.org/10.1103/PhysRev.174.1559.

[69] M. S. Morris, K. S. Thorne, and U. Yurtsever, “Wormholes, time machines, and the weak energy condition,” *Phys. Rev. Lett.* **61**, 1446–1449 (1988); https://doi.org/10.1103/PhysRevLett.61.1446.

[70] J. R. Gott, “Closed timelike curves produced by pairs of moving cosmic strings: Exact solutions,” *Phys. Rev. Lett.* **66**, 1126–1129 (1991); https://doi.org/10.1103/PhysRevLett.66.1126.

[71] J. Friedman, M. S. Morris, I. D. Novikov, F. Echeverria, G. Klinkhammer, K. S. Thorne, and U. Yurtsever, “Cauchy problem in spacetimes with closed timelike curves,” *Phys. Rev. D* **42**, 1915–1930 (1990); https://doi.org/10.1103/PhysRevD.42.1915.

[72] F. Echeverria, G. Klinkhammer, and K. S. Thorne, “Billiard balls in wormhole spacetimes with closed timelike curves: Classical theory,” *Phys. Rev. D* **44**, 1077 (1991);

https://doi.org/10.1103/PhysRevD.44.1077.
[73] C. Rovelli, "Can we travel to the past? Irreversible physics along closed timelike curves," arXiv:1912.04702 (2019).
[74] H. Price, "Does time-symmetry imply retrocausality? How the quantum world says 'maybe'," *Stud. Hist. Phil. Mod. Phys.* **43**, 75–83 (2012); https://doi.org/10.1016/j.shpsb.2011.12.003.
[75] M. S. Leifer and M. F. Pusey, "Is a time symmetric interpretation of quantum theory possible without retrocausality?" *Proc. R. Soc. A* **473**, 20160607 (2017); arXiv:1607.07871; https://doi.org/10.1098/rspa.2016.0607.
[76] W. L. Craig, "Divine Foreknowledge and Newcomb's Paradox," *Philosophia* **17**, 331–350 (1987); https://doi.org/10.1007/BF02455055.
[77] S. W. Hawking, "Chronology protection conjecture," *Phys. Rev. D* **46**, 603–611 (1992); https://doi.org/10.1103/PhysRevD.46.603.
[78] J. L. Lebowitz, "Boltzmann's entropy and time's arrow," *Phys. Today* **46**(9), 32–38 (1993); https://doi.org/10.1063/1.881363.
[79] H. D. Zeh, *The Physical Basis of the Direction of Time*, 5th ed. (Springer, Berlin, 2007).
[80] C. Rovelli, "Relational quantum mechanics," *Int. J. Theor. Phys.* **35**, 1637–1678 (1996); arXiv:quant-ph/9609002; https://doi.org/10.1007/BF02302261.
[81] M. Smerlak and C. Rovelli, "Relational EPR," *Found. Phys.* **37**, 427–445 (2007); arXiv:quant-ph/0604064; https://doi.org/10.1007/s10701-007-9105-0.
[82] A. Di Biagio and C. Rovelli, "Stable facts, relative facts," *Found. Phys.* **51**, 30 (2021); arXiv:2006.15543; https://doi.org/10.1007/s10701-021-00429-w.
[83] Č. Brukner, "A no-go theorem for observer-independent facts," *Entropy* **20**, 350 (2018); arXiv:1804.00749; https://doi.org/10.3390/e20050350.
[84] D. Frauchiger and R. Renner, "Quantum theory cannot consistently describe the use of itself," *Nat. Commun.* **9**, 3711 (2018); https://doi.org/10.1038/s41467-018-05739-8.
[85] K.-W. Bong, A. Utreras-Alarcón, F. Ghafari, Y.-C. Liang, N. Tischler, E. G. Cavalcanti, G. J. Pryde, and H. M. Wiseman, "A strong no-go theorem on the Wigner's friend paradox," *Nat. Phys.* **16**, 1199–1205 (2020); https://doi.org/10.1038/s41567-020-0990-x.
[86] E. Adlam and C. Rovelli, "Information is physical: cross-perspective links in relational quantum mechanics," *Philosophy of Physics* **1**(1), 4 (2023); arXiv:2203.13342; https://doi.org/10.31389/pop.8.
[87] E. G. Cavalcanti, A. Di Biagio, and C. Rovelli, "On the consistency of relative facts," *Eur. J. Phil. Sci.* **13**, 55 (2023); arXiv:2305.07343; https://doi.org/10.1007/s13194-023-00551-8.
[88] J. Lawrence, M. Markiewicz, and M. Żukowski, "Relative facts of relational quantum mechanics are incompatible with quantum mechanics," *Quantum* **7**, 1015 (2023); https://doi.org/10.22331/q-2023-05-23-1015.
[89] M. Markiewicz and M. Żukowski, "Relational quantum mechanics with cross-perspective links postulate: an internally inconsistent scheme," arXiv:2312.07056 (2023).
[90] W. K. Wootters, "Entanglement of formation of an arbitrary state of two qubits," *Phys. Rev. Lett.* **80**, 2245–2248 (1998); https://doi.org/10.1103/PhysRevLett.80.2245.
[91] E. H. Simpson, "The interpretation of interaction in contingency tables," *Journal of the Royal Statistical Society B* **13**(2), 238–241 (1951); https://doi.org/10.1111/j.2517-6161.1951.tb00088.x.
[92] D. A. Meyer, "Quantum strategies," *Phys. Rev. Lett.* **82**, 1052–1055 (1999); https://doi.org/10.1103/PhysRevLett.82.1052.
[93] J. Eisert, M. Wilkens, and M. Lewenstein, "Quantum games and quantum strategies," *Phys. Rev. Lett.* **83**, 3077–3080 (1999); https://doi.org/10.1103/PhysRevLett.83.3077.
[94] E. W. Piotrowski and J. Sładkowski, "An invitation to quantum game theory," *Int. J. Theor. Phys.* **42**, 1089–1099 (2003); https://doi.org/10.1023/A:1025443111388.
[95] E. W. Piotrowski and J. Sładkowski, "Quantum solution to the Newcomb's paradox," arXiv:quant-ph/0202074 (2002).
[96] D. P. Chi and K. Jeong, "Quantum Solution to the Extended Newcomb's Paradox," arXiv:quant-ph/0511097 (2005).
[97] E. G. Cavalcanti, "Causation, decision theory, and Bell's theorem: a quantum analogue of the Newcomb problem," *Brit. J. Phil. Sci.* **61**, 569–597 (2010); arXiv:0911.2500; https://doi.org/10.1093/bjps/axp050.
[98] S. Aaronson and J. Watrous, "Closed timelike curves make quantum and classical computing equivalent," *Proc. R. Soc. A* **465**, 631–647 (2009); arXiv:0808.2669; https://doi.org/10.1098/rspa.2008.0350.
[99] T. A. Brun, J. Harrington, and M. M. Wilde, "Localized closed timelike curves can perfectly distinguish quantum states," *Phys. Rev. Lett.* **102**, 210402 (2009); https://doi.org/10.1103/PhysRevLett.102.210402.
[100] Y. Aharonov, P. G. Bergmann, and J. L. Lebowitz, "Time symmetry in the quantum process of measurement," *Phys. Rev.* **134**, B1410–B1416 (1964); https://doi.org/10.1103/PhysRev.134.B1410.
[101] Y. Aharonov, D. Z. Albert, and L. Vaidman, "How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100," *Phys. Rev. Lett.* **60**, 1351–1354 (1988); https://doi.org/10.1103/PhysRevLett.60.1351.
[102] O. Oreshkov, F. Costa, and Č. Brukner, "Quantum correlations with no causal order," *Nat. Commun.* **3**, 1092 (2012); https://doi.org/10.1038/ncomms2076.
[103] O. Oreshkov and C. Giarmatzi, "Causal and causally separable processes," *New J. Phys.* **18**, 093020 (2016); https://doi.org/10.1088/1367-2630/18/9/093020.
[104] M. Araújo, P. A. Guérin, and Ä. Baumeler, "Quantum computation with indefinite causal structures," *Phys. Rev. A* **96**, 052315 (2017); https://doi.org/10.1103/PhysRevA.96.052315.
[105] Ä. Baumeler and S. Wolf, "The space of logically consistent classical processes without causal order," *New J. Phys.* **18**, 013036 (2016); arXiv:1507.01714; https://doi.org/10.1088/1367-2630/18/1/013036.